

CHAPTER 10

Unit 6

Radiation Heat Transfer

Syllabus :

Radiation Heat Transfer : Introduction, Thermal radiation, Black body radiation, radiation laws, Radiation Properties, Atmospheric and Solar radiation, The view factor, Radiation heat transfer from black surfaces, gray surface, diffuse surface, Radiation shields and the radiation effect.

10.1 Thermal Radiation

So far we have discussed the heat transfer by conduction and convection which require material media for the process of energy transfer to occur.

The heat conduction in solids is due to lattice vibrations and movement of free electrons. Whereas the heat transfer in convection is due to conduction and fluid movement.

However the thermal radiation does not require any medium for transport of energy because every surface emits electromagnetic waves continuously in all directions due to its temperature.

These electromagnetic waves while propagating carry energy and transfer thermal energy when they impinge on a surface or body. This kind of energy transfer is known as **radiation**.

Definition of Thermal Radiation

The thermal radiation or radiation can be defined as the transfer of heat energy due to electromagnetic waves without requiring any medium.

10.2 Theories of Radiation

There are two accepted theory of radiation, namely :

1. Wave theory or Maxwell theory
2. Quantum theory or Planck's theory

Both the theories are used in thermal radiation.

10.2.1 Wave or Maxwell Theory

According to this theory, the electromagnetic emissions propagate in the form of waves. Since the waves propagate through some medium, this theory assumes that universe is filled with hypothetical medium known as **ether**.

There are many types of electromagnetic waves. All types of waves carry energy and travel at different wave lengths, λ and frequency, ν in a medium at velocity of light, C . It follows the relationship as :

$$\lambda = \frac{C}{\nu} \quad \dots(10.2.1)$$

The *velocity of light in vacuum* is

$$C_0 = C = 2.998 \times 10^8 \text{ m/s. } (\approx 3 \times 10^8 \text{ m/s}).$$

Wave length is specified in μm (micrometer)

where, $1 \mu = 10^{-6}$

or in Armstrong where $\text{\AA}$ (Armstrong) = 10^{-10} m

The spectrum of electromagnetic waves is shown in Fig. 10.2.1.

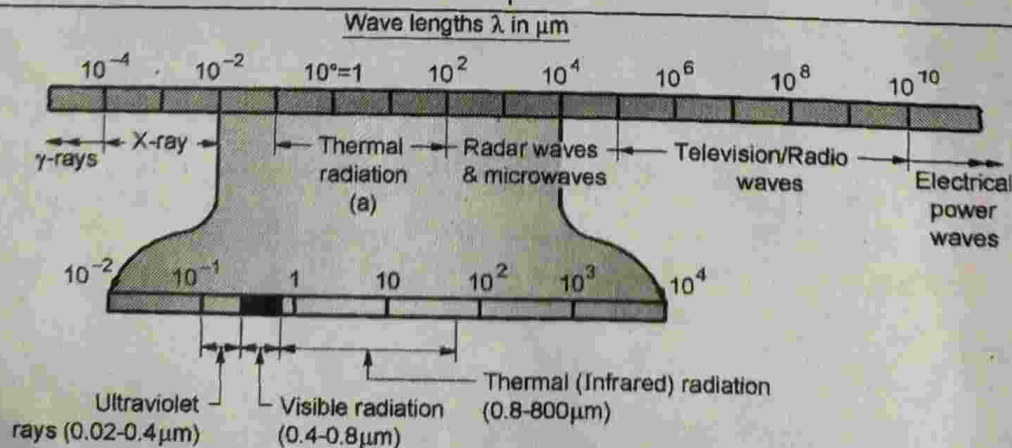


Fig. 10.2.1 : Spectrum of electromagnetic waves

The thermal radiation is considered in the wavelength of $0.1 \mu\text{m}$ to $100 \mu\text{m}$ and the amount of thermal radiation depends the characteristics of the surface and its temperature.

The thermal radiation increases with increase in temperature of a surface.

The visible portion of spectrum shown in Fig. 10.2.1 is the portion of electromagnetic waves which triggers the sensation of vision in human eye.

Short wave radiations like γ -rays and X-rays are associated with high energies with high powers of penetration.

10.2.2 Quantum Theory or Planck's Theory

This theory was developed by Max Planck. This theory is based on the concept that the radiation emissions are in the form of series of entities known as *quanta* or *photon*.

Each quanta possesses certain amount of energy which is proportional to its frequency of emission. These quanta move with the speed of light and releases its energy on impinging by some object in its route.

Thus, the **Planck's theory** states that the thermal radiation propagates in the form of discrete quanta, each quantum having an energy of,

$$E = h \cdot \nu \quad \dots(10.2.2)$$

where, **h** represents **Planck's constant** = 6.625×10^{-34} Js

ν represents frequency of photons in per second.

Using $\nu = C/\lambda$ from Equation (10.2.1), the Equation (10.2.2) reduces to

$$E = h \cdot \frac{C}{\lambda} \quad \dots(10.2.3)$$

It is evident from Equation (10.2.3) that the thermal energy, E is inversely proportional to wavelength, λ since h and C are constants in a medium. For this reason the short wave radiations have much higher photon energy as compared to long wave radiations.

10.3 Total Emissive Power of a Surface (E)

Definition : The total emissive power of a surface is defined as the total emissions of the thermal radiation by a surface per unit time per unit area in all directions into hemi-spherical half-space above the surface as shown in Fig. 10.3.1.

Consider a surface element of area, dA which emits an energy, dQ per unit time, the **emissive power**, E of the system is given as :

$$\text{Total emissive power, } E = \lim_{dA \rightarrow 0} \frac{dQ}{dA} \quad \dots(10.3.1)$$

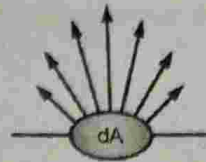


Fig. 10.3.1 : Surface emitting radiations in all direction

In order to get total emissions from the finite surface area, A per unit time, we can integrate the equation.

$$\text{Thus, } Q = \int_0^A E \cdot dA$$

The total emissive power of a surface depends upon the temperature, T of the surface and its material, surface finish etc.

10.4 Basic Concepts and Definitions

Out of the total irradiations, G or thermal radiations, Q received by a surface, a part of energy is reflected, another part of energy is absorbed and the remainder energy is transmitted across to the second medium as shown in Fig. 10.4.1.

Let : Q = Thermal radiations received by the surface

Q_r = Energy reflected from surface

Q_a = Energy absorbed by the surface

Q_t = Energy transmitted across the second medium.

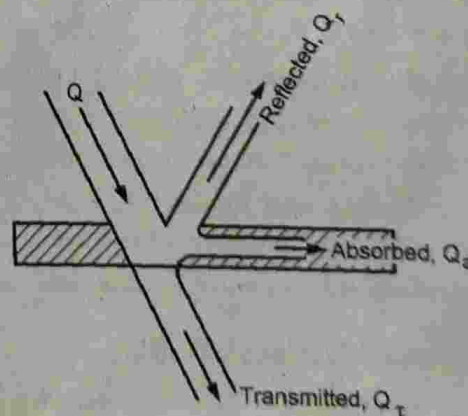


Fig. 10.4.1 : Reflection, absorption and transmission of thermal radiations

$$\text{By energy balance, } Q = Q_r + Q_a + Q_t \quad \dots(10.4.1)$$

10.4.1 Properties of Radiation

Therefore, we define the following radiation properties :

1. Reflectivity (ρ)
2. Absorptivity (α)
3. Transmissivity (τ)

1. Reflectivity (ρ)

Definition : Reflectivity is defined as the fraction of total incident radiations that are reflected by material.

$$\text{Reflectivity, } \rho = \frac{\text{Energy reflected, } Q_r}{\text{Total incident radiations, } Q} \quad \dots(10.4.2)$$

2. Absorptivity (α)

Definition : Absorptivity is defined as the fraction of total incident radiations that are absorbed by the material.

$$\text{Absorptivity, } \alpha = \frac{\text{Energy absorbed, } Q_a}{\text{Total incident radiations, } Q} \quad \dots(10.4.3)$$

3. Transmissivity (τ)

Transmissivity is defined as the fraction of total incident radiations that are transmitted through the material.

$$\text{Transmissivity, } \tau = \frac{\text{Energy transmitted, } Q_\tau}{\text{Total incident radiations, } Q} \quad \dots(10.4.4)$$

From law of conservation of energy,

$$\rho + \alpha + \tau = \frac{Q_r}{Q} + \frac{Q_a}{Q} + \frac{Q_\tau}{Q} = \frac{Q_r + Q_a + Q_\tau}{Q} = \frac{Q}{Q} = 1$$

$$\text{i.e. } \rho + \alpha + \tau = 1 \quad \dots(10.4.5)$$

The reflectivity, absorptivity and transmissivity of a body are dependent on the temperatures of the source radiation and the nature and preparation of surface.

In case of **solids and liquids**, the transmissivity is negligible ($\tau \approx 0$), hence $(\rho + \alpha) = 1$. Whereas, in case of **gases** the reflectivity is very small ($\rho \approx 0$), hence $\alpha + \tau = 1$.

10.4.2 Types of Surfaces/Bodies

Following types of surfaces are considered in thermal radiation

Types of Surfaces/Bodies

1. Black body
2. Opaque body
3. White body
4. Grey body
5. Transparent body

Fig. C10.1 : Types of Surfaces/Bodies

1. Black body

Definition : The body which absorbs entire radiations and emits maximum radiations is called a **black body**.

It implies that $\rho = \gamma = 0$ and $\alpha = 1$

A black body is also a perfect emitter or radiator of heat.

No perfect black surface has been found in nature. However, surface coated with lamp black, milk, ice, water, white paper are considered black body.

2. Opaque body

Definition : The bodies or surfaces which have negligible transmissivity of energy are called **opaque bodies** e.g. solid and liquid surfaces.

By opaque surfaces the radiant energy received is either absorbed or reflected i.e. $\rho + \alpha = 1$ and $\tau = 0$

3. White body

Definition : The surfaces which reflect the total radiant energy are called **white bodies**.

For such bodies $\alpha = \tau = 0$ and $\rho = 1$

4. Grey body

Definition : The body which has its absorptivity equal to its emissivity is called a **grey body** provided the temperature of the surface of incident radiation and the body are same.

Alternatively, the body having the same value of emissivity at all wave lengths, which is equal to the average emissivity is called as **Grey body** e.g. all metals.

5. Transparent body

The surface which transmits the entire radiations are called transparent surfaces.

In such a case : $\alpha = \rho = 0$ and $\tau = 1$.

10.4.3 Surface Emission Properties

Properties of a surface considering it as an emitter of radiation are as follows :

1. Emissive power (e)
2. Emissivity (ϵ)
3. Monochromatic emissive power (e_λ)
4. Monochromatic emissivity (ϵ_λ)
5. Radiosity (J)

1. Emissive power (e)

Definition : The radiant energy emitted per unit area per unit time from the surface of a body at a certain temperature is called the **emissive power**.

- It's units are $J/s\ m^2$ and W/m^2 or kW/m^2 .

The radiant energy/per unit area is called radiant flux.

- The emissive power of a black body will be denoted by e_b .

2. Emissivity (ϵ)

It represents the ratio of emissive power of a surface to the emissive power of a body at the same temperature. Therefore,

$$\text{Emissivity, } \epsilon = \frac{\text{Emissive power of a body/surface, } e}{\text{Emissive power of a black body, } e_b} \quad \dots(10.4.6)$$

The value of emissivity ranges between 0 to 1.

3. Monochromatic emissive power (e_λ)

Definition : Monochromatic emissive power is defined as the radiant energy emitted by the body per unit time and per unit area at particular wavelength and at a temperature.

Above definition applies that the monochromatic emissive power is wavelength (λ) dependent emissive power. Fig. 10.4.2 shows the variation of monochromatic emissive with wavelength at a temperature T .

The radiant energy emitted by the body per unit time per unit area when the wave length changes from λ to $(\lambda + d\lambda)$ is,

$$de = e_\lambda \cdot d\lambda$$

$\therefore$ Emissive power of the body,

$$e = \int_{\lambda=0}^{\lambda=\infty} e_\lambda \cdot d\lambda \quad \dots(10.4.7)$$

Monochromatic emissive power of a black body is denoted by $e_{b\lambda}$.

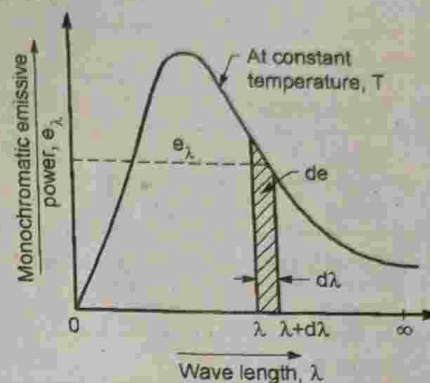


Fig. 10.4.2: Variation of monochromatic power of a body with λ at a given temperature, T

4. Monochromatic emissivity (ϵ_λ)

Definition : Monochromatic emissivity is defined as the ratio of monochromatic emissive power, e_λ of a surface to the monochromatic emissive power of a black body, $e_{b\lambda}$ for same wavelength and at same temperature.

Accordingly,

$$\text{Monochromatic emissivity, } \epsilon_\lambda = \left[\frac{\text{Monochromatic emissive power of a surface, } e_\lambda}{\text{Monochromatic emissive power of black body, } e_{b\lambda}} \right]_{T=\text{const}} \quad \dots(10.4.8)$$

Emissivity from surfaces of certain important materials is given in Table 10.4.1.

Table 10.4.1 : Emissivity from surfaces of important materials

Surface	Emissivity (ϵ)
A. Polished metal surfaces	
1. Copper polished	0.0366
2. Polished silver	0.03
3. Aluminium polished	0.049
4. Nickel polished	0.046
5. Chromium	0.07
B. Metal surfaces	

Surface	Emissivity (ϵ)
1. Copper oxidized	0.725
2. Iron (rusted)	0.85
3. Hot cast iron	0.8
4. Hot rolled C.I.	0.6
5. Iron	0.658
C. Non-metals	
1. Brick plaster	0.93
2. Glass	0.875
3. Wood	0.91
4. Porcelain	0.92
5. Clay	0.86
6. Bakelite	0.9
7. Paper	0.92
8. Ice	0.966

Note : The emissivity of polished metal surfaces is < 0.1 , for metal surfaces it is in between 0.6 to 1.0 and for non-metals it lies between 0.8 to 1.0.

5. Radiosity (J)

Definition : Radiosity is defined as the total amount of energy emitted from a surface per unit time per unit surface area denoted by J .

Radiosity differs from emissive power since the former also includes the additional energy reflected from the surface out of incident radiation on it.

6. Irradiation

Definition : Irradiation of the surface, G is defined as the total radiant energy received per unit time per unit area from all directions at all wave lengths.

It is evident from the above definition that the irradiation does not depend directly on its temperature except when the radiations are received partly by the surface from its own reflected radiations.

Though Irradiation is not a surface property but it helps in the analysis of energy transfer by radiation.

10.5 Laws of Radiation

Various laws governing the thermal radiation are :

1. Planck's law
2. Wien's law
3. Stefan Boltzmann's law
4. Kirchoff's law.

These laws are being discussed below :

10.5.1 Planck's Law

Planck's law is based on quantum theory and it gives the relationship for monochromatic emissive power of the black body, $e_{b\lambda}$ as a function of absolute temperature, T of the surface and wavelength λ , as given below :

$$e_{b\lambda} = \frac{2\pi C_1}{\lambda^5 [e^{(C_2/\lambda \cdot T)} - 1]} \text{ (W/m}^2\text{)} \quad \dots(10.5.1)$$

The constants have been determined based on experiments. These are as follows :

$$C_1 = 2\pi h C_0^2 = 3.742 \times 10^8 \text{ W} \cdot \mu\text{m}^4/\text{m}^2$$

$$= 0.596 \times 10^{-16} \text{ (W/m}^2\text{)}$$

$$C_2 = 1.4387 \times 10^{-2} \text{ (m K)}$$

$$\lambda = \text{Wavelength (m)}$$

$$T = \text{Absolute temperature (K)}$$

Fig. 10.5.1 shows the plot of emissive power of a black body varying with wavelength, λ and absolute temperature, T .

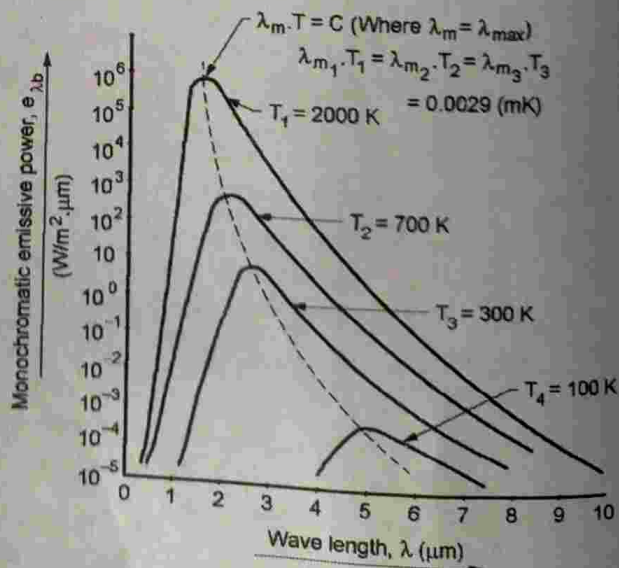


Fig. 10.5.1 : Energy distribution of a black body varying with λ and T

Following observations can be made :

1. Monochromatic emissive power of the black body at a given temperature first increases with increase in wavelength, attains a certain maximum value and then decreases.
2. For specific wavelength, the value of $e_{b\lambda}$ of black surface increases with increase in temperature.
3. Most of the thermal radiations lie in between the range of 0.2 to 10 μm .
4. The wavelength λ_m at which the maximum $e_{b\lambda}$ is obtained decreases with increase in temperature.

10.5.2 Wien's Displacement Law

Wien's law gives the relationship between the maximum wavelength, λ_m at which the maximum monochromatic emissive power is obtained and the absolute temperature, T of a black body. According to this law,

$$\lambda_m \cdot T = 0.0029 \text{ (m K)} \quad \dots(10.5.2)$$

Proof of Wien's law

According to Planck's law (Refer Equation (10.5.1))

$$e_{b\lambda} = \frac{2\pi C_1}{\lambda^5 [e^{(C_2/\lambda T)} - 1]} \quad \dots(i)$$

$$\text{Let } \frac{C_2}{\lambda T} = x \quad \dots(ii)$$

$$\therefore \lambda = \frac{C_2}{x \cdot T} \quad \dots(iii)$$

On substituting the value of λ from Equation (iii) in Equation (i),

$$e_{b\lambda} = \frac{2\pi C_1}{\left(\frac{C_2}{x \cdot T}\right)^5 (e^x - 1)} = \frac{2\pi C_1 \cdot x^5 \cdot T^5}{C_2^5} \times (e^x - 1)^{-1} \quad \dots(iv)$$

Above equation expresses the emissive power of a black body in terms of 'x'.

In order to get the wavelength (λ_m) at a specified temperature (T) at which the maximum emissive power, e_b occurs, we can differentiate the Equation (iv) with respect

'x' and equate to zero i.e. $\frac{d(e_{b\lambda})}{dx} = 0$

$$\frac{d}{dx} \left[\frac{2\pi C_1 \cdot x^5 \cdot T^5}{C_2^5} \times (e^x - 1)^{-1} \right] = 0$$

$$\frac{2\pi C_1 \cdot T^5}{C_2^5} \cdot \frac{d}{dx} [x^5 (e^x - 1)^{-1}] = 0$$

$$\text{or, } \frac{d}{dx} [x^5 (e^x - 1)^{-1}] = 0$$

$$(e^x - 1)^{-1} \cdot 5x^4 + x^5 \cdot (-1) \cdot (e^x - 1)^{-2} \cdot e^x = 0$$

$$\frac{x^4}{(e^x - 1)} \left[5 - \frac{x \cdot e^x}{(e^x - 1)} \right] = 0$$

$$5 - \frac{x \cdot e^x}{e^x - 1} = 0$$

$$\frac{5(e^x - 1) - x \cdot e^x}{e^x - 1} = 0$$

$$5(e^x - 1) - x \cdot e^x = 0$$

$$5e^x - 5 - x \cdot e^x = 0$$

$$e^x (5 - x) - 5 = 0 \quad \dots(v)$$

Above equation can only be solved by trial and error method by assuming various values of x.

On solving the above equation we get,

$$x = 4.965 \quad \dots(vi)$$

$$\text{But, } x = \frac{C_2}{\lambda_{\max} \cdot T} = 4.965 \quad \dots(vii)$$

$$\text{And, } C_2 = 1.4387 \times 10^{-2} \text{ (m)} \quad [\text{Refer Equation (10.5.1)}]$$

$$\lambda_m \cdot T = \frac{C_2}{4.965} = \frac{1.4387 \times 10^{-2}}{4.965} \quad (\text{Let, } \lambda_m = \lambda_{\max.})$$

$$\lambda_m \cdot T = 0.002898 \text{ m K} \approx 0.0029 \text{ m K} \quad \dots(10.5.3)$$

As discussed above, the Equation (10.5.3) is known as **Wien's law**. In this, $\lambda_m = \lambda_{\max}$ is the wavelength at which the monochromatic power of black body, $e_{b\lambda}$ is maximum at a particular temperature.

Maximum $e_{b\lambda}$: It is evident from above equation that the maximum value of $e_{b\lambda}$ will shift to a higher wavelength with decrease in temperature ($\because \lambda_m \propto \frac{1}{T}$).

On substituting the values of C_1 , C_2 and $\lambda_m \cdot T$ in Equation (10.5.1) we can get the maximum value $e_{b\lambda}$ as follows :

$$(e_{b\lambda})_{\max} = \frac{2\pi C_1}{\lambda_m^5 [e^{(C_2/\lambda_m \cdot T)} - 1]} = \frac{2\pi \times (0.586 \times 10^{-16})}{\left(\frac{0.0029}{T}\right)^5 \cdot [e^{4.965} - 1]}$$

$$(e_{b\lambda})_{\max} = 1.287 \times 10^{-5} \cdot T^5 = C \cdot T^5 \quad \dots(10.5.4)$$

Thus, the maximum emissive power of black body is proportional to the fifth power of the absolute temperature of the surface.

10.5.3 Stefan Boltzmann's Law

Stefan Boltzmann law states that the total emissive power of a black body is directly proportional to fourth power of its absolute temperature. Mathematically,

$$e_b \propto T^4$$

$$e_b = \sigma \cdot T^4 \quad \dots(10.5.5)$$

where, $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$ is called the **Stefan Boltzmann constant**.

Determination of Stefan Boltzmann's constant, σ

The value of ' σ ' can be determined by integrating Planck's law Equation (10.5.1) between limits $\lambda = 0$ to $\lambda = \infty$. Therefore,

$$e_b = \int_{\lambda=0}^{\lambda=\infty} e_{b\lambda} \cdot d\lambda = \int_{\lambda=0}^{\lambda=\infty} \frac{2\pi C_1}{\lambda^5 [e^{C_2/\lambda T} - 1]} \cdot d\lambda \quad \dots(i)$$

$$\text{Let, } x = \frac{1}{\lambda} \quad \dots(ii)$$

$$dx = -\frac{1}{\lambda^2} \cdot d\lambda \quad \text{or, } d\lambda = -\lambda^2 \cdot dx \quad \dots(iii)$$

When, $\lambda = 0$, $x = \infty$ and when $\lambda = \infty$, $x = 0$ from Equation (ii).

$$\text{Therefore, } \frac{d\lambda}{\lambda^5} = - \cdot dx = -\frac{1}{\lambda^3} \cdot dx = -x^3 \cdot dx$$

Using the above relations, the Equation (i) reduces to :

$$e_b = 2\pi C_1 \int_{x=\infty}^{x=0} -x^3 [e^{(C_2 \cdot x/T)} - 1]^{-1} dx$$

$$e_b = 2\pi C_1 \int_{x=0}^{x=\infty} x^3 \cdot [e^{(C_2 \cdot x/T)} - 1]^{-1} \cdot dx$$

$$e_b = 2\pi C_1 \int_{x=0}^{x=\infty} x^3 \cdot [e^{-C_2 x/T} + e^{-2C_2 x/T} + e^{-3C_2 x/T} + \dots] dx$$

$$e_b = 2\pi C_1 \int_{x=0}^{x=\infty} x^3 \cdot e^{-C_2 \cdot i \cdot x/T} \cdot dx$$

(where, $i = 1, 2, 3$) ... (iv)

$$\text{But, } \int_0^{\infty} x^n \cdot e^{-ax} \cdot dx = \frac{n!}{a^{n+1}}$$

$$e_b = 2\pi C_1 \cdot \frac{3!}{(C_2/T)^{3+1}} = \frac{2\pi C_1 \times (3 \times 2 \times 1)}{\left(\frac{C_2}{T}\right)^4}$$

$$e_b = \frac{12\pi C_1 \cdot T^4}{C_2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots \right)$$

$$e_b = \frac{12\pi C_1 \cdot T^4}{C_2^4} \left(\frac{\pi^4}{90} \right)$$

On substituting the values of C_1 and C_2 ,

$$e_b = \frac{12\pi \times (0.596 \times 10^{-16})}{(1.4387 \times 10^{-2})^4} \times \frac{\pi^4}{90} \times T^4$$

$$e_b = 5.672 \times 10^{-8} T^4 \quad \dots(v)$$

On comparing Equations (5.5.5) and (v) we have, Stefan Boltzmann constant,

$$\sigma = 5.672 \times 10^{-8} \approx 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$$

Stefan Boltzmann Equation for other surfaces having emissivity, ϵ

$$\therefore \text{Emissivity } \epsilon = \frac{e}{e_b}$$

Thus Equation (10.5.5) for all other surfaces can be modified as :

$$e = \epsilon \cdot \sigma \cdot T^4 \quad \dots(10.5.6)$$

10.5.4 Kirchhoff's Law

Kirchhoff's law states that the emissivity of the surface is equal to its absorptivity when the surface or the body is in thermal equilibrium with its surroundings.

Therefore,

Emissivity of a body,

$$\epsilon = \text{Absorptivity, } \alpha \quad \dots(10.5.7)$$

Proof of Kirchhoff's law

Consider a hollow evacuated black space maintained at temperature, T as shown in Fig. 10.5.2.

Introduce a small body of area A_1 at the same into the evacuated space. Since the body and its surroundings are at the same temperature and being under vacuum, then radiation is the only mode of heat transfer which is absorbed by the body.

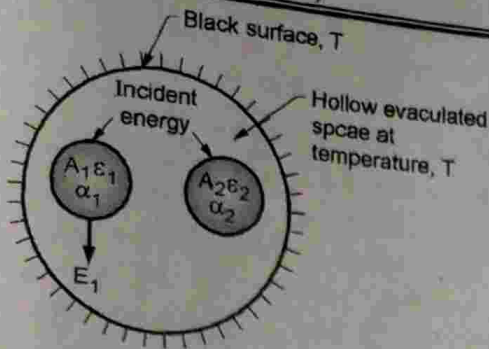


Fig. 10.5.2 : Kirchhoff's law

Let I be the intensity of radiation falling on black body in W/m^2 and E be the total emissive power of the body having its absorptivity as ' α_1 '.

Under equilibrium,

The rate of energy absorbed = Rate of energy emitted

$$I \cdot A_1 \alpha_1 = E_1 \cdot A_1$$

$$I = \frac{E_1}{\alpha_1} \quad \dots(i)$$

Similarly if body 2 is introduced of area A_2 , emissivity ϵ_2 and absorptivity, α_2 . Then,

$$I A_2 \alpha_2 = E_2 \cdot A_2 \quad \text{i.e.} \quad I = \frac{E_2}{\alpha_2} \quad \dots(ii)$$

Since I is same for each body, from Equations (i) and (ii) we get

$$\frac{E_1}{\alpha_1} = \frac{E_2}{\alpha_2} = \text{Energy incident from the black surface, } \frac{E_b}{\alpha_b}$$

$$\text{But } \alpha_b = 1$$

$$\therefore \frac{E_1}{\alpha_1} = \frac{E_2}{\alpha_2} = E_b$$

Thus any body of absorptivity α we can write in general,

$$\frac{E}{\alpha} = E_b ; \quad \alpha = \frac{E}{E_b} \quad \dots(iii)$$

$$\text{But, Emissivity, } \epsilon = \frac{E}{E_b} \quad \dots(iv)$$

Therefore from Equations (iii) and (iv) we get,

$$\text{Emissivity, } \epsilon = \text{Absorptivity, } \alpha \quad \dots(10.5.8)$$

Comments on above results

- Though we have proved the results in case the body is in thermal equilibrium with its surroundings, however these results have also been verified experimentally.

- Since absorptivity ranges between $\alpha = 0$ to $\alpha = 1$, it implies that emissivity also ranges between $\epsilon = 0$ to $\epsilon = 1$.

10.6 Emissivity of Real Surfaces

In Section 10.4.3 we have defined the emissivity of a surface, ϵ as the ratio of emissive power of the surface to emissive of a black body at the same temperature.

However, the emissivity of a real surfaces varies with :

- Temperature
- Wavelength
- Direction of emitted radiations.

Based on above facts we define various types of emissivity as follows :

- Spectral emissivity (ϵ_λ)
- Directional emissivity, ϵ_θ
- Hemispherical emissivity
- Total emissivity $\epsilon(T)$

1. Spectral emissivity (ϵ_λ)

It represents the emissivity of a surface at a certain wavelength.

2. Directional emissivity (ϵ_θ)

It represents the emissivity in a certain direction where θ is the angle between the direction of radiation and the normal to the surface.

3. Hemispherical emissivity

It represents the average emissivity of a surface over all directions.

4. Total emissivity $\epsilon(T)$

It represents the emissivity averaged over all wavelengths. Therefore,

$$\epsilon(T) = \frac{\text{Total emissive power of temperature, } T \text{ of a real surfaces } E(T)}{\text{Total emissive power of a black body at temperature } T, E_b(T)}$$

$$\text{where, } E(T) = \epsilon \cdot \sigma \cdot T^4$$

$$\text{and } E_b(T) = \sigma \cdot T^4 \quad \dots(10.6.1)$$

10.7 Intensity of Radiation (I) and Solid Angle (ω)

The radiant energy falling on a surface is from all directions. Thus the effect of radiation on the surface depends on the directional distribution of energy. This effect can be analysed by considering the concept of intensity of radiation and the solid angle.

10.7.1 Solid Angle (ω)

Definition : A solid angle, $d\omega$ is defined as the measure of space enclosed by a conical surface with the vertex of the cone whose apex is at the centre of dA_1 and whose base is area dA_2 relative to the total space of a sphere of radius 'r' as shown in Fig. 10.7.1.

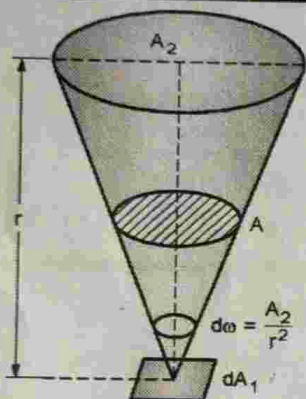


Fig. 10.7.1 : Solid angle subtended, by a finite area A_2 on elementary area dA_1

$$\text{Solid angle, } d\omega = \frac{A_2}{r^2} \quad \dots(10.7.1)$$

Unit of solid angle is steradian (Sr).

Referring to Fig. 10.7.2, consider an elemental black surface area dA_1 at 'O' with OM as normal to it.

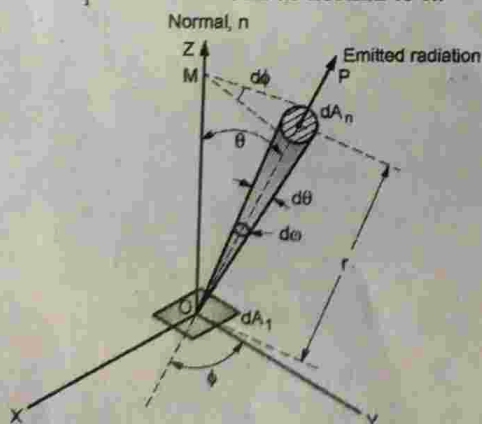


Fig. 10.7.2 : Distribution of radiation emitted from a surface

Consider any direction OP at an angle θ from the normal direction.

The normal area dA as viewed from M will be $(dA \cdot \cos \theta)$ when viewed from P.

Let, $d\omega$ = Solid angle around the line OP
 dA_n = Normal area to radius, r of a sphere making solid angle $d\omega$ through which the emitted radiations are passing through

$$\text{Then, } d\omega = \frac{dA_n}{r^2} \quad \dots(10.7.2)$$

where, $dA_n = (r \sin \theta \cdot d\phi) r d\theta$

Note : Solid angle subtended by the complete hemisphere
 $= \frac{2\pi r^2}{r^2} = 2\pi$...(10.7.3)

10.7.1.1 Solid Angle between Two Elementary Areas

The solid angle subtended by elementary area dA_2 at dA_1 can be written as :

$$d\omega = \frac{dA_2 \cdot \cos \beta_2}{r^2} \quad \dots(10.7.4)$$

Similarly, the solid angle subtended by area dA_1 at dA_2 can be written as :

$$d\omega = \frac{dA_1 \cdot \cos \beta_1}{r^2} \quad \dots(10.7.5)$$

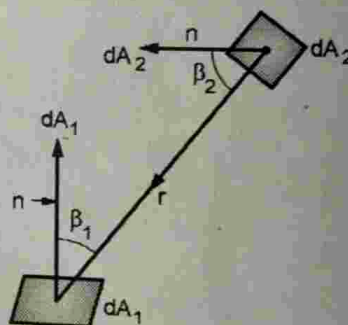


Fig. 10.7.3 : Solid angle between two elementary surfaces

10.7.2 Intensity of Radiation (i)

Definition : The intensity of radiation (i) is defined as the rate of energy leaving a surface in a given direction per unit solid angle per unit area of the emitting surface normal to the mean direction in space. Mathematically,

$$\text{Mathematically, } i = \frac{de}{d\omega} \quad \dots(10.7.6)$$

The intensity of radiation varies in different directions and it is maximum into the direction normal to the surface.

10.7.2.1 Lambert Cosine Law

Intensity of radiation in any direction is proportional to the cosine of the angle made by that direction with the normal is known as **Lambert cosine law**.

Referring to Fig. 10.7.4

$$i = i_n \cdot \cos \beta \quad \dots(10.7.7)$$

where, i_n = Intensity of radiation (maximum) in normal direction to the surface.

β = Angle made from normal direction by the direction in which i is measured.

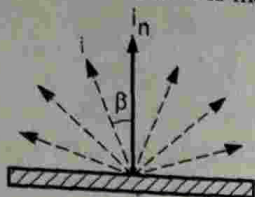


Fig. 10.7.4 : Energy emitted by a surface in different directions

Relationship between emissive power and normal intensity

The intensity of radiation from Equation (10.7.6) can be written as :

$$i = \frac{de}{d\omega} ; de = i \cdot d\omega \quad \dots(i)$$

From Lambert's principle,

$$i = i_n \cdot \cos \beta \quad \dots(ii)$$

From Equations (i) and (ii), the emissive power can be obtained by integrating the equation,

$$\int_{\text{hemisphere}} de = \int_{\text{hemisphere}} i_n \cdot \cos \beta \cdot d\omega \quad \dots(iii)$$

Using the definition of solid angle, it can be shown that for a unit hemisphere,

$$d\omega = 2\pi \sin \beta \, d\beta$$

Thus from Equation (iii),

$$\begin{aligned} \beta &= \pi/2 \\ \therefore e &= 2\pi i_n \int_{\beta=0}^{\pi/2} \cos \beta \cdot \sin \beta \cdot d\beta \\ &= \pi \cdot i_n \int_0^{\pi/2} 2 \cos \beta \sin \beta \cdot d\beta \\ e &= \pi \cdot i_n \cdot \int_0^{\pi/2} \sin 2\beta \cdot d\beta \\ &= \pi \cdot i_n \left[-\frac{\cos 2\beta}{2} \right]_0^{\pi/2} = -\pi i_n \left[\frac{-1-1}{2} \right] \end{aligned}$$

$$e = \pi \cdot i_n = \pi \cdot e_b \quad \dots(10.7.8)$$

where, e_b is the emissive power of a black body.

Thus, the total emissive power of a diffuse surface over a hemisphere is equal to π times the intensity of radiation in normal direction (i_n).

10.8 Heat Exchange between Two Finite Black Surfaces by Radiation

The radiant heat energy exchange between two surfaces depends on :

1. The geometric arrangement between the two surfaces.
2. The type of separating medium.

(Though radiation does not need any medium however, certain mediums may interfere with propagation of radiation by absorption, emission or scattering of radiation e.g. in case radiation having medium like ammonia, CO_2 , CO, water vapour etc.)

3. The characteristics of the surfaces.

In order to simplify the problem of heat exchange, we assume :

1. Heat exchange is between the black surfaces since these are perfect absorbers as well emitters.
2. Radiation is in vacuum as medium which does not interfere with radiations at low temperatures.

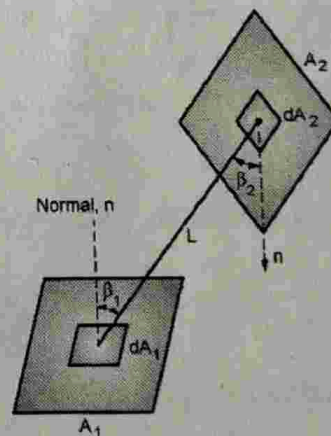


Fig. 10.8.1 : Heat exchange between two finite black surfaces

Consider two black surface A_1 and A_2 with elementary areas dA_1 and dA_2 respectively at 'L' distance apart as shown in Fig. 10.8.1.

Let β_1 and β_2 be the angles made by dA_1 and dA_2 by their respective normals to the line joining their centers.

The solid angle subtended by dA_2 on dA_1 (using Equation (10.7.4)),

$$d\omega_{1-2} = \frac{dA_2 \cdot \cos \beta_2}{L^2} \quad \dots(i)$$

The rate of intensity of radiation emitted by dA_1 in direction of dA_2 (using Equation (10.7.7)),

$$i = i_n \cdot \cos \beta_1 = i_{b1} \cos \beta_1 \quad \dots(ii)$$

But $i_n = i_b$ (radiation of black body)

Therefore the rate of radiative energy, dQ_1 leaving the area dA_1 that strikes dA_2 can be written as,

$$dQ_1 = d\omega_{1-2} \times i_{b1} \times \text{area}$$

$$dQ_1 = \left(\frac{dA_2 \cdot \cos \beta_2}{L^2} \right) \times (i_{b1} \times \cos \beta_1) \times dA_1 \quad \dots(iii)$$

$$\therefore e_{b1} = \pi \times i_n = \pi \times i_{b1};$$

$$i_{b1} = \frac{e_{b1}}{\pi} = \frac{\sigma T_1^4}{\pi} \text{ using this in Equation (iii),}$$

$$\therefore dQ_1 = \left(\frac{\cos \beta_1 \cdot \cos \beta_2}{\pi L^2} \right) \cdot dA_1 \cdot dA_2 \cdot e_{b1}$$

$$dQ_1 = \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi \cdot L^2} \times dA_1 \cdot dA_2 \cdot \sigma T_1^4 \quad \dots(10.8.1)$$

Similarly the rate of radiative energy, dQ_2 leaving dA_2 that strikes dA_1 can be written as :

$$dQ_2 = \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi L^2} \times dA_1 \cdot dA_2 \cdot \sigma \cdot T_2^4 \quad \dots(10.8.2)$$

Since surfaces considered are black, the net rate of heat transfer,

$$dQ = (\text{Energy radiated, } dQ_1 - \text{Energy absorbed } dQ_2)$$

$$dQ = \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi L^2} \cdot dA_1 \cdot dA_2 \cdot \sigma (T_1^4 - T_2^4) \quad \dots(10.8.3)$$

The heat transfer rate from A_1 to A_2 can be obtained by integrating the Equation (10.8.3) over the areas A_1 and A_2 as follows :

$$\begin{aligned} Q &= \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi \cdot L^2} \cdot dA_1 \cdot dA_2 \cdot \sigma (T_1^4 - T_2^4) \\ &= \sigma (T_1^4 - T_2^4) \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2 \cdot dA_1 dA_2}{\pi \cdot L^2} \end{aligned}$$

Multiplying and dividing by A_1 ,

$$Q = \sigma \cdot A_1 \cdot (T_1^4 - T_2^4)$$

$$\times \frac{1}{A_1} \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi L^2} dA_1 \cdot dA_2 \quad \dots(10.8.4)$$

The term $\left[\frac{1}{A_1} \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi \cdot L^2} \cdot dA_1 \cdot dA_2 \right]$ is called

the **shape factor**, F_{12} which depends on the geometric parameters of the surface.

$$\therefore \text{Shape factor, } F_{12} = \frac{1}{A_1} \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi L^2} dA_1 \cdot dA_2 \quad \dots(10.8.5)$$

The Equation (10.8.4) can be rewritten as :

$$Q = \sigma \cdot A_1 \cdot F_{12} (T_1^4 - T_2^4) \quad \dots(10.8.6)$$

Note : Above equation is applicable to black surfaces only and its equation should not be used for other surfaces since their emissivity is less than 1.

10.9 Radiant Shape Factor Algebra

Definition : Shape factor is defined as a fraction of energy emitted by one surface and directly intercepted by the other surface.

The shape factor is also called as view factor, configuration factor or angle factor.

Referring to Fig. 10.8.1, the total energy emitted by surface A_1 ,

$$Q_1 = \sigma \cdot A_1 \cdot T_1^4$$

The energy intercepted by surface 2,

$$Q'_1 = \sigma \cdot A_1 \cdot T_1^4 \cdot F_{12}$$

$$\text{Thus, } F_{12} = \frac{Q'_1}{Q_1}$$

10.9.1 Reciprocal Relation

Referring to Fig. 10.8.1 and using Equation (10.8.5), the shape factor of surface 1 with respect to surface 2 is,

$$F_{12} = \frac{1}{A_1} \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{L^2} \cdot dA_1 \cdot dA_2 \quad \dots(i)$$

Similarly, the shape factor of surface 2 w.r.t. surface 1,

$$F_{21} = \frac{1}{A_2} \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{L^2} \cdot dA_1 \cdot dA_2 \dots (ii)$$

The net heat transfer

$$Q = \sigma \cdot A_1 \cdot F_{12} (T_1^4 - T_2^4) = \sigma \cdot A_2 \cdot F_{21} (T_1^4 - T_2^4)$$

It follows from above :

$$A_1 \cdot F_{12} = A_2 \cdot F_{21} \dots (10.9.1)$$

Above relation is known as reciprocal relation or reciprocity theorem.

10.9.2 Important Points Regarding Shape Factor

1. Reciprocal relationship and its importance
2. Dependence of shape factor
3. Enclosure theorem
4. Shape factor of a surface with respect to itself, F_{11}
5. Additive rule-relation between shape factors
6. Shape factor F_{12} if convex surface 1 is kept in an enclosure of surface 2
7. Shape factor when two surfaces are large and parallel

1. Reciprocal relationship and its importance

- The shape factor F_{12} and F_{21} are related by

$A_1 \cdot F_{12} = A_2 \cdot F_{21}$ the reciprocal Equation (10.9.1).

- This equation is of great importance and utility whenever the shape factor of one of the two surfaces is unity.
 - Such a case occurs when one of the surface is enclosed in another surface and the inner surface is convex.
 - The reason is that the radiated energy of inner surface is completely intercepted by the outer surface and not vice-versa.
 - The shape factor of other surface equals to the ratio of areas of the two surfaces provided one of the surface has shape factor equal to 1.
- #### 2. Dependence of shape factor
- The parameters which affect the shape factor are :
 - (a) Shape and size of surface.
 - (b) Orientation of surfaces.
 - (c) Distance between two surfaces.
 - The shape factor cannot be calculated analytically except for simple geometry.
 - Thus, the shape factor is usually determined by numerical methods.
 - The shape factor for few standard shapes are shown in Fig. 10.9.1 to Fig. 10.9.3 respectively.

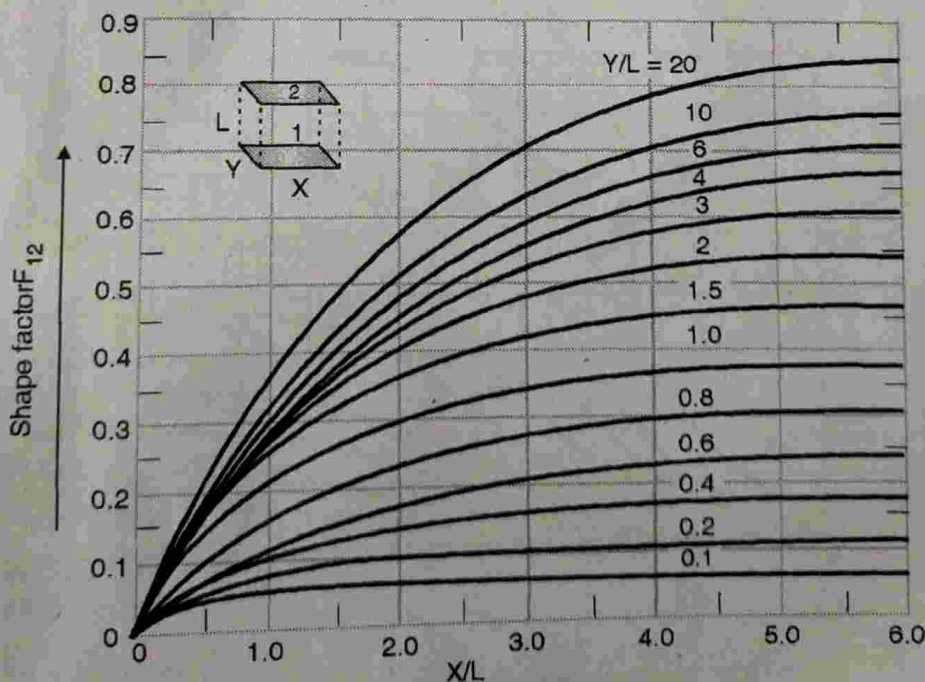


Fig. 10.9.1 : Shape factor for aligned parallel plates

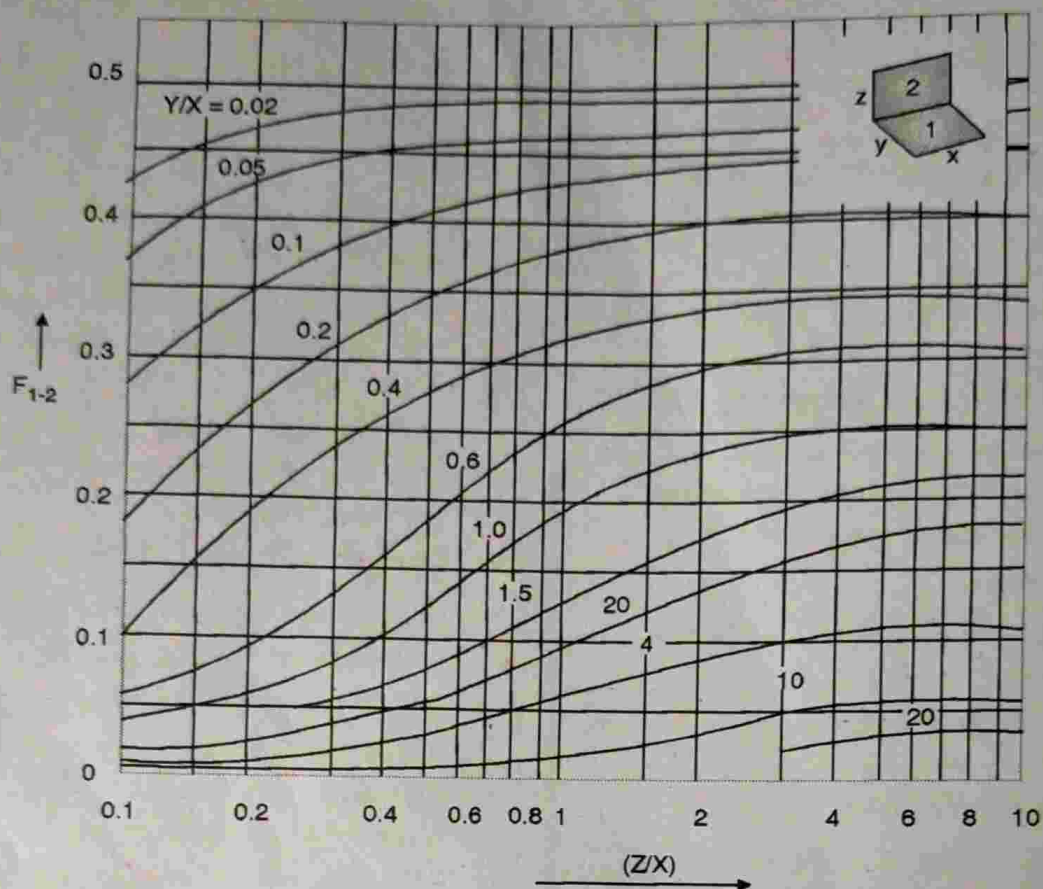


Fig. 10.9.2 : Shape factor for perpendicular rectangles with common edge

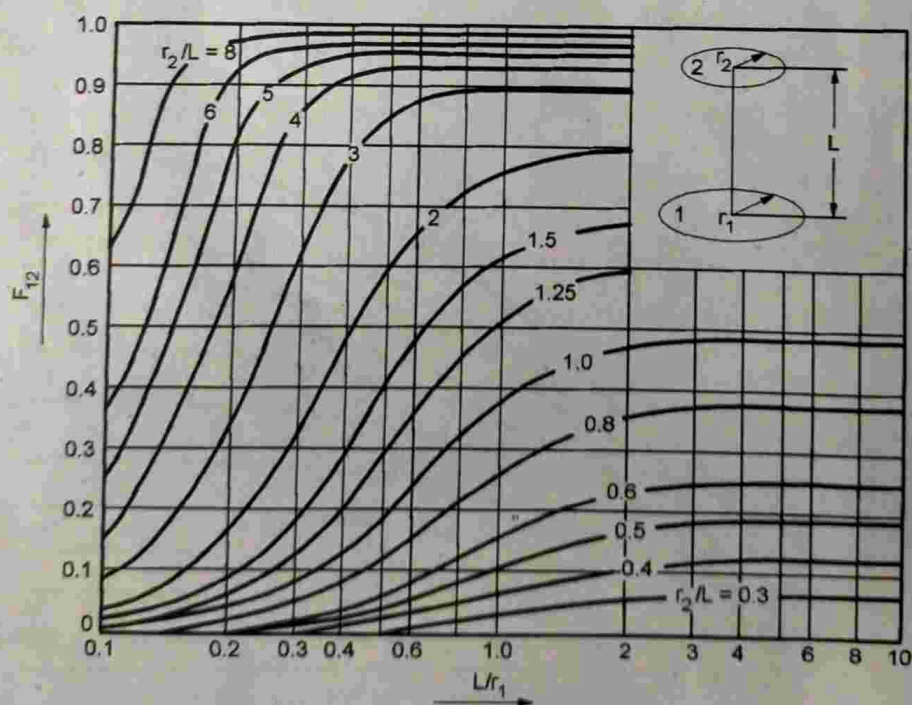


Fig. 10.9.3 : Shape factor between two coaxial parallel disks

3. Enclosure theorem

Consider an enclosure of n surfaces and each is maintained at different temperature.

Then each surface will exchange radiant energy with the remaining $(n - 1)$ surfaces.

Consider any surface say $(m)^{\text{th}}$ surface. The radiant energy leaving the $(m)^{\text{th}}$ surface, the fraction of this energy will be intercepted by n surfaces considering that the energy is also intercepted by its own surface.

Shape factor F_{mn} as the fraction of energy the surface m and heading towards surface n can be written as F_{mn} . Then,

$$\sum_{n=1}^{n=n} F_{mn} = 1 \text{ for } m = 1, 2, \dots, n$$

It follows that :

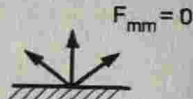
$$\left. \begin{array}{l} F_{11} + F_{12} + F_{13} + \dots + F_{1n} \\ F_{21} + F_{22} + F_{23} + \dots + F_{2n} \\ \vdots \\ F_{n1} + F_{n2} + F_{n3} + \dots + F_{nn} = 1 \end{array} \right\} \dots(10.9.2)$$

Equation (10.9.2) is called *enclosure theorem*.

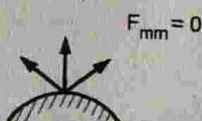
4. Shape factor of a surface with respect to itself, F_{11}

$$(a) F_{11} = F_{22} = F_{33} = \dots F_{nn} = 0$$

(For flat and convex surfaces)

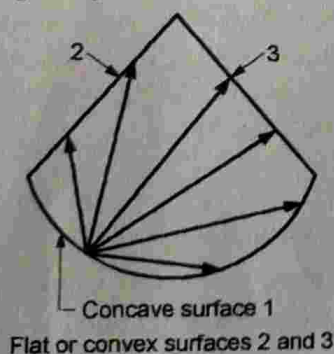


(i) Flat surface



(ii) Convex

(b) $F_{11}, F_{22}, \dots, F_{nn} \neq 0$ in case of **concave surfaces** since the energy leaving the surface 1 may also be intercepted by itself as shown in Fig. 10.9.4.



Concave surface 1
Flat or convex surfaces 2 and 3

Fig. 10.9.4 : Radiant energy of a surface 1 intercepted by itself (concave surfaces)

5. Additive rule—relation between shape factors

Refer Fig. 10.9.5. The shape factor between two surfaces A_1 and A_2 is equal to the sum of the shape factor of two areas A_3 and A_4 made up from Area A_2 . Accordingly,

$$F_{12} = F_{13} + F_{14} \dots(10.9.3)$$

Above relation is also called as **summation relation**.

However, $F_{21} \neq F_{31} + F_{41}$

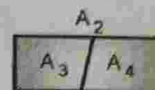


Fig. 10.9.5 : Relation between shape factors if one of the surface 2 is made up of two surfaces 3 and 4

6. Shape factor F_{12} if convex surface 1 is kept in an enclosure of surface 2

From reciprocity relation :

$$A_1 F_{12} = A_2 \cdot F_{21}, \text{ But } F_{12} = 1$$

($\because$ Total radiant energy diffused from surface 1 reaches to surface 2)

$$\therefore F_{21} = \frac{A_1}{A_2} \dots(10.9.4)$$

7. Shape factor when two surfaces are large and parallel

$\because$ All radiations emitted one surface reaches to the other surface (i.e. $A_1 = A_2$), it implies :

$$F_{12} = F_{21} = 1 \text{ (Distance between parallel surfaces is small)} \dots(10.9.5)$$

10.9.3 Application Shape Factor for Standard Configurations and Radiant Heat Exchange

1. Hemispherical shape
2. Two concentric cylinders
3. A tube with cross-section of an equilateral triangle

1. Hemispherical shape

Consider a black hemispherical cavity of radius ' r ' surface 1 having the flat plate 2 on its top as shown in Fig. 10.9.6. By enclosure theorem :

$$F_{11} + F_{12} = 1 \dots(i)$$

$$F_{21} + F_{22} = 1 \dots(ii)$$

But $F_{22} = 0$ (since a flat surface).

From Equation (ii)

$$F_{21} + 0 = 1 \quad \dots(iii)$$

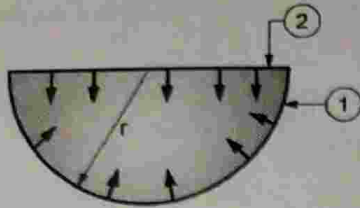


Fig. 10.9.6 : Shape factor of hemi-spherical cavity with flat plate over it

By reciprocal relation,

$$A_1 F_{12} = A_2 \cdot F_{21} \text{ [But } F_{21} = 1 \text{ from Equation (iii)]}$$

$$\therefore F_{12} = \frac{A_2}{A_1} \times 1 = \frac{A_2}{A_1} \quad \dots(iv)$$

From Equation (i) and (iv),

$$F_{11} = 1 - F_{12} = 1 - \frac{A_2}{A_1} = 1 - \frac{\pi r^2}{2\pi r^2} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\text{i.e. } F_{11} = \frac{1}{2} \text{ or } 0.5 \quad \dots(10.9.6)$$

It shows that 50% of radiations emitted by surface 1 (hemispherical surface) will fall back.

2. Two concentric cylinders

Consider two black concentric cylinders of diameters d_1 and d_2 as shown in Fig. 10.9.7.

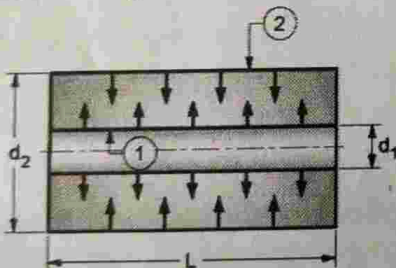


Fig. 10.9.7 : Heat exchange between two concentric cylinders

By enclosure theorem :

$$F_{11} + F_{12} = 1 \quad \dots(i)$$

$$F_{21} + F_{22} = 1 \quad \dots(ii)$$

But $F_{11} = 0$ (Since cylinder surface is convex)

(Assume cylinder 1 is completely enclosed by 2)

$$\therefore F_{12} = 1 \quad \dots(iii)$$

Using reciprocal relation :

$$A_1 F_{12} = A_2 \cdot F_{21} ;$$

$$F_{21} = \frac{A_1}{A_2} \times F_{12} = \frac{A_1}{A_2} \times 1 = \frac{d_1}{d_2} \quad \dots(iv)$$

On substituting F_{21} from Equation (iv) in Equation (ii),

$$\frac{A_1}{A_2} + F_{22} = 1 ; \quad F_{22} = 1 - \frac{A_1}{A_2}$$

$$F_{22} = 1 - \frac{\pi d_1 \cdot L}{\pi d_2 \cdot L} = 1 - \frac{d_1}{d_2} \quad \dots(10.9.7)$$

Therefore, a fraction of $\left(1 - \frac{d_1}{d_2}\right)$ radiations will fall back itself on surface 2.

3. A tube with cross-section of an equilateral triangle

Refer Fig. 10.9.8.

From summation rule,

$$F_{11} + F_{12} + F_{13} = 1 \quad \dots(i)$$

$$F_{21} + F_{22} + F_{23} = 1 \quad \dots(ii)$$

$$F_{31} + F_{32} + F_{33} = 1 \quad \dots(iii)$$

In each of the above equations :

$$F_{11} = F_{22} = F_{33} = 0 \text{ (For flat surfaces)}$$

$$\text{and } F_{12} = F_{13} ; F_{21} = F_{23} ; F_{31} = F_{32} \text{ (By geometry)}$$

$$\therefore F_{12} = F_{13} = 0.5 ; F_{21} = F_{23} = 0.5 \text{ and } F_{31} = F_{32} = 0.5$$

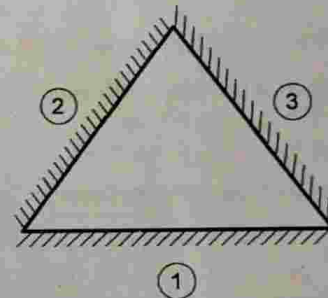


Fig. 10.9.8 : Heat exchange in an equilateral triangle

10.10 Heat Exchange Between Grey Bodies

So far we have only discussed the radiant heat exchange between black bodies in which the body absorbs the entire radiation. Now we shall discuss the radiant exchange between grey bodies. By definition, a **grey body** is the one which has its absorptivity, α equal to emissivity, ϵ in thermal equilibrium i.e. $\alpha = \epsilon$

Assumptions

1. All surfaces have uniform temperature.
2. All reflections and emissions are diffuse.
3. All surfaces are grey and opaque.

We recall the following terms :

- (a) **Radiosity, J** : represents the total amount of energy emitted from a surface per unit time and per unit area. It includes both reflected and emitted energy.

- (b) **Irradiation, G** : represents the total radiant energy received per unit time per unit area from all directions and wavelengths.

Thus by above definitions,

$$J = \rho \cdot G + e = \rho \cdot G + \epsilon \cdot e_b \quad \dots(10.10.1)$$

Where, ρ = reflectivity of surface ;

e = emissive power of surface

ϵ = emissivity of surface

e_b = emissive power of black body.

10.10.1 Net Radiation Heat Exchange from A Surface and Electrical Analogy : (Radiosity-Irradiation Approach)

Refer Fig. 10.10.1(a).

The net energy leaving the surface

$$\frac{Q}{A} = J - G \quad \dots(i)$$

From Equation (10.10.1),

$$J = \rho G + \epsilon e_b \quad \dots(ii)$$

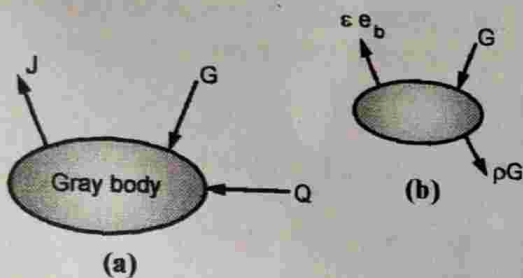


Fig. 10.10.1 : Energy transfers from a gray body

For opaque, diffuse and grey surface we can write,

$$\rho + \alpha + \tau = 1 ;$$

But $\tau = 0$ for opaque surface and $\epsilon = \alpha$ for grey surface

$$\therefore \rho + \epsilon + 0 = 1 ; \quad \rho = 1 - \epsilon \quad \dots(iii)$$

On substituting the value of ρ from Equation (iii) in Equation (ii),

$$J = (1 - \epsilon) G + \epsilon e_b ;$$

$$\therefore G = \frac{J - \epsilon e_b}{(1 - \epsilon)} \quad \dots(10.10.2)$$

On substituting the value of G from above equation in Equation (i) we get,

$$\frac{Q}{A} = J - \frac{(J - \epsilon e_b)}{(1 - \epsilon)} = \frac{J - \epsilon J - J + \epsilon e_b}{(1 - \epsilon)} = \frac{\epsilon (e_b - J)}{(1 - \epsilon)}$$

$$Q_{net} = \frac{A \cdot \epsilon (e_b - J)}{(1 - \epsilon)} = \frac{(e_b - J)}{(1 - \epsilon)/\epsilon A} \quad \dots(10.10.3)$$

Note : For a black body : $\epsilon = 1$, hence $e_b = J$

Electrical Analogy

Equation (10.10.3) provides the basis for equivalent electrical network for radiation shown in Fig. 10.10.2.

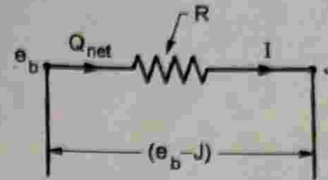


Fig. 10.10.2 : Surface resistance to radiation heat transfer

Comparing the Equation (10.10.3) with ohm's law equation,

$$I = \frac{V}{R} \text{ we have :}$$

- (a) Q_{net} corresponds to current flow, I .
- (b) $(e_b - J)$ corresponds to potential difference, V .
- (c) $(1 - \epsilon)/\epsilon A$ corresponds to resistance, R called surface resistance to radiation.

10.10.2 Net Heat Exchange between Two Grey Bodies/Surfaces

Consider two grey surfaces A_1 and A_2 having radiant heat exchange as shown in Fig. 10.10.3(a).

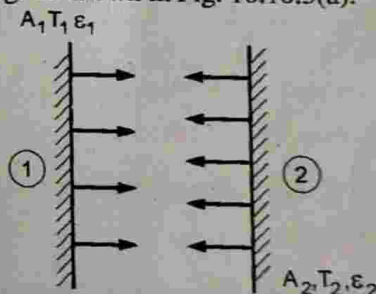
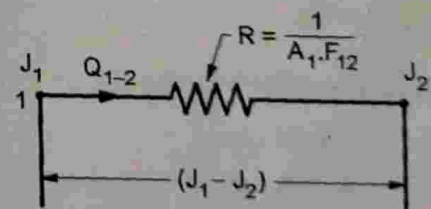


Fig. 10.10.3(a) : Radiant heat exchange



(b) Electrical network with space resistance

Fig. 10.10.3 : Radiant heat exchange between grey surfaces

Let surfaces A_1 and A_2 are maintained at uniform temperatures T_1 and T_2 having emissivity as ϵ_1 and ϵ_2 respectively with radiosity as J_1 and J_2 .



(a) Fraction of energy leaving the surface 1 and reaching to surface 2, $Q_1 = A_1 \cdot F_{12} \cdot J_1$... (i)

(b) Fraction of energy leaving the surface 2 and reaching to surface 1, $Q_2 = A_2 \cdot F_{21} \cdot J_2$... (ii)

Net radiation energy exchange between the surfaces,

$$Q_{12} = A_1 F_{12} J_1 - A_2 \cdot F_{21} \cdot J_2 \quad \dots (iii)$$

By reciprocal relation,

$$A_1 \cdot F_{12} = A_2 \cdot F_{21}$$

Hence Equation (iii) reduces to ;

$$Q_{12} = A_1 F_{12} (J_1 - J_2) = \frac{(J_1 - J_2)}{(1/A_1 \cdot F_{12})} \quad \dots (10.10.4)$$

The corresponding electrical network is shown in Fig. 10.10.3(b) in which the **space or shape resistance**, $R_{12} = (1/A_1 \cdot F_{12})$.

By combining the Equations (10.10.2) to (10.10.4), the overall equivalent electrical network takes the shape as shown in Fig. 10.10.4.

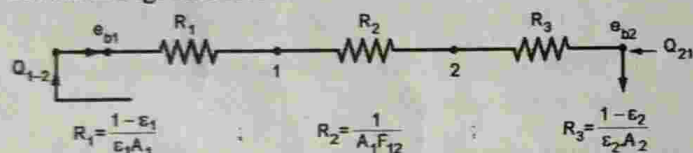


Fig. 10.10.4 : Equivalent electrical circuit for radiant heat exchanges between two grey surfaces

$$\therefore Q_{1-2} = \frac{e_{b1} - e_{b2}}{R_1 + R_2 + R_3} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{(1-\epsilon_1)}{\epsilon_1 A_1} + \frac{1}{A_1 \cdot F_{12}} + \frac{(1-\epsilon_2)}{\epsilon_2 A_2}} \quad \dots (10.10.5)$$

where, R_1 and R_3 are called **surface resistance** and R_2 as **space resistance**.

10.11 Application of Electrical Analogy to Thermal Radiation Heat Exchange Problems

10.11.1 Two Parallel Infinite Plane Surfaces

Fig. 10.11.1. In this case,

$$A_1 = A_2 = A \text{ (say) (Infinite long surfaces)}$$

By reciprocity rule,

$$A_1 F_{12} = A_2 \cdot F_{21}$$

$$\therefore F_{12} = F_{21} = 1 \text{ (Large plane surfaces closely placed)}$$

From Equation (10.10.5), the net heat exchange is :

$$\begin{aligned} Q_{12} &= \frac{\sigma \cdot (T_1^4 - T_2^4)}{\frac{(1-\epsilon_1)}{\epsilon_1 A_1} + \frac{1}{A_1 \cdot F_{12}} + \frac{(1-\epsilon_2)}{\epsilon_2 A_2}} \\ &= \frac{\sigma \cdot A \cdot (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} - 1 + 1 + \frac{1}{\epsilon_2} - 1} \\ &\therefore Q_{12} = \frac{\sigma \cdot A (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} \quad \dots (10.11.1) \end{aligned}$$

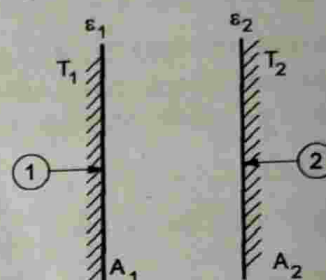


Fig. 10.11.1 : Radiant heat exchange between two infinitely long plane surfaces

10.11.2 Radiation Heat Exchange between Two Concentric Infinitely Long Grey Cylinders

Refer Fig. 10.11.2. Rewriting Equation (10.10.5),

$$Q_{1-2} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{(1-\epsilon_1)}{\epsilon_1 A_1} + \frac{1}{A_1 F_{12}} + \frac{(1-\epsilon_2)}{\epsilon_2 A_2}} \quad \dots (i)$$

Let R_2 and R_1 be the radii of outer and inner cylinders respectively having emissivity as ϵ_2 and ϵ_1 .

$$\therefore F_{11} + F_{12} = 1 \text{ (But } F_{11} = 0);$$

($\because$ Inner cylinder 1 surface is convex)

$$\therefore F_{12} = 1$$

On substituting the value of F_{12} in Equation (i) we get

$$\begin{aligned} Q_{12} &= \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{A_1} \left(\frac{1}{\epsilon_1} - 1 + 1 \right) + \frac{1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} \\ &= \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_2} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} \quad \dots (10.11.2) \end{aligned}$$

where, $A_1 = 2\pi R_1 L$ and $A_2 = 2\pi R_2 L$

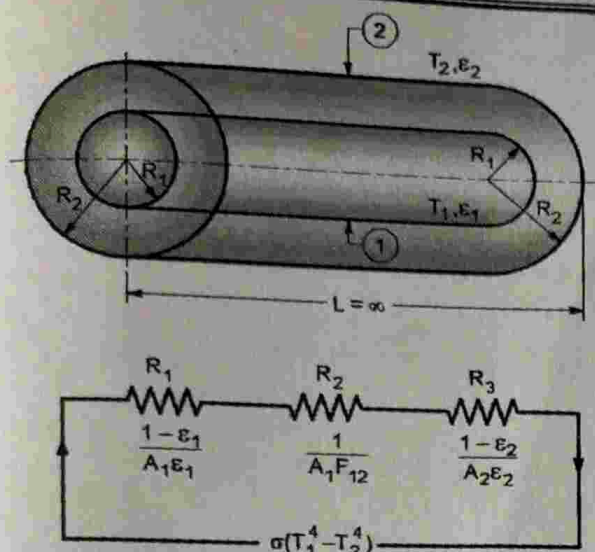


Fig. 10.11.2 : Two concentric infinitely long cylinders

10.11.3 Radiation Heat Exchange between Two Concentric Spheres

Equation (10.11.2) derived above is applicable since $F_{12} = 1$ (inner sphere being enclosed in outer sphere) except that :

$$A_1 = 4\pi R_1^2 \quad \text{and} \quad A_2 = 4\pi R_2^2$$

1. Thus it reduces heat transfer by radiation.

2. By introducing a third surface in between two grey surfaces called **radiation shields**.

These surfaces must be highly reflecting and have low thickness. These radiation shields donot either add or remove energy from the system. However, these shields add space resistances and reduce the radiative heat transfer.

Radiation shields are widely used in high temperature measurement by thermocouples to reduce error in temperature measurement and to reduce heat losses in thermos flask.

Analysis of radiation shields

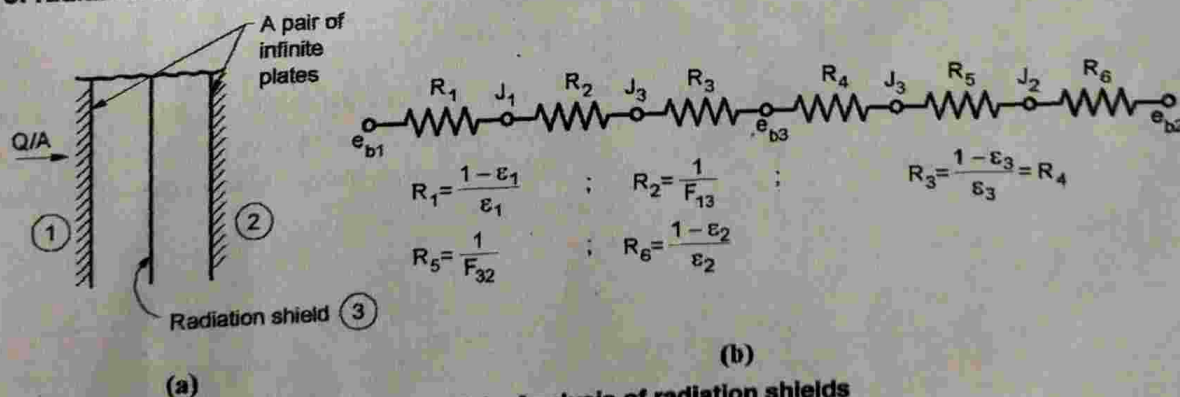


Fig. 10.12.1 : Analysis of radiation shields

Refer Fig. 10.12.1 and assuming $T_1 > T_2$, the heat flow rate without shield will be (Refer Fig. 10.11.1)

$$\text{Heat flux, } q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} \quad \dots(i)$$

Note :

1. Though Equation (10.11.2) has been derived for two concentric cylinders, however this equation is applicable to any two surfaces in which one inner surface is enclosed in the outer surface.

e.g. Two concentric spheres, eccentric cylinders etc.

2. When a small body is enclosed by a large body, the,

$\frac{A_1}{A_2} \approx 0$ and $F_{12} = 1$. Equation (10.10.5) reduces to

$$Q_{12} = \epsilon_1 \sigma A_1 (T_1^4 - T_2^4).$$

10.12 Radiation Shields

Radiative heat transfer between two surfaces can be reduced by following two methods :

1. Using the materials which are highly reflective.

In this case the emissivity of the surface is small and the surface resistance $[(1 - \epsilon) / \epsilon A]$ when $\epsilon \rightarrow 0$, approaches to infinity.

Since $\epsilon = (1 - \rho)$ for opaque materials it indicates that by using highly reflective surface the value of ρ tends to become



Now a radiation shield having both side emissivity ϵ_3 is placed between the surfaces 1 and 2. Under steady state conditions, the radiation shield will attain a temperature T_3 . Under this condition, the radiation received by shield from surface 1, the same radiations are transferred by the shield to surface 2. It implies that :

$$q_{13} = q_{32} = \frac{\sigma(T_1^4 - T_3^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1} = \frac{\sigma(T_3^4 - T_2^4)}{\frac{1}{\epsilon_3} + \frac{1}{\epsilon_2} - 1} \quad \dots(ii)$$

$$\text{Let : } \frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1 = x \quad \dots(iii)$$

$$\text{and } \frac{1}{\epsilon_3} + \frac{1}{\epsilon_2} - 1 = y \quad \dots(iv)$$

Using above substitutions in Equation (ii),

$$\frac{T_1^4 - T_3^4}{x} = \frac{T_3^4 - T_2^4}{y}$$

$$\frac{T_3^4}{x} + \frac{T_3^4}{y} = \frac{T_1^4}{x} + \frac{T_2^4}{y}$$

On taking its LCM and solving,

$$(x+y)T_3^4 = y \cdot T_1^4 + x \cdot T_2^4$$

$$T_3^4 = \frac{y \cdot T_1^4 + x \cdot T_2^4}{(x+y)} \quad \dots(v)$$

On substituting the value of T_3^4 from Equation (v) in Equation (ii),

$$q_{13} = \frac{\sigma}{x} \left[T_1^4 - \frac{(y \cdot T_1^4 + x \cdot T_2^4)}{(x+y)} \right]$$

$$= \frac{\sigma}{x(x+y)} \times [x \cdot T_1^4 + y \cdot T_1^4 - y \cdot T_1^4 - x \cdot T_2^4]$$

$$q_{13} = \frac{\sigma \cdot x}{x(x+y)} (T_1^4 - T_2^4) = \frac{\sigma(T_1^4 - T_2^4)}{(x+y)}$$

On substituting the values of 'x' and 'y' from Equations (iii) and (iv) respectively in the above expression we get,

$$q_{13} = \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1\right) + \left(\frac{1}{\epsilon_3} + \frac{1}{\epsilon_2} - 1\right)}$$

$$= \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right) + \left(\frac{2}{\epsilon_3} - 1\right)} \quad \dots(vi)$$

Since ϵ_3 is very small, hence denominator of above expression will be very large. Therefore, *there shall be very large reduction of q_{12} to q_{13}*

If $\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon$, Equation (vi) becomes :

$$q_{13} = \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon} + \frac{1}{\epsilon} - 1\right) + \left(\frac{2}{\epsilon} - 1\right)}$$

$$= \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{2}{\epsilon} - 1\right) + \left(\frac{2}{\epsilon} - 1\right)} \quad \dots(10.12.1)$$

It means $\left(\frac{2}{\epsilon} - 1\right)$ is used twice with one shield. Hence, the Equation (10.12.1) with 'n' shields can be written as,

$$q_{1n} = \frac{\sigma(T_1^4 - T_2^4)}{(n+1)\left(\frac{2}{\epsilon} - 1\right)} \quad (\text{with } n\text{-shields}) \quad \dots(10.12.2)$$

$$\therefore q_{1n} = \frac{\sigma A (T_1^4 - T_2^4)}{(n+1)\left(\frac{2}{\epsilon} - 1\right)} \quad \dots(10.12.2A)$$

With one shield

$$q_{13} = \frac{\sigma(T_1^4 - T_2^4)}{2\left(\frac{1}{\epsilon} + \frac{1}{\epsilon} - 1\right)}$$

$$= \frac{1}{2} (q_{12}) \text{ without shield} \quad \dots(10.12.3)$$

Conclusions

With one radiation shield, the net transfer is reduced to half i.e. by 50%. By using more number of shields, the space resistance increases thereby reduces the net heat transfer rate to a very large extent.

Table 10.12.1 : List of Formulae

1. Wavelength, $\lambda = \frac{\text{Velocity of light, } C (3 \times 10^8 \text{ m/s})}{\text{frequency, } \nu}$
2. Energy/quanta $= h \cdot \nu = \frac{h \cdot C}{\lambda}$ ($h = 6.625 \times 10^{-34} \text{ Js}$ called Planck's constant)
3. Emissive power of a surface, $E = \lim_{dA \rightarrow 0} \frac{dQ}{dA}$

4. Emissivity, $\varepsilon = \frac{\text{emissive power of a surface, } e}{\text{emissive power of a black body, } e_b}$
5. Emissive power, $e = \int_{\lambda=0}^{\lambda=\infty} e_\lambda \cdot d\lambda$ (e_λ represents monochromatic emissive power at particular wavelength.)
6. Planck's law : $e_{b\lambda} = \frac{2\pi C_1}{\lambda^5 [e^{(C_2/\lambda T)} - 1]}$
 $C_1 = 0.596 \times 10^{-16} \text{ W/m}^2$; $C_2 = 1.4387 \times 10^{-2} \text{ (m)} \lambda \text{ in (m) and } T \text{ in (K)}$
7. Wien's Law : $\lambda_m \cdot T = 0.0029 \text{ (m K)}$ where λ_m is the maximum wavelength at temperature, T (Kelvin);
 $(e_{b\lambda})_{\max} = 1.287 \times 10^{-5} \cdot T^5 \cdot (\text{W/m}^2)$
8. Stefan Boltzmann's equation :
 $e_b = \sigma \cdot T^4$ (For black body)
 For other surfaces : $e = \varepsilon \cdot \sigma \cdot T^4$; $E = \varepsilon \cdot A \cdot \sigma \cdot T^4$
9. Kirchhoff's law : Emissivity, $\varepsilon =$ Absorptivity, α (if surface is in thermal equilibrium with surroundings).
10. Solid angle subtended by dA_2 on $dA_1 = (dA_2 \cdot \cos \beta_2)/r^2$
11. Solid angle subtended by dA_1 on $dA_2 = (dA_1 \times \cos \beta_1)/r^2$
12. Lambert cosine rule : $i = i_n \cdot \cos \beta$ and $e = \pi \cdot e_b$ (For unit hemisphere)
13. Shape factor $F_{12} = \frac{1}{A_1} \int_{A_1} \int_{A_2} \frac{\cos \beta_1 \cdot \cos \beta_2}{\pi L^2} \cdot dA_1 \cdot dA_2$
 - A. Reciprocal relation : $A_1 \cdot F_{12} = A_2 \cdot F_{21}$
 $n = n$
 - B. Enclosure theorem : $\sum_{n=1} F_{mn} = 1$ (for $m = 1, 2, \dots, n$).
 - C. Shape factor with respect to Itself
 (a) $F_{mm} = 0$ (For straight and convex surfaces) i.e. $F_{11} = F_{22} = F_{33} = \dots = 0$
 (b) $F_{mm} \neq 0$ (For concave surfaces)
 - D. Additive Rule
 $F_{12} = F_{13} + F_{14}$ If area A_2 is equal to the sum of areas A_3, A_4
 - E. Shape factor $F_{12} = 1$ if surface 1 is flat or convex enclosed by surface 2 and $F_{21} = \frac{A_1}{A_2}$
 - F. Shape factor when two surfaces are large and parallel ($A_1 = A_2$) $\therefore F_{12} = F_{21} = 1$

Heat exchange between Grey bodies

1. $J = \rho G + \varepsilon \cdot e_b$
2. $Q = Q_{\text{net}} = \frac{(e_b - J)}{(1 - \varepsilon) \varepsilon A} = \frac{\text{Potential difference}}{\text{Radiation resistance}}$
3. $Q_{12} = A_1 \cdot F_{12} \cdot (J_1 - J_2) = \frac{J_1 - J_2}{(1/A_1 \cdot F_{12})}$
4. $Q_{1-2} = \frac{e_{b1} - e_{b2}}{R_1 + R_2 + R_3} = \frac{\sigma (T_1^4 - T_2^4)}{\left(\frac{1 - \varepsilon_1}{\varepsilon_1 A_1}\right) + \frac{1}{A_1 \cdot F_{12}} + \frac{(1 - \varepsilon_2)}{\varepsilon_2 A_2}}$



5. Between infinite parallel plates, $Q_{12} = \frac{\sigma \cdot A (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$ ($\because F_{12} = F_{21} = 1$ and $A_1 = A_2 = A$)

6. Between two infinitely long cylinders ($F_{11} = 0, F_{12} = 1$)

$$Q_{12} = \frac{\sigma \cdot A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} ; (A_1 = 2\pi R_1 L, A_2 = 2\pi R_2 L)$$

7. Between two concentric spheres

Same Equation as of srl.6, except $A_1 = 4\pi R_1^2$ and $A_2 = 4\pi R_2^2$

Radiation shields : ($\epsilon_1 = \epsilon_2 = \epsilon_3$)

1. Without shield, $q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$

2. With one shield, $q_{13} = \frac{\sigma (T_1^4 - T_2^4)}{\left(\frac{2}{\epsilon} - 1 \right) + \left(\frac{2}{\epsilon} - 1 \right)} = \frac{1}{2} q_{12}$

3. With n-shields, $q_{13} = \frac{\sigma (T_1^4 - T_2^4)}{(n+1) \left(\frac{2}{\epsilon} - 1 \right)}$

Ex. 10.12.1 : A body at 1400 K of area 10 m^2 is kept in black surroundings at 800 K of area 6 m^2 . It has an emissivity of 0.4 at 1400 K and emissivity of 0.6 at 800 K. Calculate the rate of heat loss by body at 1400 K.

Soln. :

Given : $T_1 = 1400 \text{ K}, A_1 = 10 \text{ m}^2; T_2 = 800 \text{ K},$
 $A_2 = 6 \text{ m}^2 \quad \epsilon_1 = 0.4, \epsilon_2 = 0.6$

Heat radiated by body at 1400 K,

$$Q_1 = \epsilon_1 \sigma A_1 T_1^4 = 0.4 \times (5.67 \times 10^{-8}) 10 \times 1400^4$$

$$= 8.713 \times 10^5 \text{ W}$$

Heat radiated by body at 800 K and absorbed by body at 1400 K,

$$Q_2 = \epsilon_2 \sigma A_2 T_2^4$$

$$= 0.6 \times (5.67 \times 10^{-8}) \times 6 \times 800^2$$

$$= 0.836 \times 10^5 \text{ W}$$

$$\therefore \text{Net heat loss} = Q_1 - Q_2 = 8.713 \times 10^5 - 0.836 \times 10^5$$

$$= 7.877 \times 10^5 \text{ W} \quad \dots \text{Ans.}$$

Ex. 10.12.2 : Consider a black body at a temperature of 2000 K.

- Calculate its total hemispherical emissive power;
- Calculate the wavelength at which maximum emissive power is available from this body.

State and explain the laws of radiation, which you have used to calculate the above mentioned quantities.

Soln. :

$$T = 2000 \text{ K}$$

(i) **Total emissive power**

From Stefan Boltzmann law,

$$E = \sigma T^4 = 5.67 \times 10^{-8} \times (2000)^4$$

$$= 907200 \text{ W/m}^2 \quad \dots \text{Ans.}$$

(ii) **Wavelength corresponding to max. emissive power**

From Weins displacement law :

$$\lambda_{\max} \cdot T = 2.898 \times 10^{-2} = \frac{2.898 \times 10^{-3}}{2000}$$

$$= 1.449 \times 10^{-6} \text{ mK} \quad \dots \text{Ans.}$$

Ex. 10.12.3 : The filament of 100 W incandescent light bulb radiates energy into an enclosure at 350 K. Both the bulb and enclosure act as black bodies. Determine the temperature of the filament of the bulb if it is 0.12 mm in diameter and 4 cm long by considering the heat dissipation by radiation only.

Soln. :

Given : $Q = 100 \text{ W}, T_2 = 350 \text{ K},$

$$d = 0.12 \text{ mm} = 0.12 \times 10^{-3} \text{ m}, L = 5 \text{ cm} = 0.05 \text{ m}.$$

Surface area of filament,

$$A = \pi d L$$

$$= \pi \times (0.12 \times 10^{-3}) \times (0.05)$$

$$= 1.885 \times 10^{-5} \text{ m}^2$$

$$Q = \sigma \cdot A (T_1^4 - T_2^4)$$

$$100 = (5.67 \times 10^{-8}) \times (1.885 \times 10^{-5}) (T_1^4 - 350^4)$$

$$T_1 = 3110.24 \text{ K}$$

...Ans.

Ex. 10.12.4 : One side of metallic plate absorbs a heat flux of 1000 W/m^2 . Its other side is insulated. The emissivity of surface is 0.8 and the convective heat transfer coefficient is $20 \text{ W/m}^2 \text{ K}$. Ambient is maintained at 300 K . Determine the temperature of plate under steady state conditions $\sigma = 5.67 \times 10^{-8}$

Soln. : Refer Fig. P. 10.12.4

$$T_\infty = 300 \text{ K}; \quad \epsilon = 0.8$$

$$\text{Heat flux, } q = 1000 \text{ W/m}^2 = \frac{Q}{A}$$

$$h = 20 \text{ W/m}^2 \text{ K}$$

Under steady state condition

Heat absorbed = Heat rejected

$$Q = Q_{\text{convection}} + Q_{\text{radiation}}$$

$$Q = h A (T_s - T_\infty) + \sigma A \epsilon (T_s^4 - T_\infty^4)$$

$$\frac{Q}{A} = h (T_s - T_\infty) + \sigma \epsilon (T_s^4 - T_\infty^4)$$

$$1000 = 20 \times (T_s - 300) + 5.67 \times 10^{-8}$$

$$\times 0.8 (T_s^4 - 300^4)$$

$$1000 = 20 T_s - 6000 + 4.536 \times 10^{-8} T_s^4 - 367.4$$

$$4.536 \times 10^{-8} T_s^4 + 20 T_s - 7367.4 = 0$$

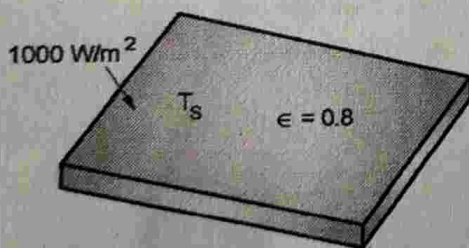


Fig. P. 10.12.4

1st approximation

$$20 T_s = 7367.4; \quad T_s \approx 368.4 \text{ K}$$

$\therefore 4.536 \times 10^{-8} T_s^4$ to be added, actual value will be

less than $T_s = 368.4 \text{ K}$

$$\text{Let, } T_s = 340 \text{ K}; \quad \text{LHS} = 38.76$$

$$\text{Let } T_s = 338 \text{ K}; \quad \text{LHS} = -15.38$$

$$\therefore T_s = 338.5 \text{ K (Approx)}$$

...Ans.

Ex. 10.12.5 : Assuming the sun to be a black body emitting radiation with a maximum intensity at $\lambda = 0.5 \mu\text{m}$.

Calculate the following :

(i) The surface temperature of the sun.

(ii) The heat flux at the surface of sun.

Soln. :

$$\lambda_m = 0.5 \mu\text{m}$$

(i) Surface temperature of sun :

$$\lambda_m T = 0.0029 \text{ mK}$$

$$T = \frac{0.0029}{0.5 \times 10^{-6}} = 5800 \text{ K}$$

...Ans.

(ii) The heat flux at the surface of sun :

$$q = \sigma T^4 = 5.67 \times 10^{-8} \times 5800^4$$

$$= 64.16 \text{ MW/m}^2$$

...Ans.

Ex. 10.12.6 : A black body emits radiation of maximum intensity at a wavelength of $0.5 \mu\text{m}$. Calculate its surface temperature and emissive power. State the laws of radiation heat transfer, which you have used to solve this problem.

Soln. :

Given : $\lambda_{\text{max}} = 0.5 \mu\text{m}$

1) The surface temperature of the body, T .

According to Wein's displacement law,

$$\lambda_{\text{max}} \cdot T = 2829 \mu\text{mK}$$

$$T = \frac{2829}{\lambda_{\text{max}}} = \frac{2829}{0.5} = 5658 \text{ K}$$

...Ans.

2) Emissive power (e_b)

$$e_b = \sigma \cdot T^4 = 5.67 \times 10^{-8} \times 5658^4$$

$$e_b = 58.107 \times 10^6 \text{ W/m}^2$$

...Ans.

The law used to calculate the temperature and emissive power are :

(1) **Wein's displacement law :** It gives the relationship between the temperature of black body and the wavelength at which the maximum value of monochromatic emissive power occurs and given as $\lambda_m \cdot T = 0.0029 \text{ mK}$

(2) **Stefan-Boltzmann law :** The emissive power of a black body is proportional to fourth power of its absolute temperature and given as $q \propto T^4$.

Ex. 10.12.7 : Effective temperature of a body having an area of 0.12 m^2 is 527°C . Calculate the following :

- Rate of radiation energy emission
- Intensity of normal radiation
- Wavelength of maximum monochromatic emissive power

Soln. :

Given : $A = 0.12\text{ m}^2$; $T = 527^\circ\text{C} = 527 + 273 = 800\text{ K}$

(a) Total emission of radiation Q

$$\begin{aligned} Q &= \sigma \cdot A \cdot T^4 \\ &= (5.67 \times 10^{-8}) \times 0.12 \times 800^4 \\ &= 2786.9\text{ W} \end{aligned}$$

...Ans.

(b) Intensity of normal radiation, I_n

$$\begin{aligned} I_n &= \frac{q_b}{\pi} = \frac{\sigma T^4}{\pi} = \frac{5.67 \times 10^{-8} \times (800)^4}{\pi} \\ &= 7392.5\text{ W/m}^2\text{ sr} \end{aligned}$$

...Ans.

(c) Wavelength of max monochromatic emissive power, λ_m

From Wien's Displacement Law;

$$\begin{aligned} \lambda_m T &= 0.0029\text{ mK} \\ \lambda_m &= \frac{0.0029}{T} = \frac{0.0029}{800} \\ \therefore \lambda_m &= 3.625 \times 10^{-4}\text{ m} \\ &= 3.625\text{ }\mu\text{m} \end{aligned}$$

...Ans.

Ex. 10.12.8 : A black surface is maintained at 5000 K . Find :

- Heat flux due to thermal radiation.
- Maximum value of monochromatic emission power and the wave length at which it occurs.
- Fraction of radiations emitted between $\lambda_1 = 0.5\text{ }\mu\text{m}$ to $\lambda_2 = 0.7\text{ }\mu\text{m}$

Soln. :

$$T = 5000\text{ K}$$

(i) Heat flux due to radiation, q_b for black body

$$\begin{aligned} q_b &= \sigma \cdot T^4 = (5.67 \times 10^{-8}) \times 5000^4 \\ &= 35437.5 \times 10^3\text{ W/m}^2 \\ &= 35437.5\text{ kW/m}^2 \end{aligned}$$

...Ans.

(ii) Maximum value of monochromatic emissive power, $e_{b\lambda}$ and wavelength, λ_m

From Wien's law, the condition for maximum emissive power is,

$$\lambda_m \cdot T = 0.00298\text{ mK}$$

$$\lambda_m = \frac{0.00298}{5000} = 0.596 \times 10^{-4}\text{ m} = 0.596\text{ }\mu\text{m} \quad \dots\text{Ans.}$$

From Planck's law :

$$\begin{aligned} e_{b\lambda} &= \frac{2\pi C_1}{\lambda^5 [e^{(C_2/\lambda T)} - 1]} \\ \therefore (e_{b\lambda})_{\max} &= \frac{2\pi C_1}{\lambda_m^5 [e^{(C_2/\lambda_m T)} - 1]} \\ &= \frac{2\pi \times (0.596 \times 10^{-4})}{(0.596 \times 10^{-4})^5 \left[e^{(1.4387 \times 10^{-2})/(0.596 \times 10^{-4} \times 5000)} - 1 \right]} \\ &= 4.018 \times 10^{13}\text{ W/m}^2 \end{aligned}$$

...Ans.

(iii) Fraction of radiations emitted between $\lambda_1 = 0.5\text{ }\mu\text{m}$ to $\lambda_2 = 0.7\text{ }\mu\text{m}$

$$\lambda_1 T = 0.5 \times 5000 = 2500\text{ }\mu\text{m K and}$$

$$\lambda_2 T = 0.7 \times 5000 = 3500\text{ }\mu\text{m K}$$

$$f_{0.7} - f_{0.5} = 0.38294 - 0.16137 = 0.22157$$

Radiation between $\lambda_2 = 0.7\text{ }\mu\text{m}$ and $0.5\text{ }\mu\text{m}$,

$$\begin{aligned} (e_{b\lambda_2} - e_{b\lambda_1}) &= (f_{0.7} - f_{0.5}) \times q_b \\ &= 0.22157 \times 35437.5 \\ &= 7851.9\text{ kW/m}^2 \end{aligned}$$

...Ans.

Ex. 10.12.9 : The sun may be assumed to be a black body having its maximum intensity of radiations at $\lambda = 0.5\text{ }\mu\text{m}$. Calculate the surface temperature of the sun and the heat flux at the surface of sun.

Soln. :

Given : Maximum wavelength, $\lambda_m = 0.5\text{ }\mu\text{m} = 0.5 \times 10^{-6}\text{ m}$

(i) Temperature of sun, T

Using Wien's displacement law,

$$\lambda_m \cdot T = 0.002898\text{ mK}$$

$$T = \frac{0.002898}{0.5 \times 10^{-6}} = 5796\text{ K}$$

...Ans.

(ii) Heat flux at the surface of sun, q_b

$$\begin{aligned} q_b &= \sigma \cdot T^4 = (5.67 \times 10^{-8}) (5796)^4 \\ &= 6.3988 \times 10^7\text{ W/m}^2 \end{aligned}$$

...Ans.

Ex. 10.12.10 : One side of metallic plate absorbs a heat flux of 1000 W/m^2 . It's other side is insulated. The emissivity of surface is 0.8 and the convective heat transfer coefficient is $20\text{ W/m}^2\text{ K}$. Ambient is maintained of 300 K . Determine the temperature of plate under steady state conditions.

Soln. :

Heat flux absorbed, $q_a = 100 \text{ W/m}^2$, $\varepsilon = 0.8$,
 $h = 0 \text{ W/m}^2 \text{ K}$, $T_2 = 300 \text{ K}$.

Let T_1 be the temperature of plate.

Rate of Heat absorbed = Rate of heat flux transmitted by convection + radiation

$$q_a = h(T_1 - T_2) + \varepsilon \cdot \sigma(T_1^4 - T_2^4)$$

$$1000 = 20(T_1 - 300) + 0.8 \times (5.67 \times 10^{-8})(T_1^4 - 300^4)$$

$$1000 = 20T_1 - 6000 + 4.536 \left[\left(\frac{T_1}{100} \right)^4 - \left(\frac{300}{100} \right)^4 \right]$$

$$\left(\because \frac{1}{100^4} = 10^{-8} \right)$$

$$1535.1 - 4.386 T_1 = \left(\frac{T_1}{100} \right)^4 - 81$$

$$\left(\frac{T_1}{100} \right)^4 + 4.286 T_1 - 1616.1 = 0;$$

On solving by trial and error method,

$$T_1 = 340 \text{ K}$$

...Ans.

Ex. 10.12.11 : The furnace of a boiler may be assumed as a black body having a temperature of 2500 K. Find the following :

- Wavelength at which emission is maximum and the magnitude of emissive power at this wave length.
- Total emissive power.
- Monochromatic emissive power at wavelength of $1 \mu\text{m}$.
- Consider the furnace as real body having emissivity as 0.92, find its emissive power.

The area of furnace is 2 m^2 .

Soln. : $T = 2500 \text{ K}$, $A = 2 \text{ m}^2$.

- Maximum wavelength, λ_m and maximum emissive power, $(E_b)_{\max}$.**

$$\lambda_m = \frac{0.002898}{2500} = 1.1592 \times 10^{-6} \text{ m} \quad \dots \text{Ans.}$$

$$(E_b)_{\max} = (e_b)_{\max} \cdot A = 1.287 \times 10^{-5} \times T^5 \times A$$

$$= 1.287 \times 10^{-5} \times (2500)^5 \times 2$$

$$= 2.5137 \times 10^{12} \text{ W} \quad \dots \text{Ans.}$$

- Total emissive power, E_b**

$$E_b = \sigma \cdot A \cdot T^4$$

$$= (5.67 \times 10^{-8}) \times 2 \times (2500)^4$$

$$= 4.42968 \times 10^6 \text{ W} \quad \dots \text{Ans.}$$

- Monochromatic emissive power at $\lambda = 1 \mu\text{m}$**
 $= 1 \times 10^{-6} \text{ m}$, $E_{b\lambda}$

From Planck's law :

$$E_{b\lambda} = A \cdot e_{b\lambda} = A \times \frac{2\pi C_1}{(\lambda)^5 [e^{(C_2/\lambda T)} - 1]}$$

where, $C_1 = 0.596 \times 10^{-6} \text{ W/m}^2$ and $C_2 = 1.4387 \times 10^{-2} \text{ m}$

$$E_{b\lambda} = 2 \times \frac{2\pi \times (0.596 \times 10^{-6})}{(1 \times 10^{-6})^5 \times [e^{(1.4387 \times 10^{-2})/(1 \times 10^{-6} \times 2500)} - 1]}$$

$$= 2.3799 \times 10^{12} \text{ W}$$

...Ans.

- Total emissive power, E when $\varepsilon = 0.92$**

$$E = \varepsilon \cdot \sigma \cdot A \cdot T^4 = 0.92 \times (5.67 \times 10^{-8}) \times 2 \times (2500)^4$$

$$= 4.0753 \times 10^6 \text{ W} \quad \dots \text{Ans.}$$

Ex. 10.12.12 : A radiator of rectangular cross-section having dimensions as $(3 \times 2 \times 1) \text{ m}$ is kept in a room to be maintained at 20°C . The surface temperature of radiator is maintained at 60°C . The emissivity of surfaces of radiator is 0.9. The average convective transfer coefficient is $10 \text{ W/m}^2 \text{ K}$ for all sides. Find the total heat transfer from radiator under steady state conditions.

Soln. :

Given : Size of radiator = $3\text{m} \times 2\text{m} \times 1\text{m}$

$$\therefore \text{Surface area, } A = 2 [3 \times 2 + 2 \times 1 + 1 \times 3] = 22 \text{ m}^2.$$

$$T_s = 60^\circ\text{C} = 333 \text{ K}; T_\infty = 20^\circ\text{C} = 293 \text{ K},$$

$$\varepsilon = 0.9, h = 10 \text{ W/m}^2 \text{ K}$$

Total heat transfer into room will take place due to heat emitted by radiation into room and by convection.

$\therefore$ Total heat transfer,

$$Q = Q_c + Q_{\text{rad.}}$$

$$= hA(T_s - T_\infty) + \varepsilon \cdot \sigma \cdot A \cdot (T_s^4 - T_\infty^4)$$

$$= 10 \times 22 \times (60 - 20) + 0.8 \times (5.67 \times 10^{-8})$$

$$\times 22 \times (333^4 - 293^4)$$

$$= 13716.07 \text{ W} \quad \dots \text{Ans.}$$

Ex. 10.12.13 : Two small surfaces A and B are placed in an isothermal enclosure maintained at constant temperature considered as black body. The irradiation to the surface by the enclosure is 8000 W/m^2 . The absorption of the surfaces A and B are 1000 W/m^2 and 6800 W/m^2 . Under the conditions of steady state find :

- Absorptivity of each surface.
- The temperature and heat flux to each surface.
- Emissive power of each surface and their emissivity.

**Soln. :**

Given : $G = 8000 \text{ W/m}^2$, $G_{\alpha A} = 1000 \text{ W/m}^2$,
 $G_{\alpha B} = 6800 \text{ W/m}^2$.

(i) Absorptivity of surface A, α_A and of surface B, α_B

$$\therefore \text{Absorptivity} = \frac{G_{\alpha}}{G}$$

$$\therefore \alpha_A = \frac{G_{\alpha A}}{G} = \frac{1000}{8000} = 0.125 \quad \dots \text{Ans.}$$

$$\alpha_B = \frac{G_{\alpha B}}{G} = \frac{6800}{8000} = 0.85 \quad \dots \text{Ans.}$$

(ii) The temperature, T and heat flux to each surface q_A and q_B

Since the enclosure is maintained at constant temperature and considered as black body it implies that,

Temperature of enclosure,

$$T = T_A = T_B.$$

Thus,

Irradiation to the surface = Emission from surface

$$8000 = \sigma \cdot T^4 = (5.67 \times 10^{-8}) \cdot T^4$$

$$T = 612.9 \text{ K} \quad \dots \text{Ans.}$$

$$q_A = q_B = 0$$

(* Enclosure and surfaces are at the same temperature)

 $\dots \text{Ans.}$ **(iii) (a) Emissivity of each surface, ϵ_A and ϵ_B** According Kirchhoff's law, $\epsilon = \alpha$

$$\therefore \epsilon_A = \alpha_A = 0.125 \quad \text{and} \quad \epsilon_B = \alpha_B = 0.85 \quad \dots \text{Ans.}$$

(b) Emissive power of each surface e_a and e_b

$$e = \alpha \cdot G$$

$$\therefore e_a = \alpha_A \cdot G$$

$$= 0.125 \times 8000 = 1000 \text{ W/m}^2 \quad \dots \text{Ans.}$$

$$e_b = \alpha_B \times G = 0.85 \times 8000 = 6800 \text{ W/m}^2 \quad \dots \text{Ans.}$$

Ex. 10.12.14 : A 0.5 m^2 plane, grey, diffuse, opaque surface with absorptivity = 0.7 is maintained at 500°C with an irradiation of 10000 W/m^2 . Determine (i) the absorbed energy

(ii) Emitted energy (iii) Total energy leaving the surface per unit area.

Take Stefan Boltzmann constant as $5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$.

Soln. :

$$\alpha = 0.7$$

$$T = 500^\circ\text{C} = 500 + 273 \text{ K} = 773 \text{ K}$$

$$G = 10000 \text{ W/m}^2$$

$$A = 0.5 \text{ m}^2$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$$

(i) Total heat absorbed = αAG

$$= 0.7 \times 0.5 \times 10000 = 3500 \text{ W} \quad \dots \text{Ans.}$$

(ii) Emitted energy = $(\sigma AT^4) \times \alpha$

$$= 5.67 \times 10^{-8} \times 0.5 \times 773^4 \times 0.7$$

$$= 7.08 \times 10^3 \text{ W} \quad \dots \text{Ans.}$$

(iii) Rate of heat reflection,For opaque surface, $\tau = 1$ (* $\alpha + \rho + \tau = 1$)

$$\therefore \rho = 1 - \alpha = 1 - 0.7$$

$$\rho = 0.3$$

$$\therefore \text{Rate of heat reflected} = \rho A G = 0.3 \times 0.5 \times 10000$$

$$= 1500 \text{ W} \quad \dots \text{Ans.}$$

Ex. 10.12.15 : The filament of a 200 W light bulb may be considered a black body radiating into a black surface at 70°C . The filament diameter is 0.15 mm and length is 6 cm. Considering radiation, determine the filament temperature.

Soln. :Given : $Q_{12} = 200 \text{ W}$;Filament 1 : $d_1 = 0.15 \text{ mm} = 0.15 \times 10^{-3} \text{ m}$; $L = 6 \text{ cm} = 0.06 \text{ m}$,

$$T_2 = 70^\circ\text{C}$$

$$= 70 + 273 = 343 \text{ K}$$

Let T_1 be the filament temperature. The radiant heat transfer is given by the equation :

$$Q_{12} = \frac{A_1 \cdot \sigma \cdot (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} \quad \dots (i)$$

Since $A_1 = \pi dL$ is very small compared to A_2 , therefore $\frac{A_1}{A_2} \approx 0$

$$\epsilon_1 = \epsilon_2 = 1 \text{ being black surfaces.}$$

Thus the Equation (i) reduces to :

$$Q_{12} = \epsilon_1 A_1 \cdot \sigma \cdot (T_1^4 - T_2^4)$$

$$200 = 1 \times (\pi \times 0.15 \times 10^{-3} \times 0.06) (5.67 \times 10^{-8}) (T_1^4 - 343^4)$$

$$T_1 = 3342.14 \text{ K or } 3341.4^\circ\text{C} \quad \dots \text{Ans.}$$

Ex. 10.12.16 : Calculate the rate of heat loss from a thermoflask if the polished silvered surfaces have emissivities of 0.05. The liquid in the flask is at 100°C and the casing is at 20°C. Calculate the heat loss if both surfaces are black.

Soln. : $A_1 = A_2$ and $\epsilon_1 = \epsilon_2 = 0.05$

$$F_{1-2} = \frac{1}{\frac{1}{\epsilon_1} + \left(\frac{1}{\epsilon_2} - 1\right) \frac{A_1}{A_2}}$$

$$= \frac{1}{\frac{1}{0.05} + \left(\frac{1}{0.05} - 1\right) \times 1} = 0.0256$$

Case I : When polished silver surfaces

$$\frac{Q}{A} = \sigma F_{1-2} (T_1^4 - T_2^4)$$

$$= (5.67 \times 10^{-8}) \times 0.0256 [(373)^4 - (293)^4]$$

$$= 17.4 \text{ W/m}^2 \quad \dots \text{Ans.}$$

Case II : When two surfaces are black,

$$\epsilon_1 = \epsilon_2 = 1$$

$$F_{1-2} = 1 \text{ and}$$

$$Q/A = (5.67 \times 10^{-8}) \times 1 [(373)^4 - (293)^4]$$

$$= 679.65 \text{ W/m}^2 \quad \dots \text{Ans.}$$

Ex. 10.12.17 : A grey opaque surface has an absorptivity = 0.7. It is maintained at 200°C. It receives an irradiation of 1000 W/m². Its surface area is 0.2 m². Calculate,

- rate of heat absorption
- rate of heat emission
- rate of heat reflection
- radiosity of the surface.

Soln. :

Given : $\alpha = 0.7$, Temperature $T = 200^\circ\text{C}$

Irradiation, $G = 1000 \text{ W/m}^2$, surface area $A = 0.2 \text{ m}^2$

(i) The total radiation absorbed

$$= \alpha \times A \times \text{irradiation}, G = 0.7 \times 0.2 \times 1000$$

$$= 140 \text{ W} \quad \dots \text{Ans.}$$

(ii) Rate of heat emission = $(\sigma \cdot A \cdot T^4) (0.7)$

$$= (5.67 \times 10^{-8}) (0.2) (200 + 273)^4 \times 0.7$$

$$= 397.33 \text{ W} \quad \dots \text{Ans.}$$

(iii) Rate of heat reflection

For opaque surface, transmissivity, $\tau = 0$

$\therefore$ Absorptivity, α + reflectivity $\rho = 1$ ($\because \alpha + \rho + \tau = 1$)

$$\rho = 1 - \alpha = 0.3$$

Rate of heat reflected, $= \rho \times A \times \text{irradiation},$

$$G = 0.3 \times 0.2 \times 1000 = 60 \text{ W} \quad \dots \text{Ans.}$$

(iv) Radiosity of the surface (J)

$$J = \rho G + e_b = \rho \cdot G + \sigma \cdot A \cdot T^4$$

$$= (0.3) (1000) + (5.67 \times 10^{-8}) \times 0.2 \times (200 + 273)^4$$

$$= 867.5 \text{ W} \quad \dots \text{Ans.}$$

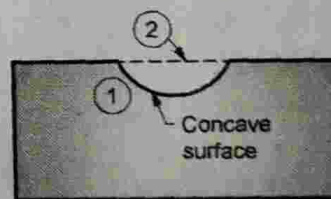
Ex. 10.12.18 :

- If the shape factor of a surface w.r.t. itself is 0.6 what may be the nature of this surface ? Explain with the help of a sketch. Also sketch and explain the types of surface which has no shape factor w.r.t. itself.
- 100 W/m² of energy is incident over an opaque surface maintained at 27°C. Emissivity of this surface = 0.2. Calculate the amount of energy flux reflected from it. Calculate its radiosity.
- The intensity of radiation in a direction normal to a black surface is 100 W/m². Calculate the intensity in a direction making an angle of 30° with the normal. Also calculate the emissive power of the surface).

Soln. :

- Shape factor with respect to itself is the fraction of incident energy emitted by a surface which gets intercepted by itself.

Since given body has shape factor with respect to itself is 0.6, it is the **concave surface** as shown in Fig. P. 10.12.18(a). ...Ans.



Hemispherical cavity

Fig. P. 10.12.18(a)

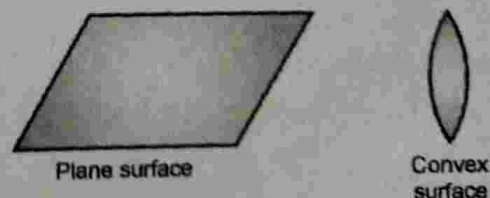


Fig. P. 10.12.18(b)

Plane or convex surfaces have shape factor with respect to itself as zero. (Refer Fig. P. 10.12.18(b)) ...Ans.

(ii) Radiosity, J Given that : $G = 100 \text{ W/m}^2$;

$$\epsilon_b = 1;$$

$$\epsilon_1 = 0.2, T = 27^\circ\text{C} = 300 \text{ K}$$

$$J = \epsilon_1 \epsilon_b + (1 - \epsilon_1) G$$

Emissive power of black body,

$$\epsilon_b = \sigma \cdot T^4 = 5.67 \times 10^{-8} \times 300^4 = 459.27 \text{ W/m}^2$$

Energy emitted $e = \epsilon \cdot \epsilon_b = 0.2 \times 459.3$

$$= 91.85 \text{ W/m}^2 \quad \dots\text{Ans.}$$

$$G = \frac{J - \epsilon \epsilon_b}{(1 - \epsilon)}$$

$$J = (1 - \epsilon) G + \epsilon \epsilon_b$$

$$= (1 - 0.2) 100 + 91.85$$

$$= 171.85 \text{ W/m}^2$$

(iii) Given : Intensity of radiation, $I_n = 100 \text{ W/m}^2$; $\beta = 30^\circ$

$$\therefore \text{Intensity, } I = I_n \cos \beta = 100 \cos 30 = 86.60 \text{ N/m}^2$$

 $\dots\text{Ans.}$ Emissive power, e

$$I_n = \frac{e}{\pi}$$

$$e = I_n \cdot \pi$$

$$e = 100 \times 3.1412 = 314.12 \text{ W/m}^2 \quad \dots\text{Ans.}$$

Ex. 10.12.19 : Prove that shape factor w.r.t. itself in case of cylindrical cavity of diameter D and depth H is given by

$$F_{11} = \frac{4H}{4H + D}$$

Soln. :

Refer Fig. P. 10.12.19.

By enclosure theorem,

$$F_{11} + F_{12} = 1 \quad \dots(i)$$

$$F_{21} + F_{22} = 1 \quad \dots(ii)$$

But, $F_{22} = 0$ (Since surface 2 is a flat surface and it cannot view itself)

$$\therefore F_{21} = 1 \quad \dots(iii)$$

By reciprocal relation,

$$A_1 \cdot F_{12} = A_2 \cdot F_{21} = A_2 \times 1 = A_2$$

$$\therefore F_{12} = \frac{A_2}{A_1} \quad \dots(iv)$$

On substituting in Equation (i)

$$F_{11} = 1 - \frac{A_2}{A_1} \quad \dots(v)$$

$$A_1 = \text{base surface area} + \text{vertical surface area}$$

$$= \frac{\pi}{4} D^2 + \pi DH$$

$$\text{and } A_2 = \frac{\pi}{4} D^2$$

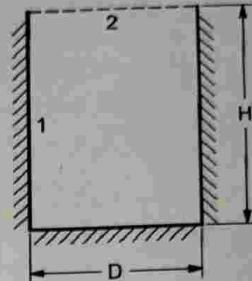


Fig. P. 10.12.19

$$\text{On substituting, } F_{11} = 1 - \frac{\frac{\pi}{4} D^2}{\pi DH + \frac{\pi}{4} D^2}$$

$$= 1 - \frac{\frac{\pi}{4} \cdot D^2}{\frac{\pi}{4} [4DH + D^2]}$$

$$= 1 - \frac{D}{4H + D} = \frac{4H + D - D}{4H + D}$$

$$= \frac{4H}{4H + D} \quad \dots\text{Proved}$$

Ex. 10.12.20 : A sphere of radius 5 cm is concentric with another sphere. Find the radius of outer sphere so that shape factor of outer sphere w.r.t. the inner sphere is 0.6.**Soln. :** Refer Fig. P. 10.12.20.

$$\text{Given : } A_1 = 4\pi r_1^2; \quad A_2 = 4\pi r_2^2 \quad F_{21} = 0.6$$

Since inside sphere (r_1) is totally enclosed in outer sphere (r_2), therefore, $F_{11} = 0$

By enclosure theorem,

$$F_{11} + F_{12} = 1 \quad \text{i.e. } F_{12} = 1$$

We know that $F_{12} A_1 = F_{21} A_2$ (Reciprocal relation)

$$\therefore A_2 = A_1 \frac{F_{12}}{F_{21}} = \frac{A_1 \times 1}{F_{21}}$$

$$4\pi r_2^2 = \frac{4\pi r_1^2 \times 1}{F_{21}}$$

$$r_2^2 = \frac{r_1^2}{F_{21}} = \frac{(0.05)^2}{0.6}$$

$$r_2 = 0.0645 \text{ m} = 6.45 \text{ cm} \quad \dots\text{Ans.}$$

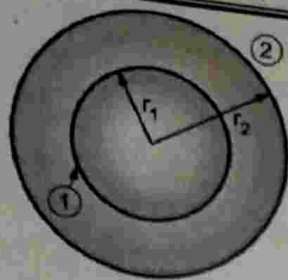


Fig. P. 10.12.20

Ex. 10.12.21 : The radiation shape factor of the circular surface of a thin hollow cylinder of 10 cm diameter and 15 cm length is 0.165. What is the shape factor of the curved surface of the cylinder with respect to itself?

Soln. : Given : $d_1 = d_2 = 10$ cm i.e. $r_1 = r_2 = 5$ cm;
 $L = 15$ cm; $F_{1-2} = 0.165$

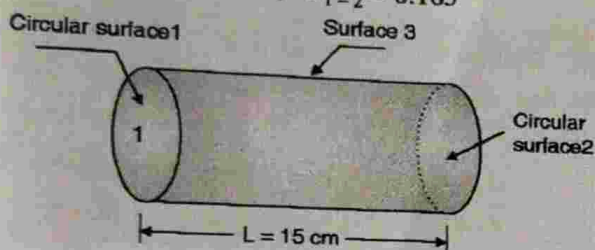


Fig. P. 10.12.21 : A cylinder

$$F_{1-2} = F_{2-1} = 0.165 \quad \text{as } (A_1 = A_2)$$

The shape factor relation among three surfaces is given by (enclosure theorem),

$$F_{1-1} + F_{1-2} + F_{1-3} = 1 \quad \dots(i)$$

$$F_{3-3} + F_{3-2} + F_{3-1} = 1 \quad \dots(ii)$$

$$\text{But, } F_{3-1} = F_{3-2} \text{ (By symmetry)} \quad \dots(iii)$$

Also, $F_{1-1} = F_{2-2} = 0$ (since it cannot view itself)

Substituting Equation (iii) into (ii),

$$F_{3-3} + (F_{3-1} + F_{3-1}) = 1$$

$$F_{3-3} = 1 - 2 \times F_{3-1} \quad \dots(iv)$$

$$A_1 \cdot F_{1-3} = A_3 F_{3-1} \text{ (Reciprocal relation)}$$

$$\begin{aligned} F_{3-1} &= F_{1-3} \times \frac{A_1}{A_3} = F_{1-3} \times \frac{\pi r^2}{2\pi r L} \\ &= F_{1-3} \times \frac{r}{2L} \quad \dots(v) \end{aligned}$$

From Equation (i),

$$\begin{aligned} F_{1-3} &= 1 - F_{1-2} \\ &= 1 - 0.165 = 0.835 \quad \dots(vi) \end{aligned}$$

Substituting the values from Equation (vi) in Equation

(v)

$$F_{3-1} = F_{1-3} \times \frac{r}{2L} = 0.835 \times \frac{r}{2L}$$

$$F_{3-1} = 0.835 \times \frac{5}{2 \times 15} = 0.139 \quad \dots(vii)$$

Substituting from Equation (vii) in Equation (iv),

$$\begin{aligned} F_{3-3} &= 1 - 2 \times F_{3-1} \\ &= 1 - 2 \times 0.139 = 0.721 \quad \dots\text{Ans.} \end{aligned}$$

Ex. 10.12.22 : The radiation shape factor of the circular surface of thin hollow cylinder of 10 cm diameter and 10 cm length is 0.1716. What is the shape factor of curved surface of cylinder with respect to itself?

Soln. : Refer Fig. P. 10.12.22

$d_1 = d_2 = 10$ cm, $r_1 = r_2 = 5$ cm

$L = 10$ cm, $F_{1-2} = 0.1716$

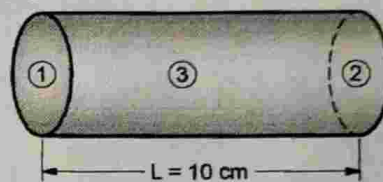


Fig. P. 10.12.22

$$F_{1-2} = F_{2-1} = 0.1716 \text{ (as } A_1 = A_2)$$

The shape factor relation among three surfaces is given by (enclosure theorem)

$$F_{1-1} + F_{1-2} + F_{1-3} = 1 \quad \dots(1)$$

$$F_{3-3} + F_{3-2} + F_{3-1} = 1 \quad \dots(2)$$

$$\text{But, } F_{3-1} = F_{3-2} \text{ (By symmetry)} \quad \dots(3)$$

$$\text{Also, } F_{1-1} = F_{2-2} = 0 \text{ (since it cannot view itself)}$$

$\therefore$ Substituting Equation (3) into Equation (2)

$$\therefore F_{3-3} + (F_{3-1} + F_{3-1}) = 1$$

$$\therefore F_{3-3} + 2F_{3-1} = 1$$

$$\therefore F_{3-3} = 1 - 2 \times F_{3-1} \quad \dots(4)$$

$$A_1 F_{1-3} = A_3 F_{3-1} \text{ (Reciprocal relation)}$$

$$F_{3-1} = \frac{A_1}{A_3} \times F_{1-3} = \frac{\pi r^2}{2\pi r L} \times F_{1-3}$$

$$F_{3-1} = \frac{r}{2L} \times F_{1-3} \quad \dots(5)$$

From Equation (1)

$$\begin{aligned} F_{1-3} &= 1 - (F_{1-1} + F_{1-2}) \\ &= 1 - F_{1-2} \\ &= 1 - 0.1716 = 0.8284 \end{aligned}$$

Substitute in Equation (5),

$$F_{3-1} = \frac{r}{2L} \times F_{1-3}$$

$$= \frac{5}{2 \times 10} \times 0.8284$$

$$F_{3-1} = 0.2071 \quad \dots(6)$$

Substitute Equation (6) in Equation (4)

$$\begin{aligned} F_{3-3} &= 1 - 2 \times F_{3-1} = 1 - 2 \times 0.2071 \\ &= 0.5858 \quad \dots \text{Ans.} \end{aligned}$$

Ex. 10.12.23 : The rectangular surfaces of equal size having dimensions $2\text{ m} \times 4\text{ m}$ are placed parallel to each other at a distance of 5 m . The temperature of surfaces are 100°C and 200°C respectively. Considering the surfaces to be black, find the heat exchange by radiation between two surfaces. Assume $F_{12} = 0.072$.

Soln. : Refer Fig. P. 10.12.23.

$$T_1 = 200^\circ\text{C} = 473\text{ K};$$

$$T_2 = 100^\circ\text{C} = 373\text{ K}$$

$$A = x \times y = 4 \times 2 = 8\text{ m}^2$$

$$\begin{aligned} Q_{1-2} &= \sigma \cdot A \cdot F_{12} \cdot (T_1^4 - T_2^4) \\ &= (5.67 \times 10^{-8}) \times 8 \times 0.072 (473^4 - 373^4) \\ &= 1002.6\text{ W} \quad \dots \text{Ans.} \end{aligned}$$

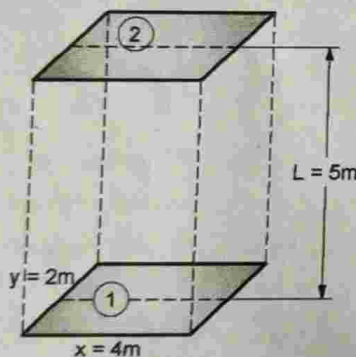


Fig. P. 10.12.23

Ex. 10.12.24 : Why does a cavity with a small hole behave as a black body? Thermal radiation strikes a surface, which has a reflectivity of 0.55 and a transmissivity, of 0.032 . The absorbed flux as measured indirectly by heating effect works out to be 95 W/m^2 . Determine the rate of incident flux.

Soln. :

Black body

The body which absorbs entire radiations and emits maximum radiations is called a black body. It implies that $\rho = \tau = 0$ and $\alpha = 1$.

$$\text{Now, } \rho = 0.55, \tau = 0.032$$

$\therefore$ we have

$$\rho + \tau + \alpha = 1$$

$$\therefore 0.55 + 0.032 + \alpha = 1$$

$$\therefore \alpha = 0.41$$

$$\text{Now, } \alpha = \frac{Q_a}{Q}$$

$$\therefore 0.41 = \frac{95}{Q}$$

$$\therefore Q = 231.70\text{ W/m}^2 \quad \dots \text{Ans.}$$

Ex. 10.12.25 : A small sphere - 1 of 0.3 m outside diameter at 600 K kept at the geometric centre of a large sphere - 2 of internal diameter of 0.6 m having a temperature of 300 K .

Calculate :

- Shape factor F_{21} and F_{12}
- Energy emitted from sphere 2 to sphere 1 assuming their surfaces to act as black body.
- Net heat exchanged between two spheres.

Soln. :

$$\begin{aligned} \text{Given : } d_1 &= 0.3\text{ m i.e. } r_1 = 0.15\text{ m } T_1 = 600\text{ K} \\ d_2 &= 0.6\text{ m i.e. } r_2 = 0.2\text{ m, } T_2 = 300\text{ K} \end{aligned}$$

Refer Fig. P. 10.12.25.

(i) **Shape factor F_{12} and F_{21}**

By summation rule (enclosure theorem),

$$F_{11} + F_{12} = 1$$

But $F_{11} = 0$ ($\because$ energy leaving surface 1 strikes itself)

$$\therefore F_{12} = 1 \quad \dots \text{Ans.}$$

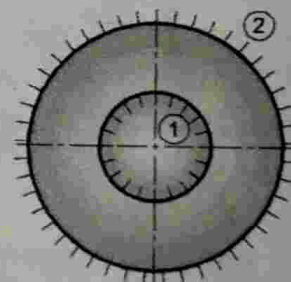


Fig. P. 10.12.25

By reciprocal relation,

$$A_1 F_{12} = A_2 \cdot F_{21};$$

$$F_{21} = \frac{A_1}{A_2} \times F_{12} = \frac{A_1}{A_2} \times 1$$

$$\therefore F_{21} = \frac{A_1}{A_2} = \frac{4\pi r_1^2}{4\pi r_2^2} = \frac{r_1^2}{r_2^2}$$

$$= \frac{(0.15)^2}{(0.3)^2} = 0.25$$

$\dots \text{Ans.}$

(ii) Energy emitted by sphere 2 to sphere 1

It shows that only 25% of radiation energy of large sphere is incident upon the small sphere and absorbed by it as black body since $F_{21} = 0.25$

...Ans.

(iii) Net heat exchanged between two spheres, Q_{net}

$$\begin{aligned} Q_{\text{net}} &= A_1 \sigma \cdot F_{12} (T_1^4 - T_2^4) \\ &= 4\pi r_1^2 \cdot \sigma \cdot F_{12} (T_1^4 - T_2^4) \\ &= 4\pi (0.15)^2 \times (5.67 \times 10^{-8}) \times 1 \times (600^4 - 300^4) \\ &= 1947.83 \text{ W} \end{aligned}$$

...Ans.

Ex. 10.12.26 : Two mutually perpendicular surfaces are shown in Fig. P. 10.12.26. Find F_{12} .

Soln. :

The radiant energy leaving the surface 1 to 2 is equal to the difference of the total energy leaving the surfaces 1 and 3 and the energy leaving the surface 3 to 2 i.e.

Assuming all surfaces to be black and at the same temperature, then

$$\therefore A_1 \cdot F_{12} = A_{(1+3)} \cdot F_{(1+3)2} - A_3 F_{32} \quad \dots(i)$$

$$X = 1.2 \text{ m,}$$

$$Z = 1.2 \text{ m, } Y_{(1+3)} = 0.9 \text{ m}$$

$$\therefore \frac{Z}{X} = \frac{1.2}{1.2} = 1,$$

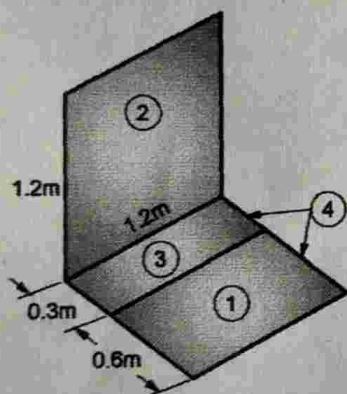


Fig. P. 10.12.26

From Fig. P. 10.12.26

$$F_{(1+3)2} = 0.24$$

For F_{32} : $X = 1.2 \text{ m, } Z = 1.2 \text{ m, } Y = 0.3 \text{ m}$

$$\frac{Z}{X} = \frac{1.2}{1.2} = 1 \text{ and } \frac{Y}{X} = \frac{0.3}{1.2} = 0.25$$

From Fig. P. 10.12.26,

$$F_{32} = 0.37$$

On substituting the values in Equation (i),

$$(0.6 \times 1.2) F_{12} = (0.9 \times 1.2) 0.24 - (0.3 \times 1.2) 0.37$$

$$F_{12} = 0.545$$

...Ans.

Ex. 10.12.27 : The dimensions of a room are 10 m $\times$ 10 m $\times$ 7 m height. The roof temperature is 60°C, while the walls and the floor are at 30°C. Calculate the rate of heat transfer from the roof to the walls and floor together. Emissivity of walls and floor = 0.8. Emissivity of roof is 0.6.

Soln. :

Given data

$$T_1 = 60 + 273 = 333 \text{ K}$$

$$T_2 = 30 + 273 = 303 \text{ K}$$

$$A_1 = 10 \times 10 = 100 \text{ m}^2 \text{ (Roof area)}$$

$$A_2 = 10 \times 10 + 4 \times 10 \times 7 = 380 \text{ m}^2$$

(floor + 4 Walls area)

$$\epsilon_1 = 0.6; \quad \epsilon_2 = 0.8$$

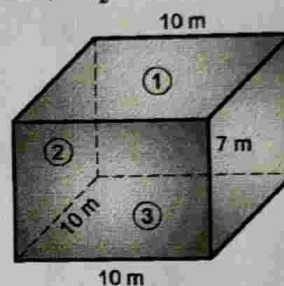


Fig. P. 10.12.27(a)

Equivalent electrical circuit

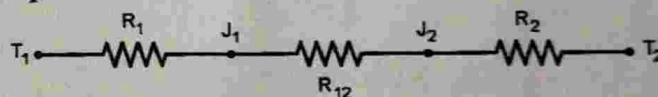


Fig. P. 10.12.27(b)

Surface Resistance

$$R_1 = \frac{1 - \epsilon_1}{A_1 \epsilon_1} = \frac{1 - 0.6}{100 \times 0.6} = 0.0067$$

$$R_2 = \frac{1 - \epsilon_2}{A_2 \epsilon_2} = \frac{1 - 0.8}{380 \times 0.8} = 0.000658$$

Since energy leaving surface 1 (roof) will all fall outside walls and floor, shape factor $F_{12} = 1$

Shape resistance

$$R_{12} = \frac{1}{A_1 F_{12}} = \frac{1}{100 \times 1} = 0.01$$

$$\begin{aligned} Q_{12} &= \frac{\sigma (T_1^4 - T_2^4)}{R_1 + R_{12} + R_2} \\ &= \frac{5.67 \times 10^{-8} (333^4 - 303^4)}{0.0067 + 0.01 + 0.000658} \end{aligned}$$

$$= \frac{5.67 \times 10^{-8} (122.96 \times 10^8 - 84.29 \times 10^8)}{0.017358}$$

$$= 12.632 \text{ kW}$$

...Ans.

Ex. 10.12.28 : Calculate the net radiant heat interchange per m^2 for two large parallel plates maintained at 800°C and 300°C . The emissivities of two plates are 0.3 and 0.6 respectively.

Soln. :

Given : $\epsilon_1 = 0.3, \epsilon_2 = 0.6,$
 $T_1 = 800 + 273 = 1073 \text{ K},$
 $T_2 = 300 + 273 = 573 \text{ K}$

Net heat transfer Q_1

$$Q_1 = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

$$= \frac{5.67 \times 10^{-8} (1073^4 - 573^4)}{\frac{1}{0.3} + \frac{1}{0.6} - 1}$$

$$= 17261.72 \text{ W}$$

...Ans.

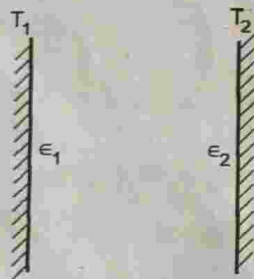


Fig. P. 10.12.28

Ex. 10.12.29 : Two opposed parallel infinite planes are maintained at 420 K and 480 K respectively. Calculate the heat flux between these planes if one has an emissivity of 0.8 and other an emissivity of 0.7. Does it matter which plane has which emissivity? How this heat flux will be affected if :

- (i) The temperature difference is doubled by raising the temperature 480 K to 540 K .
 (ii) The planes are assumed to be black?

Soln. :

Heat transfer by radiation

$$Q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{5.67 \times 10^{-8} (420^4 - 480^4)}{\left(\frac{1}{0.8} + \frac{1}{0.7} - 1\right)}$$

$$= \frac{-1245.54}{1.25 + 1.419 - 1}$$

$$= -741.83 \text{ W/m}^2 \text{ (heat flowing from plate 2 to 1)}$$

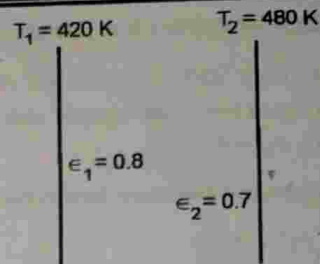


Fig. P. 10.12.29

If emissivity changes from plate 1 to 2 it shall not affect the result but it is true only for facing each other; not for enclosures / curved surfaces.

(i) $T_1 = 420 \text{ K}, T_2 = 540 \text{ K}$

$$Q_{12} = \frac{5.67 \times 10^{-8} (420^4 - 540^4)}{1.679}$$

$$= -1820.67 \text{ W/m}^2$$

...Ans.

(ii) If planes are black body

$$Q_{12} = \sigma (T_1^4 - T_2^4) = 5.77 \times 10^{-8} (420^4 - 480^4)$$

$$= -1245.54 \text{ W/m}^2$$

...Ans.

Ex. 10.12.30 : An enclosure measures $1.5 \text{ m} \times 1.75 \text{ m}$ with a height of 2 m . The walls and ceiling are maintained at 250°C and the floor at 150°C . The walls and ceiling have an emissivity of 0.82 and the floor (0.7). Determine the net radiation to the floor.

Soln. : Enclosure, $l = 1.5 \text{ m}, b = 1.75 \text{ m}, h = 2 \text{ m}$

Given :

Temperature of walls and ceiling, $T_1 = 250^\circ\text{C} = 523 \text{ K}$ Temperature of floor, $T_2 = 150^\circ\text{C} = 423 \text{ K}$ Emissivity of walls and ceiling, $\epsilon_1 = 0.82$ Emissivity of floor, $\epsilon_2 = 0.7$ A_1 = Total area of walls and ceiling

$$= [(l + b)h] + 2 \times l \times b$$

$$= [(1.5 + 1.75)2] + 2 \times 1.5 \times 1.75$$

$$= 15.625 \text{ m}^2$$

$$A_2 = \text{floor area} = 1.5 \times 1.75 = 2.625 \text{ m}^2$$

By summation rule, $F_{2-1} + F_{2-2} = 1$

$$\therefore F_{2-1} = 1 \text{ ...As floor is completely enclosed by area } A_1$$

$$A_1 F_{1-2} = A_2 F_{2-1}$$

$$F_{1-2} = A_2 / A_1 = 2.625 / 15.625 = 0.168$$

Rate of heat transfer by radiation from the walls and the ceiling to the floor

$$Q_{1-2} = \sigma A_1 F_{1-2} (T_1^4 - T_2^4)$$

$$F_{1-2} = \frac{1}{\left[\frac{1}{\epsilon_1} - 1\right] + \frac{1}{F_{1-2}} + \frac{A_1}{A_2} \left[\frac{1}{\epsilon_2} - 1\right]}$$

$$= \frac{1}{\left(\frac{1}{0.82} - 1\right) + \frac{1}{0.168} + \frac{15.625}{2.625} \times \left(\frac{1}{0.7} - 2\right)}$$

$$= 0.114$$

$$\therefore Q_{1-2} = (5.67 \times 10^{-8}) 15.625 \times 0.114 (523^4 - 423^4)$$

$$Q_{1-2} = 4.23 \times 10^3 \text{ W} = 4.23 \text{ kW} \quad \dots \text{Ans.}$$

Ex. 10.12.31 : The net radiation from the surfaces of two parallel plates maintained at temperature T_1 and T_2 is to be reduced by 69 times. Calculate the number of screens to be placed between the two surfaces to achieve this reduction in heat exchange. Assuming the emissivity of screens as 0.05 and that of the surfaces as 0.8.

Soln. :

$$\epsilon_s = 0.05 ; \quad \epsilon_1 = \epsilon_2 = 0.8,$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}$$

$$Q_1 = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{x}{\frac{1}{0.8} + \frac{1}{0.8} - 1} = \frac{x}{1.5} \quad \dots (i)$$

(Let : $\sigma (T_1^4 - T_2^4) = x$ for convenience)

$$\text{Given : } Q_{in} = \frac{1}{79} Q_1 \quad \dots (ii)$$

$$Q_{in} = \frac{\sigma (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right) + \left(\frac{2}{\epsilon_s} - 1\right)n}$$

(where n is number of shields)

$$Q_{in} = \frac{x}{1.5 + \left(\frac{2}{0.05} - 1\right)n} = \frac{x}{1.5 + 39n} \quad \dots (iii)$$

From Equation (ii) :

$$\frac{Q_1}{79} = \frac{x}{1.5 + 39n}$$

But, $Q_1 = \frac{x}{1.5}$

From Equation (i) therefore,

$$\frac{1}{79} \times \frac{x}{1.5} = \frac{x}{1.5 + 39n}$$

$$\frac{1}{79 \times 1.5} = \frac{1}{1.5 + 39n};$$

$$1.5 + 39n = 1.5 \times 79$$

$$n = 3 \text{ shields}$$

...Ans.

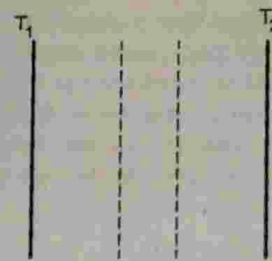


Fig. P. 10.12.31

Ex. 10.12.32 : A cubical oven has inside sides equal to 0.5 m. One of the faces of the oven forms the door. If the five other inside faces are black and maintained at 600°C , find the rate of heat loss if the oven door is kept open.

Soln. :

Given : Cubical oven with side = 0.5 m ; $\epsilon_1 = 1$;

$$T_1 = 600^\circ\text{C} = 873 \text{ K}$$

One of the faces to be a door, surfaces are black and are at 600°C .

$$\text{Surface area of door} = A_2 = 0.5 \times 0.5 = 0.25 \text{ m}^2$$

$$\text{Area of other five surfaces} = A_1 = 5 \times (0.5 \times 0.5) = 1.25 \text{ m}^2$$

$$F_{11} + F_{12} = 1 \quad \dots (i)$$

By summation rule,

$$F_{21} + F_{22} = 1 \quad \dots (ii)$$

But, $F_{22} = 0$ ($\because$ Surface cannot view itself)

$$\therefore F_{21} = 1$$

By reciprocal relation,

$$F_{12} = \frac{A_2}{A_1} \times F_{21}$$

$$= \frac{A_2}{A_1} \times 1 = \frac{A_2}{A_1} \quad \dots (iii)$$

From Equation (i)

$$F_{11} = 1 - F_{12} = 1 - \frac{A_2}{A_1}$$

$$\therefore F_{11} = 1 - \frac{0.25}{1.25} = 0.8$$

Rate of heat loss when door is open

$$Q = \sigma \cdot A_1 \cdot \epsilon_1 \cdot T_1^4 \cdot \frac{(1 - F_{11})}{1 - 1(1 - \epsilon_1) \times F_{11}}$$

$$= (5.67 \times 10^{-8}) \times (1.25) \times 1 \times (873)^4 \cdot \frac{(1 - 0.8)}{1 - 1(1 - 1) \times 0.8}$$

$$= 8.2334 \times 10^3 \text{ W} = 8.2334 \text{ kW} \quad \dots \text{Ans.}$$

Ex. 10.12.33 : Find out heat transfer rate due to radiation between two infinitely long parallel planes. One plane has emissivity of 0.4 and is maintained at 200°C . Other plane has emissivity of 0.2 and is maintained at 30°C . If a radiation shield ($\epsilon = 0.5$) is introduced between the two planes, find percentage reduction in heat transfer rate and steady state temperature of the shield.

Soln. : Refer Fig. P. 10.12.33.

Given : Plane 1 : $\epsilon_1 = 0.4$, $T_1 = 200^\circ\text{C} = 473 \text{ K}$

Plane 2 : $\epsilon_2 = 0.2$, $T_2 = 30^\circ\text{C} = 303 \text{ K}$

Radiation shield-3 : $\epsilon_3 = 0.5$

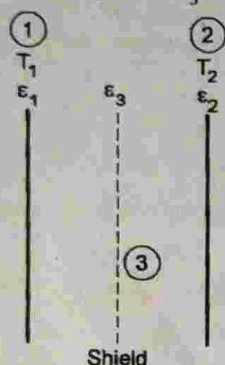


Fig. P. 10.12.33

1. Heat transfer rate without shield, Q_{12}

$$Q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right)}$$

$$= \frac{(5.67 \times 10^{-8}) (473^4 - 303^4)}{\left(\frac{1}{0.4} + \frac{1}{0.2} - 1\right)} = 363.1 \text{ W/m}^2$$

2. Heat transfer rate when shield is inserted, Q_{13}

$$Q_{13} = \frac{\sigma (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right) + \left(\frac{2}{\epsilon_3} - 1\right)}$$

$$= \frac{(5.67 \times 10^{-8}) (473^4 - 303^4)}{\left(\frac{1}{0.4} + \frac{1}{0.2} - 1\right) + \left(\frac{2}{0.5} - 1\right)} = 248.4 \text{ W/m}^2$$

3. Percentage reduction in heat transfer due to radiation shield,

$$= \frac{Q_{12} - Q_{13}}{Q_{12}} \times 100\%$$

$$= \frac{(363.1 - 248.4)}{363.1} \times 100$$

$$= 31.59\% \quad \dots\text{Ans.}$$

Ex. 10.12.34 : The size of a room is $3\text{m} \times 3\text{m} \times 3\text{m}$. The walls and ceiling are maintained at a temperature of 315 K . The floor of the room is at 308 K . Calculate the net rate of heat transfer by radiation to the floor. If floor is considered as surface '1' and (ceilings + walls) is considered as surface '2', calculate, F_{1-1} , F_{1-2} , F_{2-2} , F_{2-1} , F_{1-3} and F_{2-3} .

Soln. : Refer Fig. P. 10.12.34.

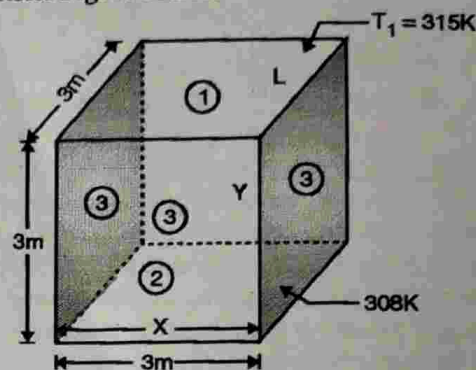


Fig. P. 10.12.34

Given : $T_1 = T_3 = 315 \text{ K}$; $T_2 = 308 \text{ K}$

$$X = Y = L = 3 \text{ m}$$

$$\frac{X}{L} = \frac{3}{3} = 1; \quad \frac{Y}{L} = \frac{3}{3} = 1$$

$\therefore$ From shape factor chart. (Fig. 10.9.1)

$$F_{12} = 0.22$$

...Ans.

By reciprocal relation

$$A_1 F_{1-2} = A_2 F_{2-1} \quad (\because A_1 = A_2)$$

$$F_{1-2} = F_{2-1} = 0.22$$

...Ans.

$$A_1 = A_2 = 3 \times 3 = 9 \text{ m}^2$$

$$A_3 = 4 \times (3 \times 3) = 36 \text{ m}^2$$

$\therefore$ Since surfaces 1 and 2 can not view themselves respective by,

$$F_{1-1} = F_{2-2} = 0$$

...Ans.

By enclosure rule :

$$F_{1-2} + F_{1-3} = 1$$

$$\text{and } F_{2-1} + F_{2-3} = 1$$

$$F_{1-3} = 1 - 0.22 = 0.78$$

...Ans.

$$F_{2-3} = 0.78$$

...Ans.

Net heat transfer rate

$$Q_{\text{net}} = A_1 \cdot F_{12} \sigma (T_1^4 - T_2^4) + A_3 \cdot F_{13} \sigma (T_1^4 - T_2^4)$$

$$= 9 \times 0.22 (5.67 \times 10^{-8}) (315^4 - 308^4)$$

$$+ 36 \times 0.78 (5.67 \times 10^{-8}) (315^4 - 308^4)$$

$$= 1442.6 \text{ W} \quad \dots\text{Ans.}$$

Ex. 10.12.35 : A small sphere (outside diameter = 60 mm) with a surface temperature of 300°C is located at the geometric centre of a large sphere (inside diameter 360 mm) with an inner surface temperature of 15°C. Calculate how much percent of emission from the inner surface of the large sphere is incident upon the outer surface of the small sphere. What is the net interchange of heat between the spheres? Assume that both surfaces approach black body behaviour.

Soln. :

Given : $D_1 = 60 \text{ mm}; r_1 = 30 \text{ mm} = 0.03 \text{ m}$
 $D_2 = 360 \text{ mm}; r_2 = 180 \text{ mm} = 0.18 \text{ m}$
 $T_1 = 300 + 273 = 573 \text{ K},$
 $T_2 = 15 + 273 = 288 \text{ K}$

Both sphere are black bodies.

(i) What % of emission from sphere 2 is falling on spheres 1 i.e. F_{12}

Shape factor $F_{12} = 1$ on sphere 1 is completely enclosed by sphere 2.

As per reciprocal theorem,

$$F_{12} \cdot A_1 = F_{21} \cdot A_2$$

$$\therefore F_{21} = F_{12} \cdot \frac{A_1}{A_2} = F_{12} \cdot \frac{4\pi r_1^2}{4\pi r_2^2}$$

$$= 1 \times \frac{0.03^2}{0.18^2} = 0.0278$$

i.e. 2.78% radiation from sphere 2 is incident on sphere 1

...Ans.

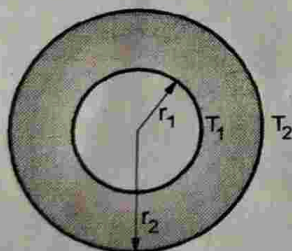


Fig P. 10.12.35

(ii) Heat exchange between spheres,

$$Q_{12} = \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)}$$

Since both spheres are black bodies, hence $\epsilon_1 = \epsilon_2 = 1$

$$\therefore Q_{12} = \sigma A_1 (T_1^4 - T_2^4)$$

$$= 5.67 \times 10^{-8} \times 4\pi \times 0.03^2 (573^4 - 288^4)$$

$$= 64.71 \text{ W}$$

...Ans.

Ex. 10.12.36 : A double-walled thermos flask may be assumed to be equivalent to two infinite parallel plates. The emissivities of wall are 0.3 and 0.7 respectively. Space between them is evacuated. Find the heat transfer rate by radiation through the flask if inside surface temperature is 60°C and outside surface at 30°C under steady state.

Soln. : Refer Fig. P. 10.12.36.

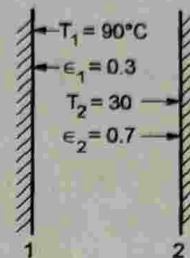


Fig. P. 10.12.36

$$T_1 = 90 + 273 = 363 \text{ K}; \epsilon_1 = 0.3$$

$$T_2 = 30 + 273 = 303 \text{ K}; \epsilon_2 = 0.7$$

$$q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{5.67 \times 10^{-8} (363^4 - 303^4)}{\frac{1}{0.4} + \frac{1}{0.7} - 1}$$

$$= \frac{5.67 \times 10^{-8} (173.6 \times 10^8 - 84.29 \times 10^8)}{3.34 + 1.429 - 1}$$

$$= 134.43 \text{ W/m}^2$$

...Ans.

Ex. 10.12.37 : Cryogenic fluid flows through a tube 30 mm diameter which is concentric with a tube of 90 mm diameter. Surface emissivities of inner and outer tubes are 0.2 and 0.5 while respective temperatures are 100 K and 300 K. Find the heat gain by fluid.

Soln. :

Given : $d_1 = 30 \text{ mm} = 0.03 \text{ m}, \epsilon_1 = 0.2, T_1 = 100 \text{ K}$

$d_2 = 90 \text{ mm} = 0.09 \text{ m}, \epsilon_2 = 0.5, T_2 = 300 \text{ K}$

Heat flow rate from inner pipe to outer tube by radiation per m length,

$$q_{12} = \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)}$$

$$(\because A_1 = \pi d_1 L; A_2 = \pi d_2 \cdot L)$$

$$= \frac{5.67 \times 10^{-8} \times (\pi \times 0.03 \times 1) (100^4 - 300^4)}{\frac{1}{0.2} + \frac{\pi \times 0.03 \times 1}{\pi \times 0.09 \times 1} \left(\frac{1}{0.5} - 1 \right)}$$

$$= -\frac{42.75}{5 + 0.334} = -8.01 \text{ W}$$

...Ans.

Negative sign indicates that heat is being received by body 1 i.e. by the fluid flowing in it.

Ex. 10.12.38 : A cylindrical heater element with diameter = 2 cm has emissivity = 0.7. It is kept at 727°C. It is located in a large room whose walls are maintained at 27°C. Find the rate of radiant heat transfer per unit length from the heater towards the walls of the room.

If the heater is enclosed in another pipe at 27°C, with diameter = 200 mm and emissivity = 0.2, what is the rate of radiant transfer?

Soln. :

Given : Heater element $d_1 = 2$ cm

$$\text{i.e. } r_1 = \frac{2}{2} = 1 \text{ cm} = 0.01 \text{ m,}$$

$$\epsilon_1 = 0.7, T_1 = 727^\circ\text{C} = 1000 \text{ K}$$

$$T_2 = 27^\circ\text{C} = 300 \text{ K. Assume length } L = 1 \text{ m}$$

$$A_1 = 2\pi r_1 L = 2\pi \times 0.01 \times 1 = 0.06283 \text{ m}^2$$

$$R_1 = \frac{1 - \epsilon_1}{A_1 \epsilon_1} = \frac{1 - 0.7}{0.06283 \times 0.7} = 6.82$$

Consider the pipe as surface (1) and enclosure as surface (2)

$$Q = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}} \quad \dots(i)$$

Case (i) Enclosure as large as room

As A_2 is very large, the resistance $\frac{(1 - \epsilon_2)}{A_2 \epsilon_2}$ will be very small and can be neglected. Also, $F_{12} = 1$ (since all the heat is absorbed by the walls of the room)

The Equation (i) reduces to :

$$\begin{aligned} \therefore Q &= A_1 \sigma \epsilon_1 (T_1^4 - T_2^4) \\ &= 0.06283 \times 0.7 \times (5.67 \times 10^{-8}) (1000^4 - 300^4) \\ &= 2473.5 \text{ W/m} \quad \dots\text{Ans.} \end{aligned}$$

Case (ii) Enclosure as pipe at 27°C

$$D_2 = 200 \text{ mm} = 0.2 \text{ m i.e. } r_2 = 0.2 \text{ m, } \epsilon_2 = 0.2$$

$$q = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

In this case also we have $F_{12} = 1$ (Since surface 1 is enclosed in surface 2)

$$A_1 = (\pi d_1 L)_{\text{heater}} = \pi \times 0.02 \times L = 0.02 \pi L$$

$$A_2 = (\pi d_2 L)_{\text{pipe}} = \pi \times 0.2 \times L = 0.2 \pi L$$

$$\therefore Q = \frac{5.67 \times 10^{-8} (1000^4 - 300^4)}{\frac{1 - 0.7}{0.02\pi L \times 0.7} + \frac{1}{0.02\pi L} + \frac{1 - 0.2}{0.2\pi L \times 0.2}}$$

$$\therefore \frac{Q}{L} = 1932.5 \text{ W/m} \quad \dots\text{Ans.}$$

Ex. 10.12.39 : Determine heat lost by radiation per meter length of 80 mm diameter pipe at 300°C, if,

- Located in a large room with red brick wall at a temperature of 27°C;
- Enclosed in a 160 mm diameter red brick conduit at a temperature of 27°C.

Take ϵ (pipe) = 0.79 and ϵ (brick conduit) = 0.93

Soln. :

$$D = 80 \text{ mm} = 0.08 \text{ m}; \quad L = 1 \text{ m}$$

$$T_1 = 300 + 273 = 573 \text{ K}; \quad \epsilon_1 = 0.79$$

$$T_2 = 27 + 273 = 300 \text{ K}; \quad \epsilon_2 = 0.93$$

(i) **Case I : Heat lost by radiation,**

$$Q = \sigma \epsilon_1 A_1 (T_1^4 - T_2^4)$$

$$= \sigma \times \epsilon_1 \times \pi D L \times (T_1^4 - T_2^4)$$

$$Q = (5.67 \times 10^{-8}) \times 0.79 \times (\pi \times 0.08 \times 1) [573^4 - 300^4]$$

$$Q = 1122.4 \text{ W/m length}$$

...Ans.

(ii) **Case II : Refer Fig. P.10.12.39**

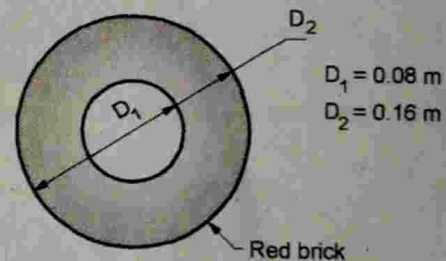


Fig. P. 10.12.39

$$D_1 = 0.08 \text{ m}; \quad D_2 = 0.16 \text{ m}; \quad L = 1 \text{ m}$$

$$A_1 = \pi D_1 L = \pi \times 0.08 \times 1 = 0.2513$$

$$A_2 = \pi D_2 L = \pi \times 0.16 \times 1 = 0.5026$$

Heat lost by radiation,

$$Q_2 = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

$$F_{12} = 1$$

$$Q_2 = \frac{5.67 \times 10^{-8} (573^4 - 300^4)}{\frac{1 - 0.79}{0.2513 \times 0.79} + \frac{1}{0.2513 \times 1} + \frac{1 - 0.93}{0.5026 \times 0.93}}$$

$$Q_2 = 997 \text{ W/m length}$$

...Ans.

Ex. 10.12.40 : Determine the heat loss rate by radiation from a sheet tube of outside diameter 70 mm and 3 m long at a temperature of 227°C. This tube is located within a square brick conduit of 0.3 m side and at 27°C. Assume, emissivity of steel, $\epsilon_1 = 0.79$; emissivity of brick, $\epsilon_2 = 0.93$.

Soln. :

Given : Tube 1 : $d_1 = 70 \text{ mm} = 0.07 \text{ m}$, $L = 3 \text{ m}$;
 $T_1 = 227^\circ\text{C} = 500 \text{ K}$
 $A_1 = \pi d_1 L = \pi \times 0.07 \times 3 = 0.65974 \text{ m}^2$;
 $\epsilon_1 = 0.79$

Brick conduit 2

Each side, $a = 0.3 \text{ m}$
Hence, $A_2 = 4 \times a \times L = 4 \times 0.3 \times 3 = 3.6 \text{ m}^2$;
 $\epsilon_2 = 0.93$
 $T_2 = 27^\circ\text{C} = 300 \text{ K}$

Heat loss rate by tube surface,

$$Q_{12} = \frac{A_1 \cdot \sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)}$$

$$= \frac{0.65974 \times (5.67 \times 10^{-8}) (500^4 - 300^4)}{\frac{1}{0.79} + \frac{0.65974}{3.6} \left(\frac{1}{0.93} - 1 \right)}$$

$$= 1591.5 \text{ W} \quad \dots \text{Ans.}$$

Ex. 10.12.41 : Consider two concentric spheres 'A' and 'B' of diameters 300 mm and 400 mm respectively. Space between these two spheres is evacuated. Liquid gas at -183°C is stored inside the sphere 'A'. The surfaces of the spheres facing each other are coated with aluminium ($\epsilon = 0.03$). Latent heat of vaporization of liquid gas is 210 kJ/kg. If the system is kept in a space vehicle having 40°C , calculate the rate of evaporation of liquid gas.

Soln. :

Given : $r_1 = 0.15 \text{ m}$, $r_2 = 0.2 \text{ m}$,

$$r_1 = \frac{d_1}{2} = \frac{300 \text{ mm}}{2} = 150 \text{ mm} = 0.15 \text{ m};$$

$$T_1 = -173 + 273 = 90 \text{ K};$$

$$T_2 = 40 + 273 = 313 \text{ K}$$

$$r_2 = \frac{d_2}{2} = \frac{400}{2} = 0.2 \text{ m}$$

$$\epsilon_1 = \epsilon_2 = \epsilon_3 = 0.03; \quad h_{fg} = 210 \text{ kJ/kg}$$

Rate of evaporation of liquid gas

The heat of flow from the inner sphere surface to the outer sphere surface is given by,

$$Q_{1-2} = \frac{A_1 \sigma (T_1^4 - T_2^4)}{\left(\frac{1 - \epsilon_1}{\epsilon_1} \right) + \frac{1}{F_{1-2}} + \left(\frac{1 - \epsilon_2}{\epsilon_2} \right) \frac{A_1}{A_2}}$$

$$Q_{12} = \frac{4\pi r_1^2 \sigma (T_1^4 - T_2^4)}{\left(\frac{1 - \epsilon_1}{\epsilon_1} \right) + 1 + \left(\frac{1 - \epsilon_2}{\epsilon_2} \right) \frac{r_1^2}{r_2^2}}$$

$$= \frac{4\pi (0.15)^2 \times (5.67 \times 10^{-8}) (90^4 - 313^4)}{\left(\frac{1 - 0.03}{0.03} \right) + 1 + \left(\frac{1 - 0.03}{0.03} \right) \left(\frac{0.15}{0.2} \right)^2}$$

$$= -2.966 \text{ W} \quad \dots \text{Ans.}$$

Negative sign indicates that heat is gained by surface 1 i.e. heat flows from outside surface to inside surface.

Rate of evaporation of liquid

$$m = \frac{Q_{12}}{h_{fg}} = \frac{2.966 \times 10^{-3}}{210} \text{ kg/s}$$

$$= \frac{2.966 \times 10^{-3} \times 3600}{210} \text{ kg/hr}$$

$$= 0.05085 \text{ kg/hr} \quad \dots \text{Ans.}$$

Ex. 10.12.42 : Calculate the net radiant heat exchange per m^2 area for two large parallel plates of temperature 427°C and 27°C respectively ϵ (hot plate) = 0.9 and ϵ (cold plate) = 0.6. If a polished aluminum shield is placed between them, find the % reduction in heat transfer : ϵ (shield) = 0.4

Soln. :

Refer Fig. P. 10.12.42(a)

$$T_1 = 427 + 273 = 700 \text{ K}$$

$$T_2 = 27 + 273 = 300 \text{ K}$$

$$\epsilon_1 = 0.9, \epsilon_2 = 0.6, \epsilon_3 = 0.4, A = 1 \text{ m}^2$$

(a) Radiant heat transfer Q without shield

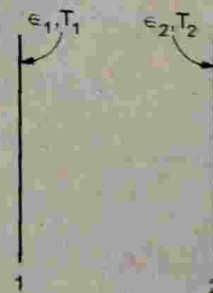


Fig. P. 10.12.42(a)

$$Q = \frac{\sigma A (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$



$$= \frac{5.67 \times 10^{-8} [700^4 - 300^4]}{\left(\frac{1}{0.9} + \frac{1}{0.6} - 1\right)}$$

$$= 7399.35 \text{ W/m}^2$$

...Ans.

(b) Radiant heat transfer Q_{1-3} with shield

Refer Fig. P.10.12.42(b)

When a shield is placed between the plates 1 and 2, then

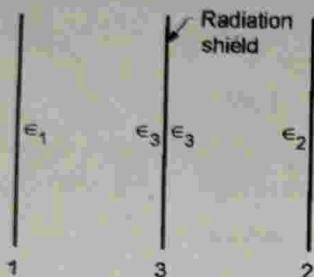


Fig P. 10.12.42(b)

$$(Q_{13})_{\text{net}} = (Q_{32})_{\text{net}}$$

$$\frac{\sigma A (T_1^4 - T_3^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1} = \frac{\sigma A (T_3^4 - T_2^4)}{\frac{1}{\epsilon_2} + \frac{1}{\epsilon_3} - 1}$$

$$\frac{700^4 - T_3^4}{\frac{1}{0.9} + \frac{1}{0.4} - 1} = \frac{T_3^4 - 300^4}{\frac{1}{0.4} + \frac{1}{0.6} - 1}$$

$$\frac{700^4 - T_3^4}{2.6111} = \frac{T_3^4 - 300^4}{3.1666}$$

$$3.1666 \times \frac{(700^4 - T_3^4)}{2.6111} = T_3^4 - 300^4$$

$$\therefore 1.21276 (700^4 - T_3^4) = T_3^4 - 300^4$$

$$\therefore 2.9118 \times 10^{11} - 1.2126 T_3^4 = T_3^4 - 300^4$$

$$\therefore 2.992 \times 10^{11} = T_3^4 + 1.21276 T_3^4$$

$$\therefore T_3 = 606.4 \text{ K}$$

$$(Q_{13})_{\text{net}} = \frac{\sigma A (T_1^4 - T_3^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1}$$

$$= \frac{5.67 \times 10^{-8} \times 1 [700^4 - 606.4^4]}{\left(\frac{1}{0.9} + \frac{1}{0.4} - 1\right)}$$

$$= 2277.495 \text{ W/m}^2$$

...Ans.

(c) % Reduction of radiant with shield

$$= \frac{7399.35 - 2277.495}{7399.35} \times 100 \%$$

$$= 69.22 \%$$

...Ans.

Ex. 10.12.43 : Two large parallel planes with emissivities of 0.8 and 0.4 have temperature of 427°C and 27°C respectively. If they are kept facing each other, find the rate of radiation transfer between them. If this rate of heat transfer is to be reduced to $\frac{1}{4}$ of original value, calculate the

emissivity of radiation shield to be inserted in between them. The shield has same emissivity on both sides.

Soln. :**Given :** $\epsilon_1 = 0.8$, $\epsilon_2 = 0.4$

$$T_1 = 427 + 273 = 700 \text{ K}$$

$$T_2 = 27 + 273 = 300 \text{ K}$$

1. Radiation heat transfer, Q_1

Using Equation (10.11.1) as applied two large parallel planes,

$$Q_1 = \frac{\sigma \cdot (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{5.67 \times 10^{-8} (700^2 - 300^4)}{\frac{1}{0.8} + \frac{1}{0.4} - 1}$$

$$= 4783.4 \text{ W}$$

...Ans.

2. Emissivity of radiation shield if the rate of heat transfer to be reduced to $\frac{1}{4}$, ϵ_3

$$Q_2 = \frac{Q_1}{4} = \frac{4783.4}{4} = 1195.8 \text{ W}$$

$$Q_2 = 1195.8 \text{ W} = \frac{\sigma \cdot (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{2}{\epsilon_3} + \frac{1}{\epsilon_2} - 2}$$

$$1195.8 = \frac{5.67 \times 10^{-8} (700^4 - 300^4)}{\frac{1}{0.8} + \frac{2}{\epsilon_3} + \frac{1}{0.4} - 2}$$

$$\epsilon_3 = 0.216$$

...Ans.

Ex. 10.12.44 : Two large parallel planes 'A' and 'D' are maintained at temperatures of 1500 K and 600 K respectively. Their emissivities are 0.9 and 0.4 respectively.

Two radiation shields 'B' with emissivity = 0.5 and 'C' with emissivity = 0.2 are inserted in between them, such that A, B, C and D are placed one after the other. Calculate

- Heat transfer rate without radiation shields,
- Heat transfer rate with radiation shields,
- Temperatures attained by planes 'B' and 'C'.

Soln. :**Given :**

$$\text{Plane A} \rightarrow \epsilon_1 = 0.9, \quad T_1 = 1500 \text{ K} \quad \epsilon_1 = \epsilon_A = 0.9$$

$$\text{Plane D} \rightarrow \epsilon_2 = 0.4, \quad T_2 = 600 \text{ K} \quad \epsilon_2 = \epsilon_D = 0.4$$

Shield B $\rightarrow \epsilon_3 = 0.5, T_3 = ?$ Shield C $\rightarrow \epsilon_4 = 0.2, T_4 = ?$

$\epsilon_3 = \epsilon_B = 0.5$

$\epsilon_4 = \epsilon_D = 0.2$

(i) Heat transfer rate without shields

$$q = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{(5.67 \times 10^{-8}) (1500^4 - 600^4)}{\frac{1}{0.9} + \frac{1}{0.4} - 1}$$

$$= 107117 \text{ W/m}^2$$

...Ans.

(ii) Heat transfer with shields

$$q_1 = \frac{\sigma (T_A^4 - T_B^4)}{\frac{1}{\epsilon_A} + \frac{1}{\epsilon_D} - 1 + \left(\frac{2}{\epsilon_B} - 1 + \frac{2}{\epsilon_C} - 1 \right)}$$

$$= \frac{5.67 \times 10^{-8} (1500^4 - 600^4)}{\frac{1}{0.9} + \frac{1}{0.4} - 1 + \left(\frac{2}{0.5} - 1 \right) + \left(\frac{2}{0.2} - 1 \right)}$$

$$q_1 = 18906.3 \text{ W}$$

...Ans.

Shields do not deliver or remove heat,

$q_1 = q_{AB} = q_{BC} = q_{CD}$

(iii) Temperature of shield B and C

$$\therefore \text{For plane A and shield B, } q_1 = \frac{\sigma (T_A^4 - T_B^4)}{\frac{1}{\epsilon_A} + \frac{1}{\epsilon_B} - 1}$$

$$189050.3 = \frac{(5.67 \times 10^{-8}) (1500^4 - T_B^4)}{\frac{1}{0.9} + \frac{1}{0.5} - 1}$$

$T_B = 1454.9 \text{ K}$

...Ans.

For surface C and D

$$q_1 = \frac{\sigma (T_C^4 - T_D^4)}{\frac{1}{\epsilon_C} + \frac{1}{\epsilon_D} - 1}$$

$$18906.3 = \frac{5.67 \times 10^{-8} (T_C^4 - 600^4)}{\frac{1}{0.2} + \frac{1}{0.4} - 1}$$

$T_C = 1231.08 \text{ K}$

...Ans.

Ex. 10.12.45 : A long steel rod 20 mm in diameter is to be heated from 427 °C to 538 °C. It is placed concentrically in a long cylindrical furnace which has an inside diameter of 160 mm. The inner surface of the furnace is at a temperature of 1093 °C and has an emissivity of 0.85. If the surface of the rod has an emissivity of 0.6. Estimate the time required for heating operation. Density of steel = 7000 kg/m³; Specific heat = 0.57 kJ/kg-K.

Soln. :**Steel rod :** $d_1 = 20 \text{ mm} = 0.02 \text{ m}, T_1 = 427^\circ\text{C} = 700 \text{ K};$

$T_3 = 538^\circ\text{C} = 811 \text{ K}$

Furnace : $d_2 = 160 \text{ mm} = 0.16 \text{ m}; T_2 = 1093^\circ\text{C} = 1366 \text{ K}$

The configuration corresponds to a relatively large body enclosing a smaller body. The inner body sees only the outer surface and not itself. Therefore, $F_{12} = 1$.

Hence, the radiant heat transfer rate per unit length of tube is given by,

$$Q_{12} = \frac{\sigma \cdot A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \left(\frac{1}{\epsilon_2} - 1 \right) \times A_1/A_2}$$

$$= \frac{(5.67 \times 10^{-8}) \times (\pi \times 0.02 \times 1) (T_1^4 - T_2^4)}{\frac{1}{0.6} + \left(\frac{1}{0.85} - 1 \right) \times \frac{\pi \times 0.02 \times 1}{\pi \times 0.16 \times 1}}$$

$$Q_{12} = 0.211 \left[\left(\frac{T_1}{100} \right)^4 - \left(\frac{T_2}{100} \right)^4 \right]$$

At the beginning of the heating period,

$T_1 = 700 \text{ K and at the end } T_2 = 811 \text{ K}$

Therefore,

Initially, $Q_{12} = 0.211 \left[\left(\frac{1366}{100} \right)^4 - \left(\frac{700}{100} \right)^4 \right] = 6840 \text{ W}$

Finally, $Q_{12} = 0.211 \left[\left(\frac{1366}{100} \right)^4 - \left(\frac{811}{100} \right)^4 \right] = 6433.8 \text{ W}$

 $\therefore$ Mean rate of heating per unit length,

$$= \frac{1}{2} [6840 + 6433.8] = 6637 \text{ W} = 6.627 \text{ kJ/s}$$

Heat required to raise the temperature of rod from 700 K to 811 K.

$= m C_p \Delta T$

$$= \left[\frac{\pi}{4} (0.02)^2 \times 7000 \right] \times 0.67 \times (811 - 700)$$

$= 163.55 \text{ kJ}$

 $\therefore$ Time required for the desired heating,

$$t = \frac{163.55}{6.637} = 24.64 \text{ sec}$$

...Ans.

Ex. 10.12.46 : Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500 °C.

(Stefan Boltzmann constant = 5.67×10^{-8}).(i) Monochromatic emissive power at 1.2 μm .

(ii) Wave length at which the emission is maximum.

(iii) Total emissive power of the furnace if it is assumed as a real surface with emissivity equal to 0.8.

Soln. :

Given : $T_s = 2500^\circ \text{C} = 2773 \text{ K}$; black body(i) Monochromatic emissive power at $\lambda = 1.2 \mu\text{m}$

$$\begin{aligned}
 e_{\lambda} &= \frac{2\pi C_1}{\lambda^5 (e^{C_2/\lambda T} - 1)} \\
 &= \frac{2\pi \times 0.596 \times 10^{-16}}{(1.2 \times 10^{-6})^5 \left(e^{\frac{0.014387}{1.2 \times 10^{-6} \times 2773} - 1} \right)} \\
 &= \frac{3.743 \times 10^{-16}}{2.488 \times 10^{-30} \times (e^{4.32} - 1)} \\
 &= \frac{3.743 \times 10^{-16}}{74.2 \times 2.488 \times 10^{-30}} \\
 &= 2 \times 10^{12} \text{ W/m}^2 \quad \dots \text{Ans.}
 \end{aligned}$$

(ii) Wavelength, at which emission is maximum,

$$\begin{aligned}
 \lambda_m \cdot T &= 0.0029 \text{ mK} \\
 \lambda_m &= \frac{0.0029}{2773} = 1.05 \times 10^{-4} \text{ m} = 1.05 \mu\text{m} \quad \dots \text{Ans.}
 \end{aligned}$$

(iii) Total emissive power of furnace when it is assumed as a real surface with $\epsilon = 0.8$.

$$\begin{aligned}
 E &= \epsilon \sigma AT^4 = 0.8 \times 5.67 \times 10^{-8} \times 1 \times (2773)^4 \\
 &= 2.68 \times 10^6 \text{ W/m}^2 \quad \dots \text{Ans.}
 \end{aligned}$$

Ex. 10.12.47 : 10 mm OD pipe carries a cryogenic fluid at 80 K. This pipe is encased by another pipe of 15 mm OD, and the space between them is evacuated. The outer pipe is at 280 K. Emissivity of inner and outer surfaces is 0.2 and 0.3 respectively. (i) Determine the radiant heat flow rate over a pipe length of 5 m. (ii) If a radiation shield of diameter 12 mm and emissivity 0.05 on both sides is placed between the pipes, determine the percentage reduction in flow. (iii) What is the equilibrium temperature of the shield?

Soln. :

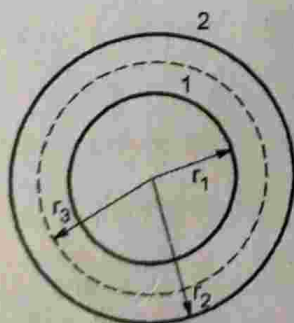


Fig. P. 10.12.47

Refer Fig. P. 10.12.47

 $d_1 = 10 \text{ mm} = 0.01 \text{ m}$; $T_1 = 80 \text{ K}$; $\epsilon_1 = 0.2$, $d_2 = 15 \text{ mm} = 0.015 \text{ m}$; $T_2 = 280 \text{ K}$; $\epsilon_2 = 0.3$; $l = 5 \text{ m}$ $d_3 = 12 \text{ mm} = 0.012 \text{ m}$; $\epsilon_3 = 0.05$ (i) Radiant heat flow rate, Q_{1-2}

$$\begin{aligned}
 Q_{12} &= \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} \\
 Q_{12} &= \frac{5.67 \times 10^{-8} \times \pi \times 0.01 \times 5 (80^4 - 280^4)}{\frac{1}{0.2} + \frac{\pi \times 0.01 \times 5}{\pi \times 0.015 \times 5} \left(\frac{1}{0.3} - 1 \right)} \\
 Q_{12} &= \frac{-54.35}{5 + 1.56} = -8.29 \text{ W} \quad \dots \text{Ans.}
 \end{aligned}$$

Negative sign indicates that heat is flowing from pipe 2 to pipe 1.

(ii) Percentage heat reduction when shield is used

Heat flow from pipe 1 to shield 3 ;

$$\begin{aligned}
 Q_{13} &= \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right) + \frac{A_1}{A_3} \left(\frac{2}{\epsilon_3} - 1 \right)} \\
 &= \frac{-54.35}{6.56 + \frac{\pi \times 0.01 \times 5}{\pi \times 0.012 \times 5} \left(\frac{2}{0.05} - 1 \right)} \\
 &= \frac{-54.35}{6.56 + 32.5} = -1.39 \text{ W}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \% \text{ reduction} &= \frac{Q_{12} - Q_{13}}{Q_{12}} \times 100 \\
 &= \frac{-8.29 - (-1.39)}{-8.29} \times 100 \\
 &= 83.2 \% \quad \dots \text{Ans.}
 \end{aligned}$$

(iii) Equilibrium temperature of shield, T_3

$$\begin{aligned}
 Q_{13} &= \frac{\sigma A_1 (T_1^4 - T_3^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_3} \left(\frac{1}{\epsilon_3} - 1 \right)} \\
 -1.39 &= \frac{5.67 \times 10^{-8} \times \pi \times 0.01 \times 5 (80^4 - T_3^4)}{\frac{1}{0.2} + \frac{\pi \times 0.01 \times 5}{\pi \times 0.012 \times 5} \left(\frac{1}{0.05} - 1 \right)} \\
 -1.39 &= \frac{8.9 \times 10^{-9} (4.096 \times 10^7 - T_3^4)}{5 + 15.833} \\
 -\frac{1.39 \times 20.833}{8.9 \times 10^{-9}} &= 4.096 \times 10^7 - T_3^4 \\
 -325.374 \times 10^7 &= 4.096 \times 10^7 - T_3^4 \\
 T_3 &= 239.58 \text{ K} \quad \dots \text{Ans.}
 \end{aligned}$$

Ex. 10.12.48 : The net radiation from the surface of two parallel plates maintained at temp. T_1 and T_2 is to be reduced by 99%. Calculate the numbers of screens to be placed between the two surfaces to achieve this reduction in heat exchange. Assuming the emissivity of screens as 0.04 and that of surfaces as 0.7.

Soln. :

$$\epsilon_1 = \epsilon_2 = 0.7; \quad \epsilon_s = 0.04$$

Heat transfer without shield

$$Q = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{\sigma (T_1^4 - T_2^4)}{1.857} = 0.538 \sigma (T_1^4 - T_2^4)$$

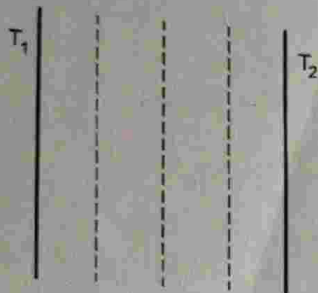


Fig. P. 10.12.48

Since 99% reduction is desired,

$$Q_{\text{with screen}} = 0.01 Q_{\text{without screen}} = 0.00538 \sigma (T_1^4 - T_2^4)$$

$$Q_{\text{with screen}} = \frac{\sigma (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1 \right) + n \left(\frac{2}{\epsilon_s} - 1 \right)}$$

$$= \frac{\sigma (T_1^4 - T_2^4)}{1.857 + n \left(\frac{2}{0.04} - 1 \right)}$$

$$= 0.00538 \sigma (T_1^4 - T_2^4)$$

or

$$1.857 + n \left(\frac{2}{0.04} - 1 \right) = 185.87$$

$$n \times (50 - 1) = 184$$

$$n = 3.755 \approx 4 \text{ screens}$$

...Ans.

Ex. 10.12.49 : A pipe carrying steam having an outside diameter of 20 cm passes through a large room and is exposed to air at temperature of 30°C. Pipe surface temperature is 200°C. Find the heat loss per meter length of pipe both by convection and radiation taking emissivity of the pipe surface as 0.8.

Use following relations :

$Nu = 0.53 (Gr Pr)^{0.25}$ for horizontal pipe Air properties :

Temp °C	k (W/mK)	$\nu \times 10^{-6}$ (m ² /s)	Pr
30	0.0267	16.00	0.701
115	0.033	24.93	0.687
200	0.0393	34.85	0.680

Soln. :

Given : Pipe diameter, $d = 20 \text{ cm} = 0.2 \text{ m}$;

$T_{\infty} = 30^\circ\text{C} = 303 \text{ K}$

$T_s = 200^\circ\text{C} = 473 \text{ K}, \quad \epsilon = 0.8$

$Nu = 0.53 (Gr Pr)^{0.25}$

$$\text{Mean temperature, } T_m = \frac{T_s + T_{\infty}}{2} = \frac{200 + 30}{2} = 115^\circ\text{C}.$$

Therefore,

$$k = 0.033 \text{ W/mK}; \quad \nu = 24.93 \times 10^{-6} \text{ m}^2/\text{s}, \quad Pr = 0.687$$

Room area, $A_2 \gg \gg$ pipe area, A_1 , therefore $\frac{A_1}{A_2} \approx 0$.

and $A_1 = \pi d \cdot L = \pi \times 0.2 \times 1 = 0.2 \pi \text{ m}^2/\text{length of pipe}$.

Heat is lost both by radiation and convection.

(i) **Heat loss by radiation**

$$Q_r = \frac{\sigma \cdot A_1 (T_s^4 - T_{\infty}^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} = \frac{\sigma \cdot A_1 (T_s^4 - T_{\infty}^4)}{\frac{1}{\epsilon_1} + 0}$$

$$Q_r = \epsilon_1 \sigma A_1 (T_s^4 - T_{\infty}^4)$$

$$= 0.8 \times (5.67 \times 10^{-8}) \times (0.2 \pi) (473^4 - 303^4)$$

$$Q_r = 1185.6 \text{ W/m of pipe length} \quad \dots(i)$$

(ii) **Heat loss by convection, Q_c**

$$\beta = \frac{1}{T_m} = \frac{1}{115 + 273} = \frac{1}{388}$$

$$Gr = \frac{g \cdot \beta \cdot \Delta T \cdot L^3}{\nu^2}$$

$$= 9.81 \times \frac{1}{388} \times \frac{(200 - 30) \times 1^3}{(24.93 \times 10^{-6})^2}$$

$$= 5.53 \times 10^7$$

$$Gr Pr = 5.53 \times 10^7 \times 0.687 = 38 \times 10^6$$

$$Nu = 0.53 (Gr Pr)^{0.25} = \frac{h \cdot d}{k}$$

$$0.53 (38 \times 10^6)^{0.25} = \frac{h \times 0.2}{0.033}$$

$$h = 6.87 \text{ W/m}^2 \text{ K}$$

$$Q_c = h \cdot A_1 (T_s - T_{\infty})$$

$$= 6.87 \times (\pi \times 0.2 \times 1) (200 - 30)$$

$$= 773.8 \text{ W/m length of pipe} \quad \dots(ii)$$

(iii) Total heat loss, Q

$$Q = Q_r + Q_c = 1185.6 + 773.8 \\ = 1919.4 \text{ W/m length} \quad \dots\text{Ans.}$$

Ex. 10.12.50 : Three hollow thin walled cylinders having diameters 10 cm, 20 cm and 30 cm are arranged concentrically. The temperature of innermost and outermost cylindrical surfaces are 80 K and 280 K respectively. Assume vacuum between annular spaces, find the steady state temperature attained by the cylindrical surface having diameter of 20 cm.

Take $\epsilon_1 = \epsilon_2 = \epsilon_3 = 0.05$.

Soln. : Refer Fig. P. 10.12.50

$$D_1 = 10 \text{ cm} = 0.1 \text{ m}$$

$$D_2 = 20 \text{ cm} = 0.2 \text{ m}$$

$$D_3 = 30 \text{ cm} = 0.3 \text{ m}$$

$$\therefore r_1 = \frac{D_1}{2} = \frac{0.1}{2} = 0.05 \text{ m}$$

$$r_2 = \frac{D_2}{2} = \frac{0.2}{2} = 0.1 \text{ m}$$

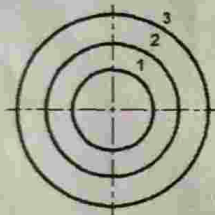


Fig. P. 10.12.50

$$r_3 = \frac{D_3}{2} = \frac{0.3}{2} = 0.15 \text{ m}$$

$$T_1 = 80 \text{ K and } T_3 = 280 \text{ K}$$

$$\epsilon_1 = \epsilon_2 = \epsilon_3 = 0.05$$

Under steady state conditions,

$$Q_{1-2} = Q_{2-3}$$

$$\frac{A_1 \sigma (T_1^4 - T_2^4)}{\left(\frac{1-\epsilon_1}{\epsilon_1}\right) + \frac{1}{F_{12}} + \left(\frac{1-\epsilon_2}{\epsilon_2}\right) \frac{A_1}{A_2}} = \frac{A_2 \sigma (T_2^4 - T_3^4)}{\left(\frac{1-\epsilon_2}{\epsilon_2}\right) + \frac{1}{F_{23}} + \left(\frac{1-\epsilon_3}{\epsilon_3}\right) \frac{A_2}{A_3}}$$

$$(\because F_{12} = F_{23} = 1 \text{ and } \epsilon_1 = \epsilon_2 = \epsilon_3 = 0.05)$$

$$\left(\therefore \frac{A_1}{A_2} = \frac{2\pi r_1 L}{2\pi r_2 L} = \frac{r_1}{r_2} = \frac{0.05}{0.1} = 0.5\right)$$

$$\frac{A_2}{A_3} = \frac{2\pi r_2 L}{2\pi r_3 L} = \frac{r_2}{r_3} = \frac{0.1}{0.15} = 0.6666$$

$$\begin{aligned} \therefore \frac{A_1 (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_2} - \frac{\epsilon_1}{\epsilon_1}\right) + 1 + \left(\frac{1}{\epsilon_2} - 1\right) \frac{A_2}{A_3}} &= \frac{A_2 (T_2^4 - T_3^4)}{\left(\frac{1}{\epsilon_2} - 1\right) + 1 + \left(\frac{1}{\epsilon_3} - 1\right) \frac{A_2}{A_3}} \\ \therefore \frac{2\pi r_1 L (T_1^4 - T_2^4)}{\frac{1}{\epsilon_2} + \left(\frac{1}{\epsilon_2} - 1\right) \frac{A_1}{A_2}} &= \frac{2\pi r_2 L (T_2^4 - T_3^4)}{\frac{1}{\epsilon_2} + \left(\frac{1}{\epsilon_3} - 1\right) \frac{A_2}{A_3}} \\ \therefore \frac{r_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_2} + \left(\frac{1}{\epsilon_2} - 1\right) \frac{A_1}{A_2}} &= \frac{r_2 (T_2^4 - T_3^4)}{\frac{1}{\epsilon_2} + \left(\frac{1}{\epsilon_3} - 1\right) \frac{A_2}{A_3}} \\ \therefore \frac{0.05 (T_1^4 - T_2^4)}{0.05 + \left(\frac{1}{0.05} - 1\right) 0.5} &= \frac{0.1 (T_2^4 - T_3^4)}{0.05 + \left(\frac{1}{0.05} - 1\right) 0.666} \\ \frac{T_1^4 - T_2^4}{29.5} &= \frac{2(T_2^4 - T_3^4)}{32.6654} \\ \frac{T_1^4 - T_2^4}{29.5} &= \frac{T_2^4 - T_3^4}{16.3327} \\ \frac{80^4 - T_2^4}{29.5} &= \frac{T_2^4 - 280^4}{16.3327} \\ \therefore T_2 &= 251 \text{ K} \quad \dots\text{Ans.} \end{aligned}$$

Ex. 10.12.51 : Cryogenic fluid flows through a tube of 90 mm outer diameter and 30 mm inner. Surface emissivities of inner and outer tubes are 0.2 and 0.5, while respective temperatures are 100 K and 300 K. Find :

- Heat gain rate by the fluid per meter length of tube.
- Percentage reduction in heat gain, if a radiation shield of tubular shape having 45 mm diameter and emissivities of 0.2 on inner surface and 0.05 on outer surface is introduced between the two tubes.

Soln. : Refer Fig. P. 10.12.51.

$$d_2 = 90 \text{ mm} = 0.09 \text{ m},$$

$$d_1 = 30 \text{ mm} = 0.03 \text{ m}$$

$$\epsilon_1 = 0.2, \quad \epsilon_2 = 0.5,$$

$$T_1 = 100 \text{ K}, \quad T_2 = 300 \text{ K}$$

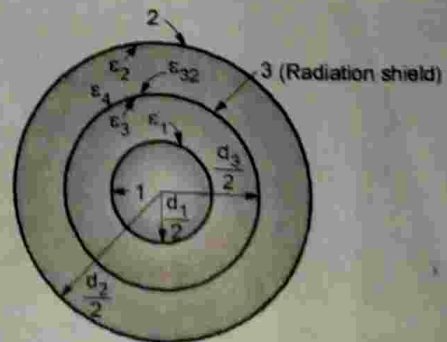


Fig. P. 10.12.51

Consider length of tube, $L = 1 \text{ m}$

$$A_1 = \pi d_1 L; A_2 = \pi d_2 L$$

(a) Heat gain rate by fluid/m length of tube, Q_{12}

$$Q_{12} = \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right)} \quad \dots \text{Equation (10.11.2)}$$

$$= \frac{(5.67 \times 10^{-8}) \times (\pi \times 0.03 \times 1) (100^4 - 300^4)}{\frac{1}{0.2} + \frac{(\pi \times 0.03 \times 1)}{(\pi \times 0.09 \times 1)} \times \left(\frac{1}{0.5} - 1 \right)}$$

$$= -8.016 \text{ W}$$

$$Q_{12} = 8.016 \text{ W}$$

i.e.

...Ans.

Negative sign shows that the radiant heat is received from surface 2 to surface 1 i.e. received by fluid.

(b) Percentage reduction in heat gain if radiation shield of diameter, $d_3 = 45 \text{ mm} = 0.045 \text{ m}$ and emissivities $\epsilon_{31} = 0.1$ on inner surface and $\epsilon_{32} = 0.05$ on outer surface is provided

$$\frac{A_1}{A_2} = \frac{d_1}{d_2} \text{ and } \frac{A_1}{A_3} = \frac{d_1}{d_3} \quad (\because A = \pi dL)$$

Heat transfer with shield,

$$Q_{13} = \frac{\sigma A_1 (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right) + \frac{A_1}{A_3} \left(\frac{1}{\epsilon_{31}} + \frac{1}{\epsilon_{32}} - 1 \right)}$$

$$= \frac{(5.67 \times 10^{-8}) \times (\pi \times 0.03 \times 1) (100^4 - 300^4)}{\frac{1}{0.2} + \frac{0.03}{0.09} \left(\frac{1}{0.5} - 1 \right) + \frac{0.03}{0.045} \left(\frac{1}{0.1} + \frac{1}{0.05} - 1 \right)}$$

$$= -1.733 \text{ W}$$

$$\text{Percentage reduction} = \frac{Q_{12} - Q_{13}}{Q_{12}} \times 100 \%$$

$$= \frac{-8.016 - (-1.733)}{-8.016} \times 100$$

$$= 78.38 \%$$

...Ans.

Ex. 10.12.52 : In a boiler, heat is radiated from the burning fuel (A_1) to the side walls (A_3) and the boiler tubes (A_2) at the top. Assume black body behaviour and side walls perfectly insulated. Derive the equation for the temperature of the side walls T_3 as a function of the temperatures for the fuel bed T_1 and the boiler tubes T_2 and the corresponding areas A_1 and A_2 . Calculate also net radiative heat transfer to the boiler tubes if $A_1 = A_2 = 0.35 \text{ m}^2$.

$$F_{12} = F_{23} = F_{13} = 0.5; \quad T_1 = 1700^\circ\text{C} \quad T_2 = 300^\circ\text{C}$$

Comments on the results.

Soln. : Refer Fig. P. 10.12.52(a).

Since surface A_3 is insulated; $Q_3 = 0$, Hence $Q_{13} + Q_{23} = 0$

$$\therefore A_1 F_{13} (T_1^4 - T_3^4) + A_2 F_{23} (T_2^4 - T_3^4) = 0$$

$$\text{or } A_1 F_{13} T_1^4 - A_1 F_{13} T_3^4 + A_2 F_{23} T_2^4 - A_2 F_{23} T_3^4 = 0$$

$$\text{or } T_3^4 (A_1 F_{13} + A_2 F_{23}) = A_1 F_{13} T_1^4 + A_2 F_{23} T_2^4$$

$$\text{or } T_3^4 = \frac{A_1 F_{13} T_1^4 + A_2 F_{23} T_2^4}{A_1 F_{13} + A_2 F_{23}}$$

$$\text{or } T_3 = \left[\frac{A_1 F_{13} T_1^4 + A_2 F_{23} T_2^4}{A_1 F_{13} + A_2 F_{23}} \right]^{1/4}$$

Required temperature Expression

Given : $F_{12} = F_{23} = F_{13} = 0.5$; $T_1 = 1700^\circ\text{C}$, $T_2 = 300^\circ\text{C}$.

Equivalent Electrical Circuit (Refer fig. P. 10.12.52(b))

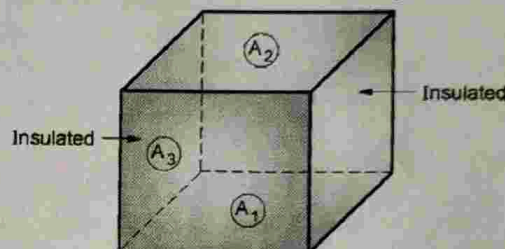


Fig. P. 10.12.52(a)

Since $Q_3 = 0 \Rightarrow R_2$ and R_3 can be taken in series.

Equivalent resistance between T_1 and T_2

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2 + R_3}$$

$$R_1 = \frac{1}{A_1 F_{12}} = \frac{1}{0.25 \times 0.5} = 8;$$

$$R_2 = \frac{1}{A_1 F_{13}} = \frac{1}{0.25 \times 0.5} = 8; \quad R_3 = 8$$

$$\therefore \frac{1}{R_e} = \frac{1}{8} + \frac{1}{8+8} = 0.125 + 0.0625 = 0.1875$$

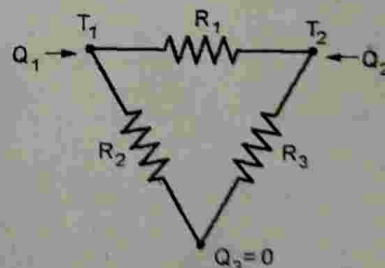


Fig. P. 10.12.52(b)

$$R_e = 5.334 \text{ K/W}$$

$$Q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{R_e} = \frac{5.67 \times 10^{-8} (1700^4 - 300^4)}{5.334}$$

$$= 160 \text{ kW}$$

...Ans.

Ex. 10.12.53 : An enclosure measures $1.5 \text{ m} \times 1.75 \text{ m}$ with a height of 2 m . Under steady state equilibrium conditions, the wall and ceiling are maintained at 525 K and floor at 400 K . Determine the net radiation to floor.

ϵ_1 (emissivity of ceiling and wall) = 0.85

ϵ_2 (emissivity of floor) = 0.75

Take $\sigma_b = 4.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$.

Soln. :

Enclosure $l = 1.5 \text{ m}$, $b = 1.75 \text{ m}$, $h = 2 \text{ m}$

Temperature of wall and ceiling, $T_1 = 525 \text{ K}$

Temperature of floor $T_2 = 400 \text{ K}$

Emissivity of wall and ceiling $\epsilon_1 = 0.85$

Emissivity of floor $\epsilon_2 = 0.7$

A_1 = Total area of walls and ceiling

$$= [(l + b) h] + 2 \times l \times b = 11.75 \text{ m}^2$$

$$A_2 = \text{Floor area} = 1.5 \times 1.75 = 2.625 \text{ m}^2$$

By summation rule

$$F_{2-1} + F_{2-2} = 1$$

$$\therefore F_{2-1} = 1 \quad \text{so floor is completely enclosed by } A_1$$

$$\therefore A_1 F_{1-2} = A_2 F_{2-1}$$

$$F_{1-2} = A_2 / A_1 = 0.2234$$

Rate of heat transfer by radiation

$$Q_{1-2} = \sigma A_1 F_{1-2} (T_1^4 - T_2^4)$$

$$\begin{aligned} \therefore Q_{1-2} &= (5.67 \times 10^{-8}) \times 11.75 \times 0.2234 \\ &\quad \times (525^4 - 400^4) \\ &= 7496.67 \text{ kW} \end{aligned}$$

...Ans.

Ex. 10.12.54 : An electric wire of 0.25 mm diameter, $\epsilon = 0.4$ is placed within a tube of 2.5 mm diameter, $\epsilon = 0.6$ having negligible thickness. This tube in turn is placed concentrically within a tube of 5 mm diameter, $\epsilon = 0.6$. Annular spaces can be assumed to be evacuated completely. If the surface temperature of the outer tube is maintained at 5°C , What must be the temperature of wire so to maintain the temperature of inner tube at 120°C ?

Soln. :

Given :

$$D_1 = 0.25 \text{ mm}; \quad \epsilon_1 = 0.4; \quad T_1 = ?$$

$$D_2 = 2.5 \text{ mm}; \quad \epsilon_2 = 0.6; \quad T_2 = 120^\circ\text{C} + 273 = 393 \text{ K}$$

$$D_3 = 5 \text{ mm}; \quad \epsilon_3 = 0.7; \quad T_3 = 5^\circ\text{C} + 273 = 278 \text{ K}$$

For steady state heat flow conditions ;

Heat loss rate from wire 1 to tube 2 is equal to heat loss rate from tube 2 to tube 3.

$$\therefore \frac{\sigma (T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}} = \frac{\sigma (T_2^4 - T_3^4)}{\frac{1 - \epsilon_2}{A_2 \epsilon_2} + \frac{1}{A_2 F_{23}} + \frac{1 - \epsilon_3}{A_3 \epsilon_3}}$$

$$A_1 = \pi D_1 L = \pi \times 0.25 \times 10^{-3} = 0.785 \times 10^{-3} \text{ m}^2/\text{meter length of wire}$$

$$A_2 = \pi D_2 L = \pi \times 2.5 \times 10^{-3} \times 1 = 7.85 \times 10^{-3} \text{ m}^2/\text{m length of tube}$$

$$A_3 = \pi D_3 L = \pi \times 5 \times 10^{-3} \times 1 = 15.7 \times 10^{-3} \text{ m}^2/\text{m length of tube}$$

$$F_{12} = 1 = F_{23}$$

Substituting values in above expression

$$\begin{aligned} \frac{5.67 \times 10^{-8} (T_1^4 - 393^4)}{\frac{1 - 0.4}{0.785 \times 10^{-3} \times 0.4} + \frac{1}{0.785 \times 10^{-3}} + \frac{1 - 0.6}{8.85 \times 10^{-3} \times 0.6}} &= \frac{5.67 \times 10^{-8} (393^4 - 278^4)}{\frac{1 - 0.6}{7.85 \times 10^{-3} \times 0.6} + \frac{1}{7.85 \times 10^{-3}} + \frac{1 - 0.7}{15.7 \times 10^{-3} \times 0.6}} \\ \frac{T_1^4 - 238.5 \times 10^8}{1910.8 + 1273.9 + 84.93} &= \frac{238.5 \times 10^8 - 59.73 \times 10^8}{84.93 + 127.39 + 27.3} \\ \frac{T_1^4 - 238.5 \times 10^8}{3269.63} &= \frac{179.11 \times 10^8}{239.62} \end{aligned}$$

$$T_1^4 = 238.5 \times 10^8 + 2443.97 \times 10^8$$

$$= 2682.47 \times 10^8$$

$$T_1 = 719.67 \text{ K} = 446.7^\circ\text{C}$$

...Ans.

Ex. 10.12.55 : The net radiation from the surface of two parallel plates maintained at T_1 and T_2 is to be reduced by 80 times. Calculate the numbers of screens to be placed between the two surfaces to achieve this reduction in heat exchange. Assuming the emissivity of screens as 0.05 and that of surfaces as 0.8.

Soln. :

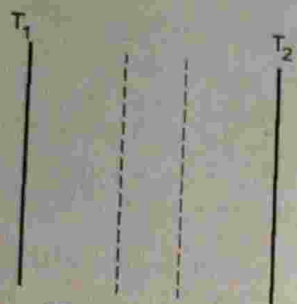


Fig. P. 10.12.55

$$\epsilon_s = 0.05 ; \epsilon_1 = \epsilon_2 = 0.8$$

$$Q_1 = \frac{\sigma(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} = \frac{x}{\frac{1}{0.8} + \frac{1}{0.8} - 1} = \frac{x}{1.5}$$

(assuming $\sigma(T_1^4 - T_2^4) = x$ for convenience)

$$Q_{II} = 0.8 Q_1$$

$$Q_{II} = \frac{\sigma(T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right) + \left(\frac{2}{\epsilon_s} - 1\right)n}$$

where n is number of shields

$$= \frac{x}{1.5 + \left(\frac{2}{0.05} - 1\right)n}$$

$$\frac{x}{1.4 + 39n} = \frac{0.2x}{1.5}$$

$$\frac{1}{1.4 + 39n} = \frac{0.2}{1.5}$$

$$0.2(1.5 + 39n) = 1.5$$

$$0.3 + 7.8n = 1.5$$

$$7.8n = 1.2$$

$$n = 0.15 \approx 1$$

...Ans.

Ex. 10.12.56 : A cylindrical enclosure is formed by 3 surfaces and the details of their shape factors, the emissivity and the temperatures are as given below.

Surface	Shape	Emissivity	Temperature (°C)
1	Curved surface	0.8	500
2	Top end closing disc	0.85	400

Surface	Shape	Emissivity	Temperature (°C)
3	Other end closing disc	0.85	400

Diameters of two closing flat surfaces and the interspacing between them are 25 mm and 100 mm respectively. Shape factor between two identical flat surfaces is 0.05. Calculate net rate of radiant heat transfer leaving from curved surface and reaching the closing flat circular surfaces.

Soln. :

Given :

$$\epsilon_1 = 0.8, \epsilon_2 = \epsilon_3 = 0.85$$

$$T_1 = 500^\circ\text{C} = 773 \text{ K}, T_2 = T_3 = 400^\circ\text{C} = 673 \text{ K}$$

$$D_1 = D_2 = D_3 = 25 \text{ mm} = 0.025 \text{ m}$$

$$l = 100 \text{ mm} = 0.1 \text{ m}$$

$$F_{23} = F_{32} = 0.05$$

We know,

$$F_{11} + F_{12} + F_{13} = 1$$

$$F_{33} + F_{31} + F_{32} = 1$$

$$F_{33} = 0 \text{ (flat surface)}$$

$$F_{12} = F_{13} \text{ (due to symmetry)}$$

$$\text{Since } F_{32} = 0.05$$

$$\therefore 0 + F_{31} + 0.05 = 1$$

$$F_{31} = 0.95$$

and

$$F_{21} = 0.95 \text{ due to symmetry}$$

$$F_{13} \cdot A_1 = F_{31} \cdot A_3$$

$$F_{13} = 0.95 \times \frac{\pi D_3^2}{4\pi D_1 l}$$

$$= 0.95 \times \frac{D_3}{4l} \text{ (as } D_1 = D_2 = D_3)$$

$$= \frac{0.95 \times 0.025}{4 \times 0.1} = 0.0594$$

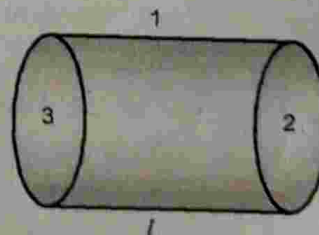


Fig. P. 10.12.56

$$F_{12} = 0.0594 \text{ (by symmetry)}$$

Since $F_{23} = F_{32}$ and surface 2 and 3 are at same temperature, hence no heat flow between these two surfaces. Hence heat transfer between two surfaces 1 and (2 or 3) can be written as.

$$Q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

$$A_1 = \pi D l = \pi \times 0.025 \times 0.1 = 0.00785 \text{ m}^2$$

$$A_2 = \frac{\pi}{4} D^2 = \frac{\pi}{4} \times 0.025^2 = 0.00049 \text{ m}^2$$

$$Q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1 - \epsilon_1}{A_1 \epsilon_1} + \frac{1}{A_1 F_{12}} + \frac{1 - \epsilon_2}{A_2 \epsilon_2}}$$

$$= \frac{5.67 \times 10^{-8} (773^4 - 673^4)}{\left(\frac{1 - 0.8}{0.00785 \times 0.8} \right) + \left(\frac{1}{0.00785 \times 0.0594} \right) + \left(\frac{1 - 0.85}{0.00049 \times 0.85} \right)}$$

$$= 3.376 \text{ W on each surface 2 and 3 ...Ans.}$$

Summary

- **Thermal radiation** is defined as the transfer of heat due to electromagnetic waves without requiring any medium.
- Theories of radiation are :

1. **Wave or Maxwell theory** says that electromagnetic emissions propagate in the form of waves in a hypothetical medium known as **ether**.

$$\text{Wavelength, } \lambda = \frac{\text{Velocity of light, } C (3 \times 10^8 \text{ m/s})}{\text{Frequency, } \nu (\text{per s})}$$

$\lambda = 0.02 - 0.04 \text{ } \mu\text{m}$ (ultraviolet rays), visible radiation ($0.4 - 0.8 \text{ } \mu\text{m}$)

Infrared radiation ($0.8 - 800 \text{ } \mu\text{m}$)

Thermal radiation ($\lambda = 0.1 \text{ } \mu\text{m}$ to $100 \text{ } \mu\text{m}$)

2. **Quantum or Planck's theory** : According to this theory, emissions are in the form of series of entities known as quanta or photon having certain amount of energy which is proportional to its frequency of emission. These quanta move at velocity of light. Thus, Energy/quanta = $h \cdot \nu = h \cdot \frac{c}{\lambda}$; ' h ' = $6.625 \times 10^{-34} \text{ Js}$ (Planck's constant)

- **Total emissive power of a surface, E** is defined as the total emissions of thermal radiation by a surface per unit time per unit area in all directions into hemi-spherical half space above the surface.

$$\left(E = \lim_{dA \rightarrow 0} \frac{dQ}{dA} \right)$$
- **Irradiation, G** is defined as the total energy received per unit time per unit area from all directions at all wavelengths.
- **Total irradiations, G or Q** = Energy reflected, Q_r + Energy absorbed, Q_a + Energy transmitted, Q_T
- **Reflectivity, ρ** = $\frac{\text{Energy reflected, } Q_r}{\text{Total incident energy, } Q}$
- **Absorptivity, α** = $\frac{\text{Energy absorbed, } Q_a}{\text{Total incident energy, } Q}$
- **Transmissivity, τ** = $\frac{\text{Energy transmitted, } Q_T}{\text{Total incident energy, } Q}$
- $\rho + \alpha + \tau = 1$
 - (a) For solids and liquids : $\tau \approx 0$ i.e. $\rho + \alpha = 1$
 - (b) For gases : $\rho \approx 0$ i.e. $\alpha + \tau = 1$
- A **black body** is one which absorbs entire radiations i.e. $\alpha = 1$. It is also a perfect emitter or radiator of heat.
- A body having negligible transmissivity is called **opaque body**.
- A body which reflects the radiant energy is called **white body** ($\alpha = \tau = 0$ and $\rho = 1$)
- A body having same value of emissivity at all wavelengths is called **gray body** (e.g. metals)
- A body which transmits entire radiations is called **transparent body**. ($\tau = 1$)
- The radiant energy per unit area is called **radiant flux**.
- The radiant energy emitted per unit area per unit time is called **emissive power of a surface, e** (W/m^2)
- **Emissive power of a black body** is by e_b .
- **Emissivity, ϵ** = $\frac{\text{Emissive power of a surface, } e}{\text{Emissive power of a black body, } e_b}$
- **Monochromatic emissive power, e_λ** represents the radiant energy emitted by the body per unit time per unit area at a particular wavelength.

$$e = \int_{\lambda=0}^{\lambda=\infty} e_{\lambda} \cdot d\lambda$$

- **Radiosity (J)** is defined as the total amount of energy emitted from a surface per unit time per unit surface area.
- **Laws of radiation** are Planck's, Wien's, Stefan Boltzmann and Kirchoff's law.
- **According to Planck's law** : $e_{b\lambda} = \frac{2\pi C_1}{\lambda^5 [e^{(C_2/\lambda T)} - 1]}$ W/m²
 $e_{b\lambda} \rightarrow$ Monochromatic emissive power of black body at wavelength λ
 $C_1 = 0.596 \times 10^{-16}$ (W/m²),
 $C_2 = 1.4387 \times 10^{-2}$ (m), λ in m, T in Kelvin.
- **Wien's displacement law** gives the relationship between the maximum wavelength λ_m at which maximum monochromatic emissive power is obtained at a given absolute temperature
 $\lambda_m \cdot T = 0.0029$ mK
 $(e_{b\lambda})_{\max} = 1.287 \times 10^{-5} \cdot T^5$
- **Stefan Boltzmann's law** state that the total emissive power of a black body is directly proportional to fourth power of its absolute temperature i.e. $e_b \propto T^4$
 $e_b = \sigma \cdot T^4$
 $e = \sigma \cdot \epsilon \cdot T^4$
 (where, σ = Stefan Boltzmann constant = 5.67×10^{-8} W/m² K⁴)
- **Kirchoff's law** states that the emissivity, ϵ of the surface is equal to its absorptivity, α when body is in thermal equilibrium with its surroundings i.e. ($\epsilon = \alpha$)
- Emissivity at certain wavelength is called **spectral emissivity** ϵ_{λ} .
- Emissivity in certain direction is called **directional emissivity**, ϵ_{θ} .
- Average emissivity in all direction is called **hemispherical emissivity**.
- Emissivity averaged at all wavelength is called **total emissivity**, $\epsilon(T)$.

- A **solid angle**, $d\omega$ is the measure of space enclosed by a conical surface with the vertex of the cone whose apex is at the centre of small area dA_1 relative to the total space area, A_2 of a sphere of a radius, r .

$$\left[d\omega = \frac{A_2}{r^2} \right]$$

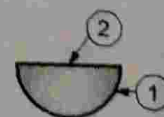
- Solid angle between two small areas dA_2 at dA_1 ,

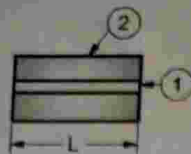
$$d\omega = \frac{dA_2 \cdot \cos \beta_2}{r^2}$$

(where β_2 is the angle made from normal to dA_2 with line joining dA_1 and dA_2)

- **Intensity of radiation (i)** represents the rate of energy leaving a surface in a given direction per unit solid angle per unit area of emitting surface normal to the mean direction in space. $\left[i = \frac{de}{d\omega} \right]$.
- **Lambert cosine law** states that intensity of radiation in any direction is proportional to the cosine of angle made by that direction with the normal. ($i = i_n \cdot \cos \beta$) and $i = \frac{de}{d\omega}$; $e = \pi \cdot e_b$ (for hemisphere)
- **Shape factor** is defined as a fraction of energy emitted by one surface and directly intercepted by the other surface. It is also called as **view factor/configuration factor/angle factor**.
 - (a) **Reciprocity relation** $A_1 F_{12} = A_2 F_{21}$
 - (b) **Enclosure theorem** (i) $F_{11} = 0$ if flat or convex surface (ii) $F_{11} \neq 0$ (concave surface)
 - (c) **Additive rule** : $F_{12} = F_{13} + F_{14}$
 - (d) **Shape factor, $F_{12} = 1$** (if surface is flat or convex) and $F_{21} = \frac{A_1}{A_2}$
 - (e) **Shape factor when two surfaces are large and parallel** ($A_1 = A_2$)
 $\therefore F_{12} = F_{21} = 1$.

1. Hemispherical shape



2. Two concentric cylinders

$$F_{12} = \frac{A_2}{A_1}; F_{11} = 0.5$$

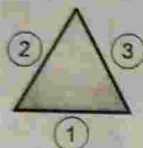
$$F_{12} = 1; F_{22} = 1 - \frac{A_1}{A_2} = 1 - \frac{d_1}{d_2}$$

$$F_{11} = 0; F_{21} = \frac{A_1}{A_2} = \frac{d_1}{d_2}$$

3. Equilateral triangle

$$F_{11} = F_{22} = F_{33} = 0; F_{12} = F_{13} = 0.5; F_{21} = F_{23} = 0.5$$

$$F_{31} = F_{32} = 0.5$$

**Equilateral triangle**

- Radiation shields** are used to reduce the radiative heat transfer between two surfaces by using thin and highly reflective surfaces.

$$q_{12} = \frac{\sigma (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1} \text{ (without shield);}$$

$$q_{13} = \frac{\sigma (T_1^4 - T_3^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1}; q_{32} = \frac{\sigma (T_3^4 - T_2^4)}{\frac{1}{\epsilon_3} + \frac{1}{\epsilon_2} - 1}$$

Exercise

[Note : For answers, please refer the section numbers indicated in bracket.]

- Q. 1** Write a short note on radiation.
[Sections 10.1 and 10.2]
- Q. 2** Define radiation. State the range of wavelengths for ultraviolet, visible and thermal radiations.
[Sections 10.1 and 10.2]
- Q. 3** State the speed of radiant energy propagation between two bodies when there exists a vacuum between bodies. [Section 10.2]

- Q. 4** State the Planck's theory of radiation. Explain how the short wave radiations have much higher energy than long wave radiations. [Section 10.2.2]

- Q. 5** Define following terms :

- (i) Total emissive power of a surface.
[Section 10.3]
- (ii) Irradiation [Section 10.4.3(6)]
- (iii) Reflectivity, absorptivity and transmissivity.
[Section 10.4.1]
- (iv) Black body and Grey body [Section 10.4.2]
- (v) Opaque body [Section 10.4.2 (2)]
- (vi) Emissive power [Section 10.4.3 (1)]
- (vii) Emissivity [Section 10.4.3 (2)]
- (viii) Monochromatic emissive power
[Section 10.4.3 (3)]
- (ix) Monochromatic emissivity [Section 10.4.3 (4)]
- (x) Radiosity [Section 10.4.3 (5)]

- Q. 6** State the Planck's law of radiation and explain.
[Section 10.5.1]

- Q. 7** State and Prove Wien's displacement law.
[Section 10.5.2]

- Q. 8** State and explain Stefan Boltzmann's law. State the value of Stefan Boltzmann's constant, σ .
[Section 10.5.3]

- Q. 9** What is Kirchhoff's law ? Prove that emissivity is equal to absorptivity in isothermal black surroundings. [Section 10.5.4]

- Q. 10** Explain various types of emissivities of grey surfaces. [Section 10.6]

- Q. 11** Define intensity of radiation. [Section 10.7.2]

- Q. 12** Define solid angle. Derive an expression for solid angle between two elementary areas.
[Sections 10.7.1 and 10.7.1.1]

- Q. 13** State and explain Lambert cosine law.
[Section 10.7.2.1]

- Q. 14** Derive an expression for radiant heat exchange between two finite black bodies at a distance L having the areas dA_1 and dA_2 making angles β_1 and β_2 from their respective normals. Hence, define shape factor and explain its meaning.
[Section 10.8]

Q. 15 Define shape factor and discuss the shape factor algebra. [Sections 10.9, 10.9.1 and 10.9.2]

Q. 16 Derive an expression for radiations emitted in the following cases :

(i) Hemispherical shape of radius, R.

[Section 10.9.3 (1)]

(ii) Two concentric cylinders. [Section 10.9.3 (2)]

Q. 17 Explain the various relations to determine shape factor. [Sections 10.9.1 and 10.9.2]

Q. 18 Prove that in case of grey body : $J = \rho \cdot G + \epsilon \cdot E_b$ where notations have their used meaning

[Section 10.10.1]

Q. 19 Explain the surface and space resistances. [Sections 10.10 and 10.10.2]

Q. 20 Show that the radiant heat exchange between two parallel infinite plane surfaces is given as :

$$Q_{12} = \frac{\sigma \cdot A (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right)} \quad [\text{Section 10.11.1}]$$

Q. 21 Show that the expression for radiation heat exchange between two concentric, very long cylinders is given as : $Q_{12} = \frac{\sigma \cdot A_1 \cdot (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)}$

(where, A_1 is the surface area of inside cylinder)

[Section 10.11.2]

Q. 22 What are radiation shields ? Explain why using radiation shields the heat transfer rates are reduced by analytical analysis. [Section 10.12]